ELECTRIC ENGINES

POWERTRAIN CONTROLLER FOR DIY ELECTRIC VEHICLES

LEAF ECU

Installation manual

Installation and setup guide for the powertrain controller built around Nissan Leaf components for DIY electric vehicles.



Revision 2.0 · October 2026
Interactive version with circuit highlighting:
https://diag.electricengines.com.ua/manual/en/

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1 General Information

Leaf ECU is the master powertrain controller: it drives the inverter and the traction-battery contactors, handles charging through the PDM and CHAdeMO, controls cooling and peripherals, and exchanges data with the BMS, the display and the gear selector over CAN.



Fig. 1 Leaf ECU controller

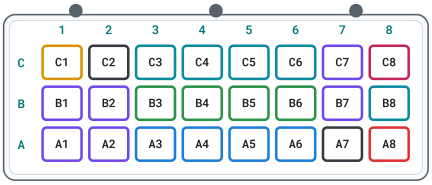
1.1 Controller Connectors

The controller has two 24-way connectors:

- **X1** — black;
- **X2** — grey.

Pin numbering is moulded on the connectors: rows A–C and positions 1–8. Pins can be inserted into the mating connector from the back of the housing without taking it apart — just pull out the yellow lock at the front of the connector first.

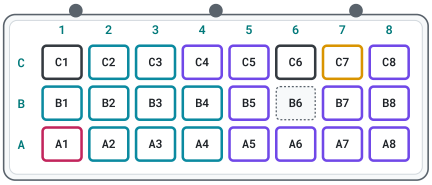
The pinout is compiled from the wiring diagrams in this manual. The "Diagrams" column lists the figures where the pin is used; hover a pin or a table row to see it on the connector. In the text pins are written as X1-B2 — hover such a tag and the same connector pops up.



Wire-side view (rear of the harness connector): pins 1–8 left to right, row C on top, keys on top

PIN	WIRE	FUNCTION	DIAGRAMS
X1-A1		Current sensor — input SIG 2 (LEM sensor channel 1, pin D)	43
X1-A2		Accelerator pedal — sensor 2 signal (not used by firmware yet)	26
X1-A3		CAN0 Powertrain — High, 500 kbps (inverter, PDM, OBD2, display, selector)	7 9 11 13
X1-A4		CAN1 — High: BMS (no CHAdeMO) or CHAdeMO	29 33 41 47 53
X1-A5		CAN2 — High: BMS when CAN1 is used by CHAdeMO	
X1-A6		CAN3 — High: expansion	
X1-A7		Ground	7 9 11 13 43
X1-A8		+12 V permanent from the battery (KL30)	7 9 11 13
X1-B1		Current sensor — input SIG 1 (LEM sensor channel 2, pin A)	43
X1-B2		Accelerator pedal — primary sensor signal	26
X1-B3		CAN0 Powertrain — Low	7 9 11 13
X1-B4		CAN1 — Low	29 33 41 47 53
X1-B5		CAN2 — Low	
X1-B6		CAN3 — Low	
X1-B7		Charge-port PP line to the PDM (pin 29)	7 9 11
X1-B8		+12 V output to the IGN relay coil	7 9 11 13
X1-C1		+5 V output (pedal, sensors)	26 43
X1-C2		Ground (pedal, sensors)	26
X1-C3		Radiator fan relay — high speed	51
X1-C4		CHAdeMO output 0 — enable relay	53
X1-C5		CHAdeMO output 1 — charge permission (inlet pin 4)	53
X1-C6		Coolant pump relay	51
X1-C7		PP (proximity) from the charge inlet	7 9 11
X1-C8		+12 V from the ignition switch (KL15)	7 9 11 13

Fig. 2 ECU — connector X1 (black)



Wire-side view (rear of the harness connector): pins 1–8 left to right, row C on top, keys on top

PIN	WIRE	FUNCTION	DIAGRAMS
X2-A1		+12 V after the ignition relay	
X2-A2		Brake vacuum pump relay	23
X2-A3		Main contactor HV− coil	24 33
X2-A4		Main contactor HV+ coil	24 33
X2-A5		Start button input (+12 V = pressed)	22
X2-A6		Brake booster vacuum sensor signal	23
X2-A7		Analog input (via 10 kΩ) — reserved	
X2-A8		CHAdeMO input 1 — charge sequence 1 (pin 2)	53
X2-B1		Status LED / IGN (D28)	
X2-B2		Power output via MOSFET — reserved	
X2-B3		Brake-light relay (regeneration)	52
X2-B4		Precharge relay coil	24 33
X2-B5		Drive button input (+12 V = on)	22
X2-B6		Not connected (reserved)	
X2-B7		CHAdeMO input 0 — charge sequence 2 (pin 10)	53
X2-B8		Brake-light sense — brake pedal pressed	52
X2-C1		Ground	
X2-C2		Radiator fan relay — low speed	51
X2-C3		Brake-light relay (duplicate of X2-B3)	
X2-C4		CHAdeMO input 2 — proximity (pin 7) / AC gun	53
X2-C5		Reverse button input (+12 V = on)	22
X2-C6		Ground	29
X2-C7		+5 V output (sensors)	23 29
X2-C8		Digital input (optocoupler) — reserved	

Fig. 3 ECU — connector X2 (grey)

1.2 Dimensions and Mounting

The controller is built into an aluminium enclosure with two 24-way connectors. It is fixed with two M6 bolts through the ears on the long sides.

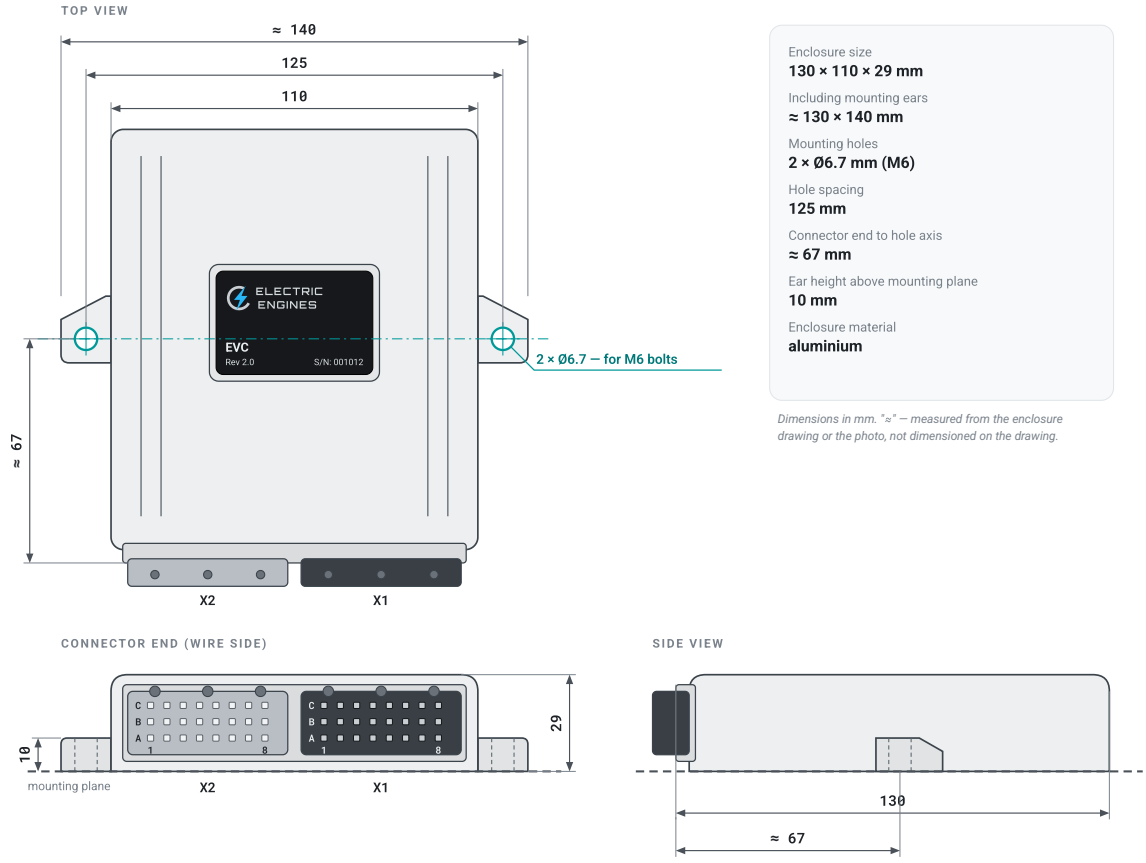


Fig. 4 ECU enclosure dimensions, mounting holes and label: top, connector-end and side views

- The mounting holes lie on one line across the enclosure, 125 mm apart.
- Leave free space in front of the connector end: the plugs are inserted and removed along the enclosure, and the harness needs room to bend.

1.3 CAN Bus

> **WARNING**
> All CAN bus wires must be twisted into twisted pairs!

The controller has four CAN buses; the main one is CAN0 Powertrain (X1-A3 / X1-B3), which connects the inverter, the PDM, the OBD2 socket, the display and the selector. The BMS connects to CAN1 (X1-A4 / X1-B4), or to CAN2 (X1-A5 / X1-B5) when CAN1 is used by CHAdeMO.

Every CAN bus must be terminated with 120 Ω at both ends. **The controller already has 120 Ω built in on every CAN channel**, so the controller sits at one end of the bus and the second 120 Ω must be at the far end. The Leaf inverters also have 120 Ω built in, so CAN0 with a Leaf inverter needs no separate resistors. If the device at the far end has no terminator of its own, fit a 120 Ω resistor next to it. Check: with the bus unpowered, ≈ 60 Ω between CAN-H and CAN-L.

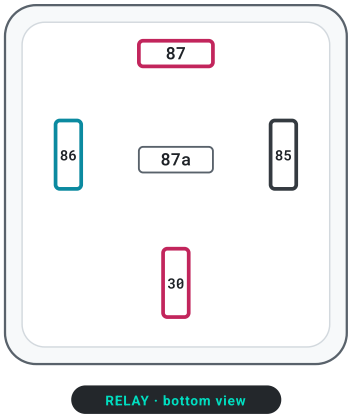
1.4 How to Read the Diagrams

All diagrams in this manual share one style. A wire's colour shows its purpose, not the insulation colour:

COLOUR	PURPOSE
	Traction battery power cables (HV)
	+12 V — supply from the battery / vehicle 12 V system
	+12 V switched — from the ignition switch or through a relay
	+5 V — sensor supply from the controller
	Ground
/	CAN-H / CAN-L
	Signals into the controller (sensors, buttons, pilot lines)
	Controller outputs (relay coils, LEDs)

Relay contacts are numbered **1–2**, the coil **4–3**: pin 4 receives the drive signal from the controller, pin 3 is ground.

The diagrams use ordinary 30–40 A automotive relays (mini ISO). Their terminals carry the standard numbers: **1 → 30**, **2 → 87**, **4 → 86**, **3 → 85**; terminal 87a (changeover relays only) is not used. Click a relay on any diagram to open this connector with the terminal highlighted.



Bottom view, onto the relay terminals

PIN	WIRE	FUNCTION	DIAGRAMS
1		30 — common contact (on the diagrams — 1); the supply may go to 30 or to 87	7 9 11 13 23 51 52 53 62
2		87 — normally open contact (on the diagrams — 2); closes to 30 while the coil is powered	7 9 11 13 23 51 52 53 62
4		86 — coil "+", drive from the ECU (on the diagrams — 4)	7 9 11 13 23 51 52 53 62
3		85 — coil "−", ground (on the diagrams — 3)	7 9 11 13 23 51 52 53 62
87a		87a — normally closed (changeover relays only), not used	

Fig. 5 Automotive relay 30–40 A (mini ISO) — terminals

> **NOTE**
>
> If the relay has a built-in diode or resistor across the coil, coil polarity matters: **86 is "+"** (from the controller), **85 is "−"** (ground).

In the web edition of this manual the diagrams are interactive: click a wire to highlight the whole circuit and open a card listing its connections; click a pin or a device to open its connector view with the pinout. The Expand button opens a diagram full-screen with zoom.

2 12 V Power System

Version 4 controller boards have a new power supply circuit. The controller needs:

- **permanent +12 V** from the battery;
- separately, **+12 V from the ignition switch** (IGN KEY).

After the ignition key is turned off the controller stays on until the motor stops, then it opens the battery contactors and switches itself off.

The full wiring — 12 V supply, IGN relay, CAN0, PDM, charge inlet, inverter and motor resolver — is given in section 3, separately for each inverter: Fig. 7, Fig. 9, Fig. 11.

> **NOTE**
> The Leaf Type 1 charge inlet has resistor R5 right at the socket, under the rubber boot. It must be relocated and wired as shown in the section 3 diagrams (for example Fig. 9).

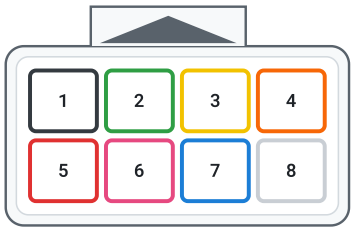
The PP line between the PDM and the charge inlet must also be broken: the inlet PP goes to the controller (X1-C7), and the controller drives its own PP signal to the PDM (X1-B7). When a charging gun is plugged into the inlet, the controller wakes up and starts charging.

3 Inverter

To connect the inverter and the PDM it is best to reuse the factory Leaf harness: strip everything unnecessary, keeping the connectors for the inverter, the PDM and the motor resolver.

Every inverter connects to the controller the same way: CAN-H and CAN-L to CAN0 Powertrain (● X1-A3 / ● X1-B3) as a twisted pair, 12 V supply from the IGN relay, ground to the chassis; the resolver and the motor temperature sensor go over the factory harness between the inverter and the motor. A full diagram is given below for each inverter: only the pin numbers differ.

The resolver connector on the motor itself is the same on all three generations — only the inverter-side pins differ:



Harness-side view of the harness connector (H.S.), per the Nissan drawing; the contacts sit the same way when looking into the socket on the motor

PIN	WIRE	FUNCTION	DIAGRAMS
1		S1 — resolver, output winding S1–S3 · black (B) wire · inverter: Gen 1 — 1, Gen 2 — 20, Gen 3 — 19	7 9 11
8		S3 — resolver, output winding S1–S3 · white (W) wire · inverter: Gen 1 — 6, Gen 2 — 21, Gen 3 — 14	7 9 11
7		S2 — resolver, output winding S2–S4 · blue (L) wire · inverter: Gen 1 — 20, Gen 2 — 18, Gen 3 — 3	7 9 11
6		S4 — resolver, output winding S2–S4 · pink (P) wire · inverter: Gen 1 — 27, Gen 2 — 17, Gen 3 — 9	7 9 11
5		R1 — resolver, excitation R1–R2 · red (R) wire · inverter: Gen 1 — 34, Gen 2 — 19, Gen 3 — 22	7 9 11
2		R2 — resolver, excitation R1–R2 · green (G) wire · inverter: Gen 1 — 35, Gen 2 — 27, Gen 3 — 16	7 9 11
3		TM — motor temperature sensor · yellow (Y); Gen 1: black/pink (B/P) wire · inverter: Gen 1 — 32, Gen 2 — 45, Gen 3 — 7	7 9 11
4		TMGND — motor temperature sensor ground · orange (O) wire · inverter: Gen 1 — 31, Gen 2 — 44, Gen 3 — 20	7 9 11

Fig. 6 Leaf traction motor — resolver connector (Gen 1, Gen 2, Gen 3)

> **NOTE**
> The inverter's permanent-supply pins (fed from the battery in the factory Leaf wiring) are fed from the IGN relay with the Leaf ECU, like the ignition pin: the controller keeps the relay on until the motor stops.

3.1 Leaf Inverter Gen 1 (ZE0)

Pins 4 and 10 (battery in the Nissan) and 33 (ignition) are the +12 V supply, pins 2 and 8 are ground; CAN-H is 12, CAN-L is 19. The resolver goes to the resolver connector on the motor: 34/35 — excitation R1/R2, 1/6 — S1/S3, 20/27 — S2/S4; motor temperature sensor — 32/31. The old table had the roles of pins 4/10 and 33 swapped; it does not affect the wiring — all three are fed from the IGN relay.

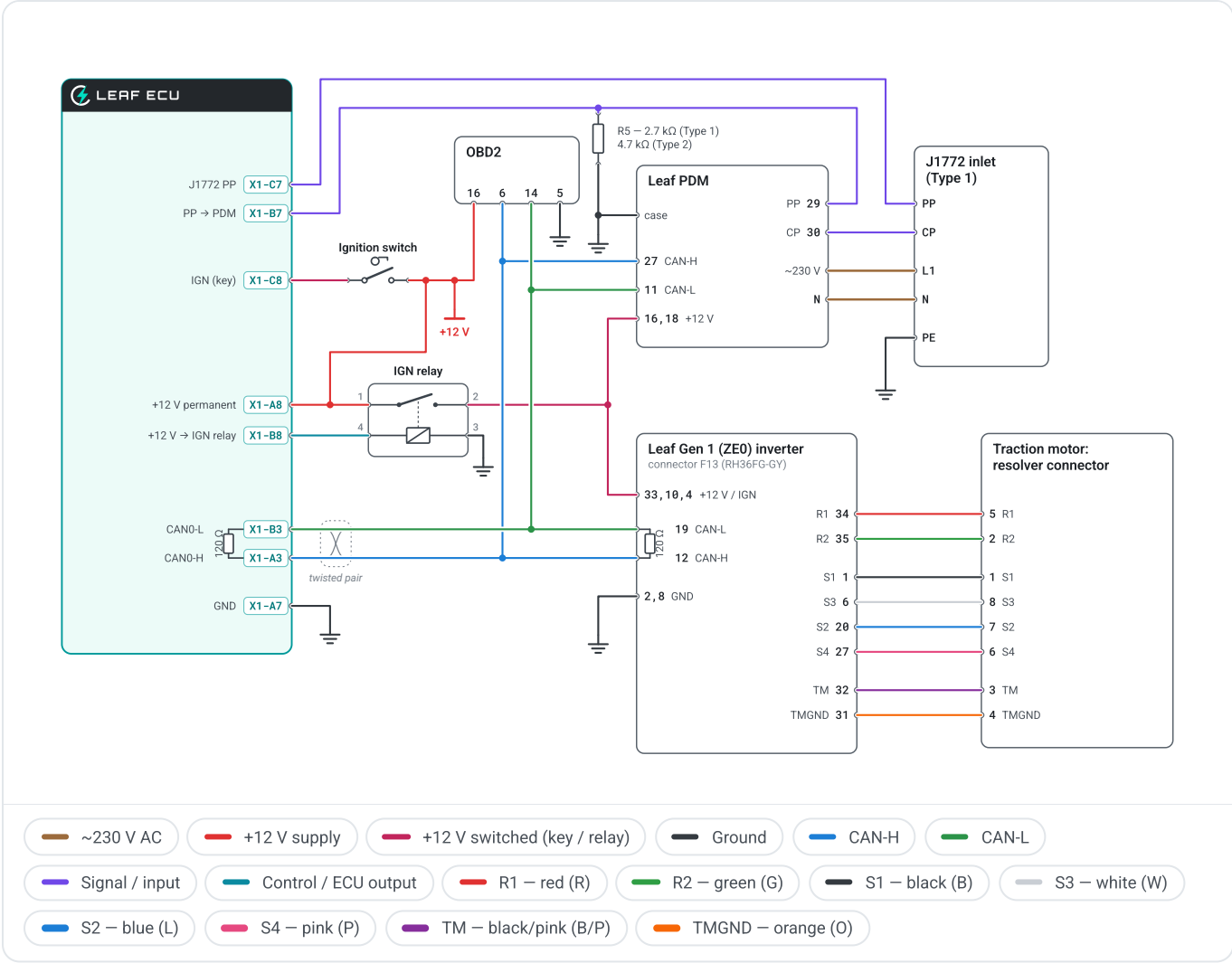
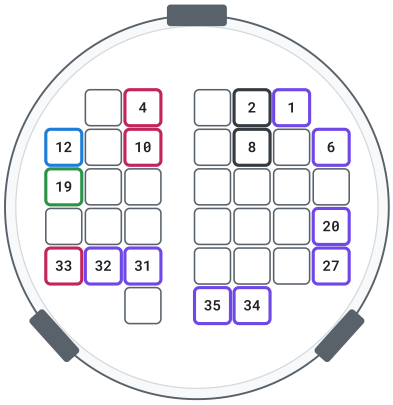


Fig. 7 Wiring with the Leaf Gen 1 (ZE0) inverter: 12 V supply, IGN relay, CAN0, PDM, J1772 inlet, inverter and motor resolver



Harness-side view of the harness connector (H.S.), per the Nissan drawing

PIN	WIRE	FUNCTION	DIAGRAMS
4		+12 V "Power supply (battery)" (battery in the Nissan; IGN relay with the Leaf ECU) · green (G) wire	
10		+12 V "Power supply (battery)" (battery in the Nissan; IGN relay with the Leaf ECU) · green (G) wire	
33		+12 V "Power supply (IGN)" — ignition (IGN relay with the Leaf ECU) · light green (LG) wire	
2		Ground (GROUND) · black (B) wire	
8		Ground (GROUND) · black (B) wire	
12		EV system CAN-H (Powertrain, CAN0) · blue (L) wire	7
19		EV system CAN-L (Powertrain, CAN0) · green (G) wire	7
34		Resolver R1 — excitation → motor resolver connector, pin 5 · red (R) wire	7
35		Resolver R2 — excitation → motor resolver connector, pin 2 · green (G) wire	7
1		Resolver S1 → motor resolver connector, pin 1 · black (B) wire	7
6		Resolver S3 → motor resolver connector, pin 8 · white (W) wire	7
20		Resolver S2 → motor resolver connector, pin 7 · blue (L) wire	7
27		Resolver S4 → motor resolver connector, pin 6 · pink (P) wire	7
32		Motor temperature sensor → motor resolver connector, pin 3 · black/pink (B/P) wire	7
31		Motor temperature sensor ground → motor resolver connector, pin 4 · orange (O) wire	7

Fig. 8 Leaf Gen 1 (ZE0) inverter — traction-motor-inverter connector

3.2 Leaf Inverter Gen 2 (AZE0)

Pins 46 and 48 (VB1, VB2) and 42 (IGN) are the +12 V supply, pins 47 and 49 are ground; CAN-H is 14, CAN-L is 15. The resolver and the motor temperature sensor run over the factory harness to the resolver connector on the motor: resolver pairs 19/27 (excitation), 20/21 and 18/17, temperature sensor 45/44. The resolver pair shields go to body ground.

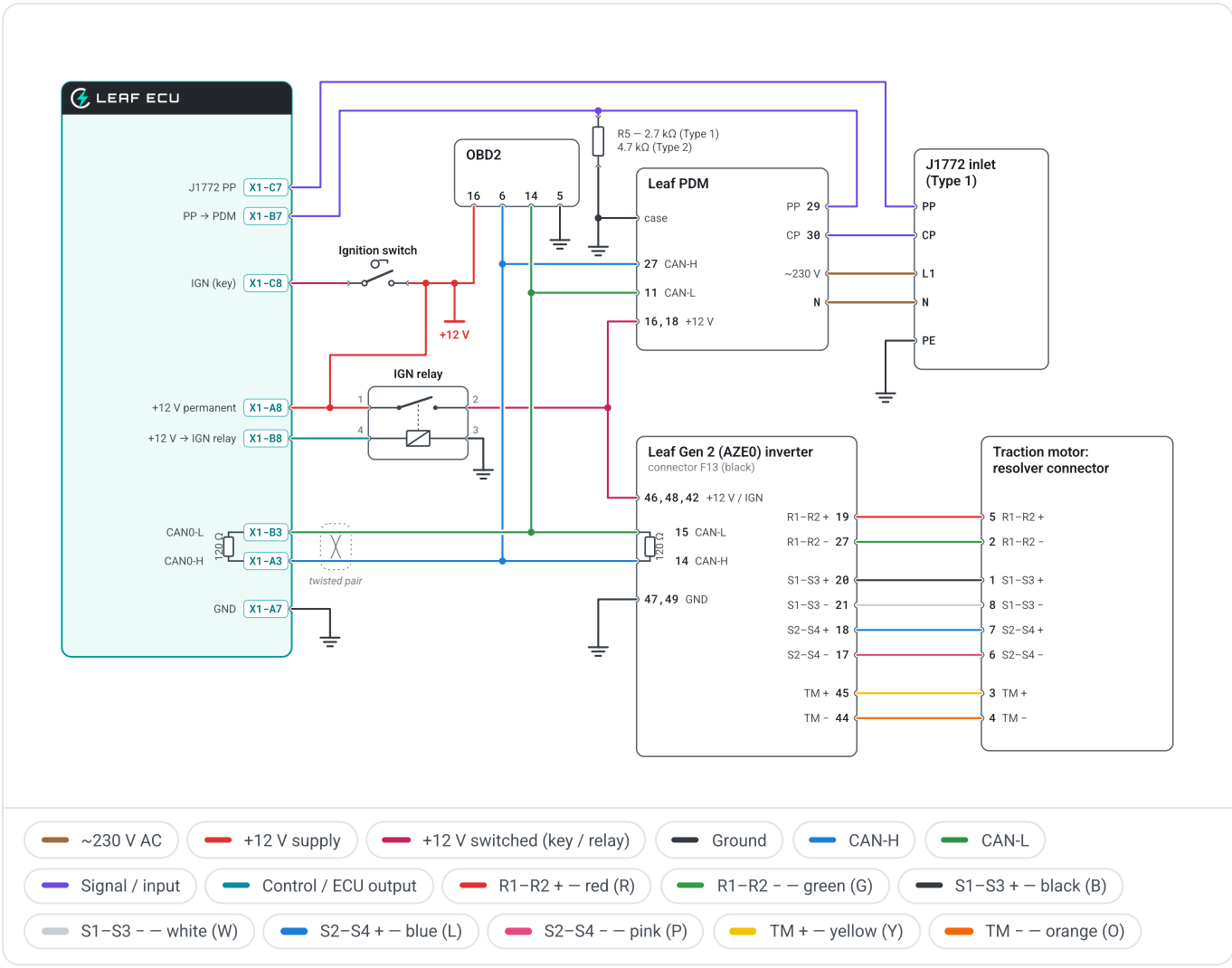
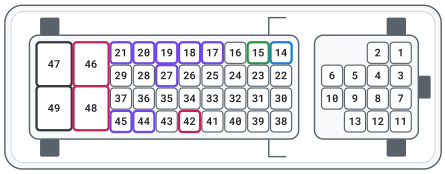


Fig. 9 Wiring with the Leaf Gen 2 (AZE0) inverter: 12 V supply, IGN relay, CAN0, PDM, J1772 inlet, inverter and motor resolver



H.S. — harness (wire) side view

PIN	WIRE	FUNCTION	DIAGRAMS
14		EV SYSTEM CAN-H — CAN-H (Powertrain, CAN0) · wire L (blue)	9
15		EV SYSTEM CAN-L — CAN-L (Powertrain, CAN0) · wire G (green)	9
17		REZ_S2 — resolver S2–S4 − → motor resolver connector, pin 6 · wire P (pink)	9
18		REZ_S4 — resolver S2–S4 + → motor resolver connector, pin 7 · wire L (blue)	9
19		REZ_R1 — resolver R1–R2 +, excitation → motor resolver connector, pin 5 · wire R (red)	9
20		REZ_S1 — resolver S1–S3 + → motor resolver connector, pin 1 · wire B (black)	9
21		REZ_S3 — resolver S1–S3 − → motor resolver connector, pin 8 · wire W (white)	9
27		REZ_R2 — resolver R1–R2 −, excitation → motor resolver connector, pin 2 · wire G (green)	9
42		IGN_SW — +12 V ignition · wire LG (light green)	
44		TMGND — motor temperature sensor − → motor resolver connector, pin 4 · wire O (orange)	9
45		TM — motor temperature sensor + → motor resolver connector, pin 3 · wire Y (yellow)	9
46		VB1 — +12 V, supply 1 (battery in the Nissan; IGN relay with the Leaf ECU) · wire G (green)	
47		VBGND1 — supply ground 1 · wire B (black)	
48		VB2 — +12 V, supply 2 (battery in the Nissan; IGN relay with the Leaf ECU) · wire G (green)	
49		VBGND2 — supply ground 2 · wire B (black)	

Fig. 10 Leaf Gen 2 (AZE0) inverter — connector F13 (black)

3.3 Leaf Inverter Gen 3 (ZE1)

Numbering is Nissan's (harness side of the harness connector). Pins 1 and 6 are the +12 V supply, 18 is ignition (Power ON), 4 and 10 are ground; CAN-H is 5, CAN-L is 11. Resolver: 22/16 — excitation, 19/14 and 3/9 — output windings; motor temperature sensor — 7/20 (all to the resolver connector on the motor); coolant temperature sensor — 12/15.

WARNING

Earlier versions of this manual showed a picture with a different numbering of this same connector for Gen 3 (CAN-H 22, CAN-L 17, ground 3/9/16/21). That numbering does not match Nissan's, and it wrongly marks pins 3 and 9 as ground. Use the numbering below.

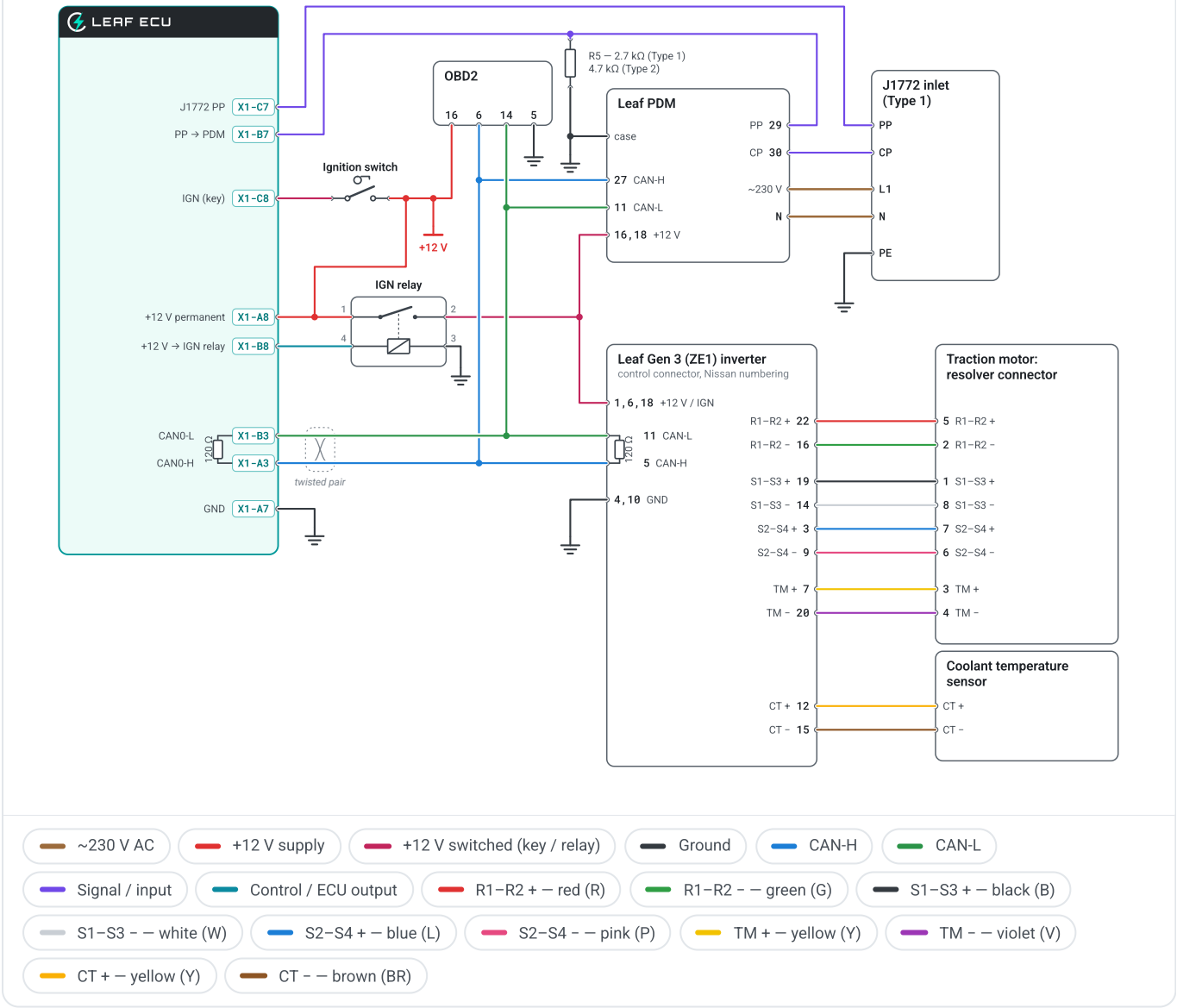
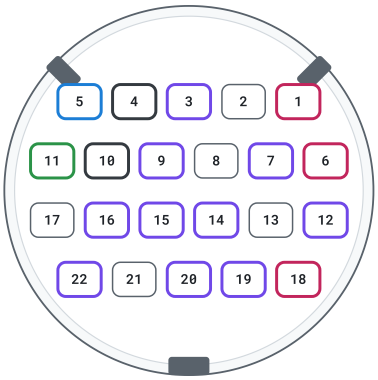


Fig. 11 Wiring with the Leaf Gen 3 (ZE1) inverter: 12 V supply, IGN relay, CAN0, PDM, J1772 inlet, inverter and motor resolver



Harness-side view of the harness connector (H.S.), Nissan numbering

PIN	WIRE	FUNCTION	DIAGRAMS
1		+12 V "12V battery power supply" (battery in the Nissan; IGN relay with the Leaf ECU) · green (G) wire	
6		+12 V "12V battery power supply" (battery in the Nissan; IGN relay with the Leaf ECU) · green (G) wire	
18		+12 V "Power ON power supply" — ignition (IGN relay with the Leaf ECU) · light green (LG) wire	
4		Ground (GND) · black (B) wire	
10		Ground (GND) · black (B) wire	
5		EV system CAN-H (Powertrain, CAN0) · blue (L) wire	11
11		EV system CAN-L (Powertrain, CAN0) · green (G) wire	11
22		Resolver R1–R2 + — excitation → motor resolver connector, pin 5 · red (R) wire	11
16		Resolver R1–R2 − — excitation → motor resolver connector, pin 2 · green (G) wire	11
19		Resolver S1–S3 + → motor resolver connector, pin 1 · black (B) wire	11
14		Resolver S1–S3 − → motor resolver connector, pin 8 · white (W) wire	11
3		Resolver S2–S4 + → motor resolver connector, pin 7 · blue (L) wire	11
9		Resolver S2–S4 − → motor resolver connector, pin 6 · pink (P) wire	11
7		Motor temperature sensor + → motor resolver connector, pin 3 · yellow (Y) wire	11
20		Motor temperature sensor − → motor resolver connector, pin 4 · violet (V) wire	11
12		Coolant temperature sensor + · yellow (Y) wire	11
15		Coolant temperature sensor − · brown (BR) wire	11

Fig. 12 Leaf Gen 3 (ZE1, 2018+) inverter — control connector

3.4 Tesla Model 3 / Model Y Rear Inverter

The controller works with the Tesla Model 3 / Model Y rear inverter (Drive Inverter Rear) running the Highland firmware. In the web configurator set **Inverter type** to **Tesla M3** (see 17.3). The controller sends every CAN message the inverter needs by itself — no other Tesla modules are required.

Only the supply, CAN and the resolver are connected to the inverter, through the logic connector X090 (30 contacts):

X090 PIN	WIRE	TO
3	red (RD)	+12 V after the IGN relay
13	black (BK)	Ground
12	blue/white (BU/WH)	CAN0-H ● X1-A3
2	blue (BU)	CAN0-L ● X1-B3
26 / 16	yellow / black	Resolver: excitation + / − (resolver pins 3 / 8)

X090 PIN	WIRE	TO
17 / 7	green / red	Resolver: sine + / − (pins 4 / 9)
8 / 27	violet / blue	Resolver: cosine + / − (pins 5 / 10)
10	green (GN)	Oil pump, pin 2 (status) — factory harness

- **HVIL and the second CAN (PRTY-CAN) are not connected** — pins 4, 23 and 20, 30 stay free, as does Unswitched +12V (22).
- The accelerator pedal and the brake switch go to the controller, not to the inverter: X090 pins 5, 6, 11, 14, 15, 21, 24, 25 are free.
- The oil pump stays connected to the inverter as in the car (pin 10 ↔ pump pin 2). The pump supply (pin 1) is +12 V after the IGN relay (in the Tesla the Body Controller switches it), pin 3 is ground. The pump connector pinout is below.
- The Tesla inverter already has 120 Ω between CAN+ and CAN−, just like the controller — CAN0 needs no separate resistors.

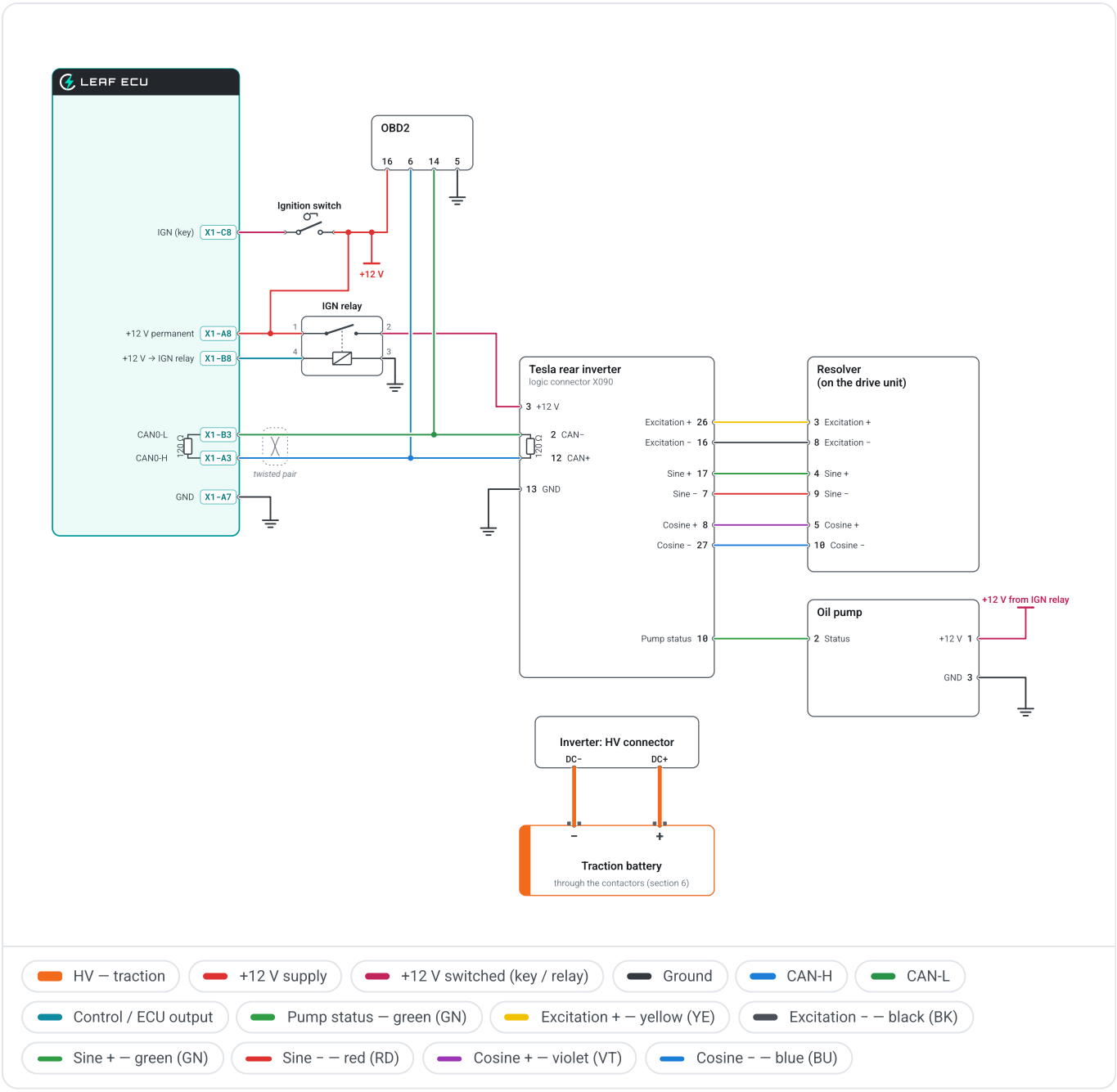


Fig. 13 Wiring the Tesla Model 3 / Model Y rear inverter (Highland firmware): 12 V supply, IGN relay, CAN0, resolver and oil pump



The source does not state the view side (Tesla drawing) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
3		+12 V switched (Switched +12V) — from the IGN relay · red (RD) wire	13
13		Ground (Ground-Chassis) · black (BK) wire	13
12		CAN+ (Vehicle-CAN) → ECU X1-A3, CAN0-H · blue/white (BU/WH) wire	13
2		CAN− (Vehicle-CAN) → ECU X1-B3, CAN0-L · blue (BU) wire	13
26		Resolver: excitation + → resolver, pin 3 · yellow (YE) wire	13
16		Resolver: excitation − → resolver, pin 8 · black (BK) wire	13
17		Resolver: sine + → resolver, pin 4 · green (GN) wire	13
7		Resolver: sine − → resolver, pin 9 · red (RD) wire	13
8		Resolver: cosine + → resolver, pin 5 · violet (VT) wire	13
27		Resolver: cosine − → resolver, pin 10 · blue (BU) wire	13
10		Oil pump status (LIN) → pump, pin 2 — factory harness · green (GN) wire	13

Fig. 14 Tesla Model 3 / Model Y rear inverter — logic connector X090

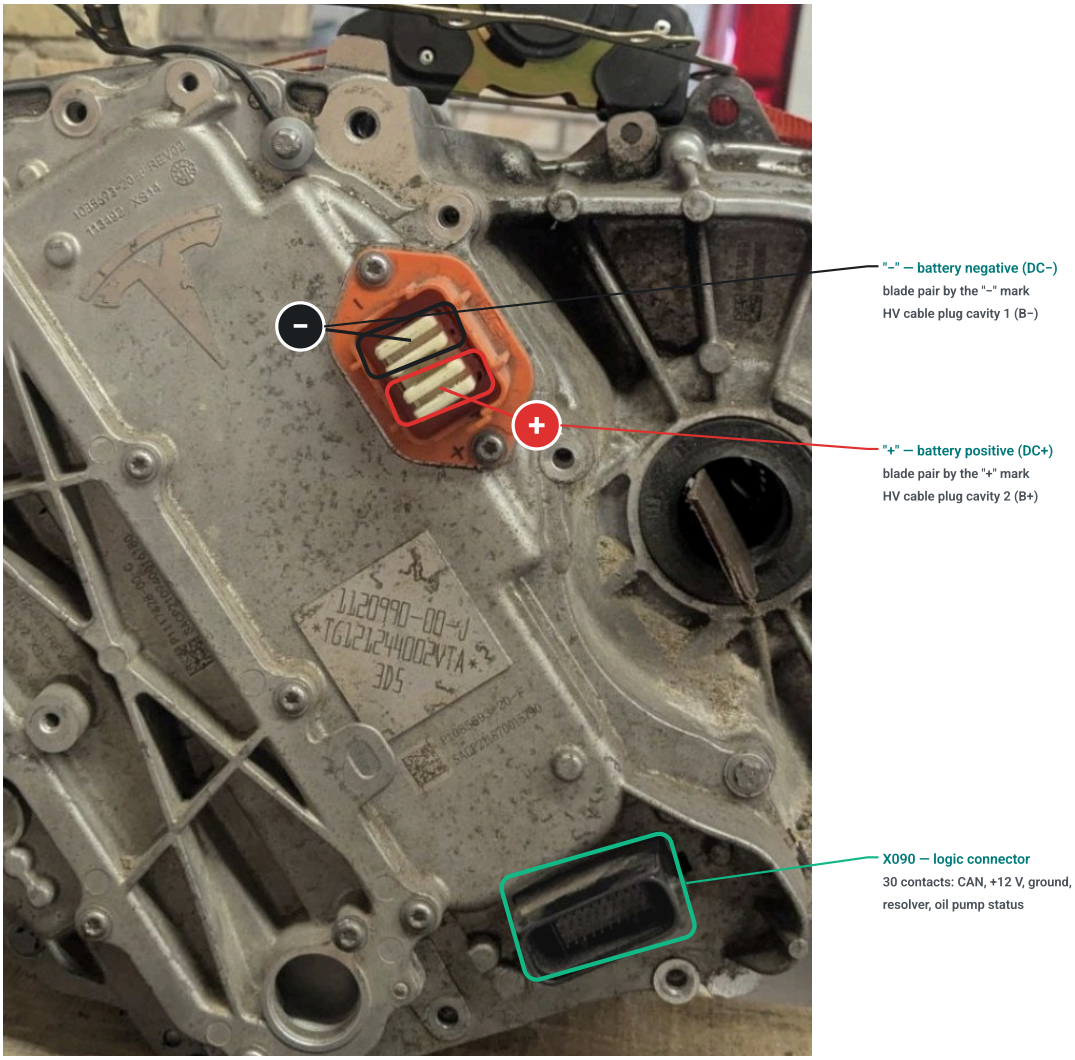


The source does not state the view side (Tesla drawing) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
3		Resolver: excitation + ← inverter pin 26 · yellow (YE) wire	13
8		Resolver: excitation − ← inverter pin 16 · black (BK) wire	13
4		Resolver: sine + ← inverter pin 17 · green (GN) wire	13
9		Resolver: sine − ← inverter pin 7 · red (RD) wire	13
5		Resolver: cosine + ← inverter pin 8 · violet (VT) wire	13
10		Resolver: cosine − ← inverter pin 27 · blue (BU) wire	13

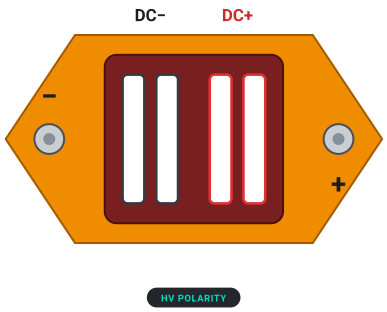
Fig. 15 Tesla rear drive unit resolver — connector X091

The high-voltage supply (through the contactors, see section 6) goes to the orange HV connector on the inverter cover. Each pole is a pair of blades: the pair by the "−" mark is the battery negative, the pair by "+" the positive. On the Tesla HV cable plug these are cavities 1 (B−) and 2 (B+); the numbers are moulded on the plug.



Rear 3DU drive unit. The "−" and "+" marks are written next to the HV connector; check the polarity with a multimeter before the first power-up.

Fig. 16 Tesla Model 3 / Model Y rear inverter: HV connector polarity and the X090 logic connector (photo)



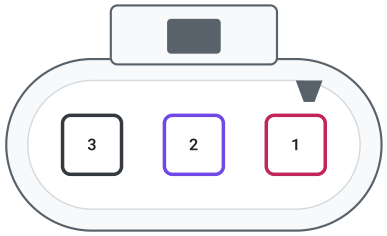
Outside view of the header on the inverter, screws left and right; the "−" and "+" marks as next to the header on the unit. On the HV cable plug: cavity 1 — B−, cavity 2 — B+

PIN	WIRE	FUNCTION
N1		DC− — traction battery negative (pair by the "−" mark; plug cavity 1)
N2		DC− — traction battery negative (pair by the "−" mark; plug cavity 1)
P1		DC+ — traction battery positive (pair by the "+" mark; plug cavity 2)
P2		DC+ — traction battery positive (pair by the "+" mark; plug cavity 2)

Fig. 17 Tesla rear inverter — HV connector (DC input)

DANGER — HIGH VOLTAGE

Reversed HV polarity will damage the inverter. Before closing the contactors for the first time, check with a multimeter that the battery "+" arrives at the blades by the "+" mark. The inverter's own polarity can be checked without the battery with a diode test (as in the Tesla service manual): red probe on the "−" pair, black probe on the "+" pair — the meter reads about 0.45–2.0 V; "OL" means the probes are swapped.



The source does not state the view side (Tesla drawing) — check against the numbers on the connector housing

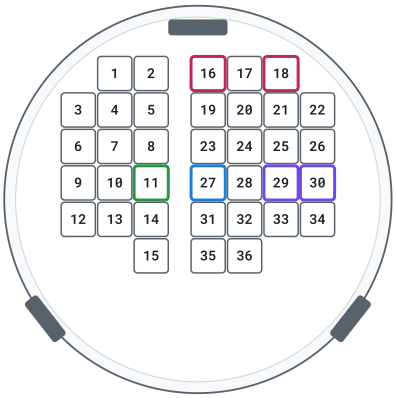
PIN	WIRE	FUNCTION	DIAGRAMS
1		Switched +12 V — from the IGN relay (Body Controller in the Tesla) · red (RD) wire	13
2		Pump status (LIN) → inverter pin 10 · yellow or green wire (depends on the harness version)	13
3		Ground · black (BK) wire	13

Fig. 18 Tesla rear drive unit oil pump — connector X092

WARNING

CAN-H and CAN-L wires must be twisted into a twisted pair.

4 PDM Module



Harness side view (H.S.)

PIN	WIRE	FUNCTION	DIAGRAMS
11		CAN-L (Powertrain, CAN0)	7 9 11
16		+12 V PDM supply (V4 board: from the IGN relay)	
18		+12 V PDM supply (V4 board: from the IGN relay)	
27		CAN-H (Powertrain, CAN0)	7 9 11
29		Charge-inlet PP (V4 board: from ECU X1-B7)	7 9 11
30		Charge-inlet CP (Control Pilot)	7 9 11

Fig. 19 PDM module interface connector (F23)

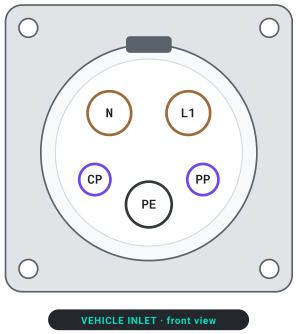
PDM pins 30 and 29 (charge inlet) connect to the two-pin connector on the inlet itself: white wire to white, green to brown (when using the Leaf harness). With the V4 board wire the PP line as shown in the section 3 diagrams (Fig. 9).

The PDM case must be connected to ground — the case is the negative output of the DC-DC converter. The stud on the side of the module is the +12 V output of the DC-DC converter.

4.1 Type 1 (J1772) Charge Inlet

The vehicle inlet and the charging gun are mirror images: on the left of the figure is the inlet (front view, from the side the gun plugs into), on the right the gun (view onto its face). The tag under each connector says which side you are looking at.

Type 1 (J1772) — vehicle inlet



Front view of the vehicle inlet (the side the gun plugs into); the mounting flange with its holes is schematic

Type 1 (J1772) — charging gun



View onto the face of the charging gun — the mirror image of the inlet

PIN	WIRE	FUNCTION	DIAGRAMS
L1		1 — phase L1 (~230 V) → PDM	7 9 11
N		2 — neutral N (L2) → PDM	7 9 11
PE		3 — protective earth → chassis	7 9 11
CP		4 — Control Pilot → PDM (pin 30)	7 9 11
PP		5 — Proximity Pilot → ECU X1-C7 (V4 board; in the basic wiring — PDM 29)	7 9 11

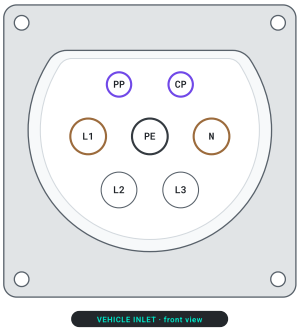
Fig. 20 Type 1 (J1772): vehicle inlet and charging gun

Resistor R5 of the stock Leaf inlet (2.7 kΩ) is relocated and wired as shown in the section 3 diagrams (Fig. 9).

4.2 Type 2 Charge Inlet

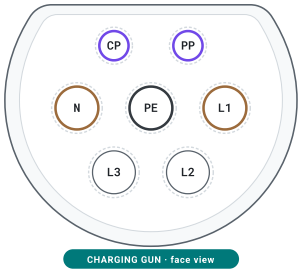
Type 2 is a three-phase connector; the single-phase Leaf PDM uses only L1, N, PE, CP and PP. With a Type 2 inlet, resistor R5 must be changed to 4.7 kΩ (Fig. 9).

Type 2 (IEC 62196-2) — vehicle inlet



Front view of the vehicle inlet (the side the gun plugs into); the mounting flange with its holes is schematic

Type 2 (IEC 62196-2) — charging gun



View onto the face of the charging gun — the mirror image of the inlet

PIN	WIRE	FUNCTION
L1		L1 — phase L1 (~230 V) → PDM
L2		L2 — phase L2 (not used by the single-phase PDM)
L3		L3 — phase L3 (not used by the single-phase PDM)
N		N — neutral → PDM
PE		PE — protective earth → chassis
CP		CP — Control Pilot → PDM (pin 30)
PP		PP — Proximity Pilot → ECU X1-C7

Fig. 21 Type 2 (IEC 62196-2): vehicle inlet and charging gun

TYPE 2 CABLE — 32 A ONLY

With a Type 2 inlet use only a **32 A** charging cable (gun) — its plug has a **220 Ω** resistor between PP and PE. Cables of other ratings have a different resistor (13 A — 1.5 kΩ, 20 A — 680 Ω, 63 A — 100 Ω) and are not suitable for this wiring.

5 Controls

5.1 Selector Buttons

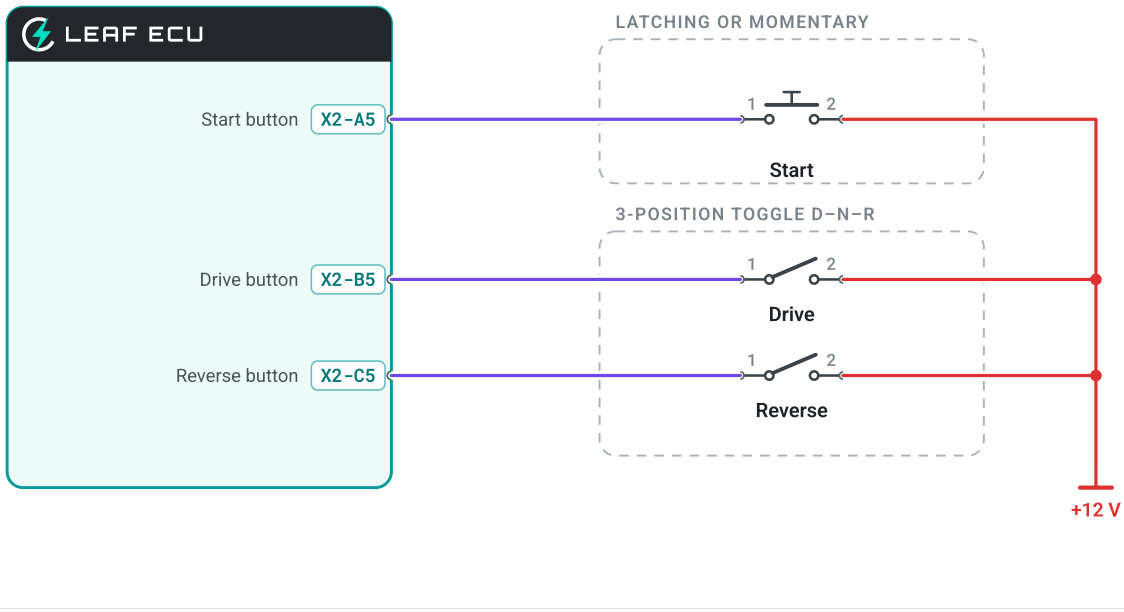


Fig. 22 Start, Drive and Reverse control buttons

Driving direction (the selector) can be chosen with buttons or with the original Nissan Leaf selector. The selector type must be set in the web configurator (**ECU → Settings**, see 17.3).

- **Drive and Reverse buttons** — latching.
- **Start button** — latching or momentary; its mode is set in the configurator.

How it works:

1. Pressing **Start** closes the contactors; the inverter is ready, the selector is in neutral.
2. To drive off, switch on **Drive** or **Reverse**.
3. If Drive and Reverse are active at the same time, the controller goes to neutral.

RECOMMENDATION

Use a 3-position toggle switch for Drive and Reverse: centre position = neutral, end positions = Drive and Reverse.

5.2 Brake Vacuum Booster

The controller can run the vacuum pump of the brake booster. Connect the vacuum sensor to the controller and the pump through a relay.

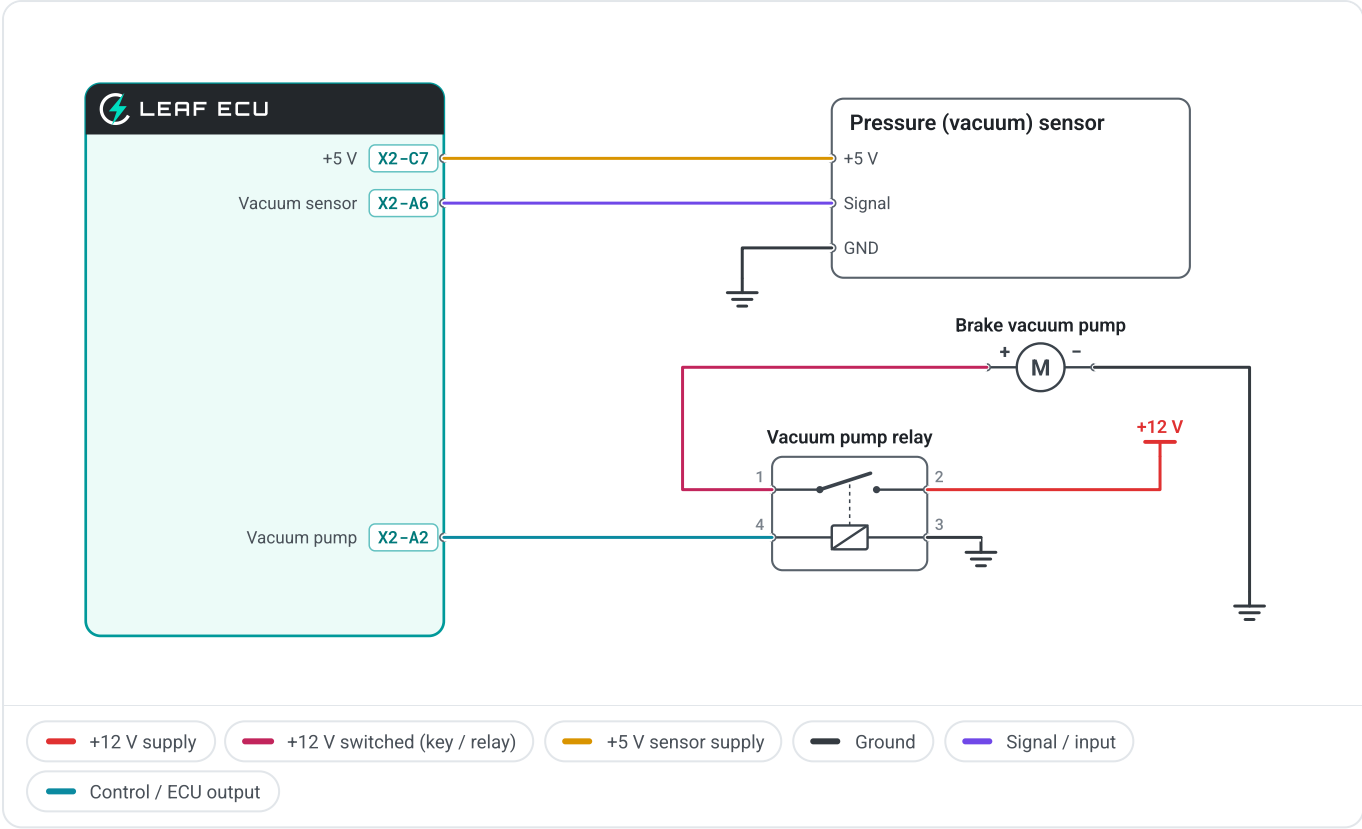


Fig. 23 Brake vacuum booster: vacuum sensor and vacuum pump relay

The sensor voltage thresholds that switch the pump on and off are set in the web configurator on **ECU → Brake** (values are entered in mV).

6 Power Relays (Contactors)

Choose power relays with ratings close to those used in the Nissan Leaf.

TESLA CONTACTORS ARE NOT SUITABLE

They have a high coil holding current and need an economiser. Similar contactors with a built-in current-reduction circuit exist — those can be used.

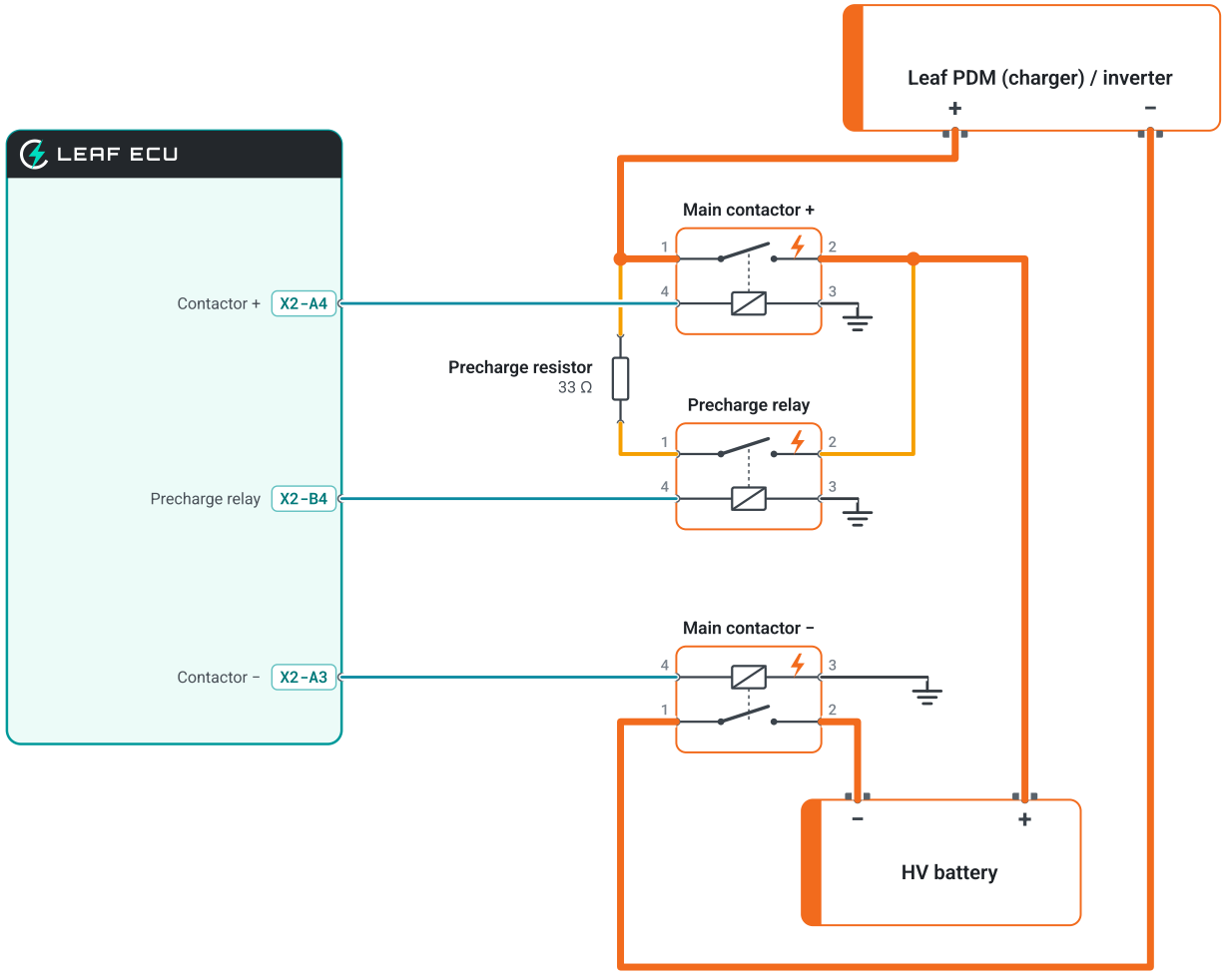


Fig. 24 Power contactors and precharge: main contactors + and −, precharge relay and resistor

Wire cross-sections in the diagram:

WIRE	CROSS-SECTION	PURPOSE
Power	35 mm²	Main traction battery cables
Precharge	2.5 mm²	Precharge path through the 33 Ω resistor
Control	0.75 mm²	Relay coils

Observe polarity when connecting the traction battery to the PDM:

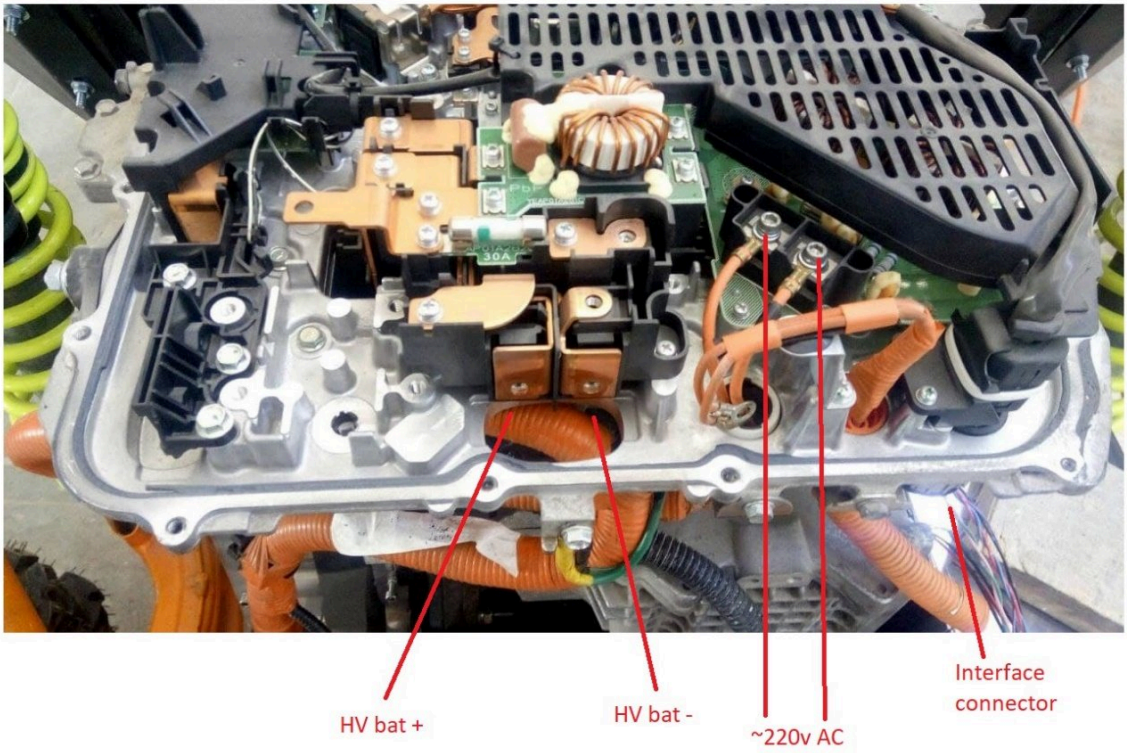


Fig. 25 Traction battery polarity at the PDM: HV bat + / HV bat −, ~230 V input and interface connector

7 Accelerator Pedal

The pedal signal wire connects to pin X1-B2.

PIN	FUNCTION
X1-B2	Pedal signal — primary sensor
X1-A2	Pedal signal — second sensor (not used by the firmware yet)
X1-C1	+5 V pedal supply
X1-C2	Pedal ground

The signal voltage must swing within 0–5 V — check this before connecting to the controller.

TIP

If your pedal has a second sensor, wire it to X1-A2 right away.

Both resistive and Hall-effect pedals can be used.

WARNING

When connecting a Hall-effect pedal, double-check the sensor supply polarity and which wire is the signal output — otherwise you can damage the sensor or the controller's 5 V regulator.

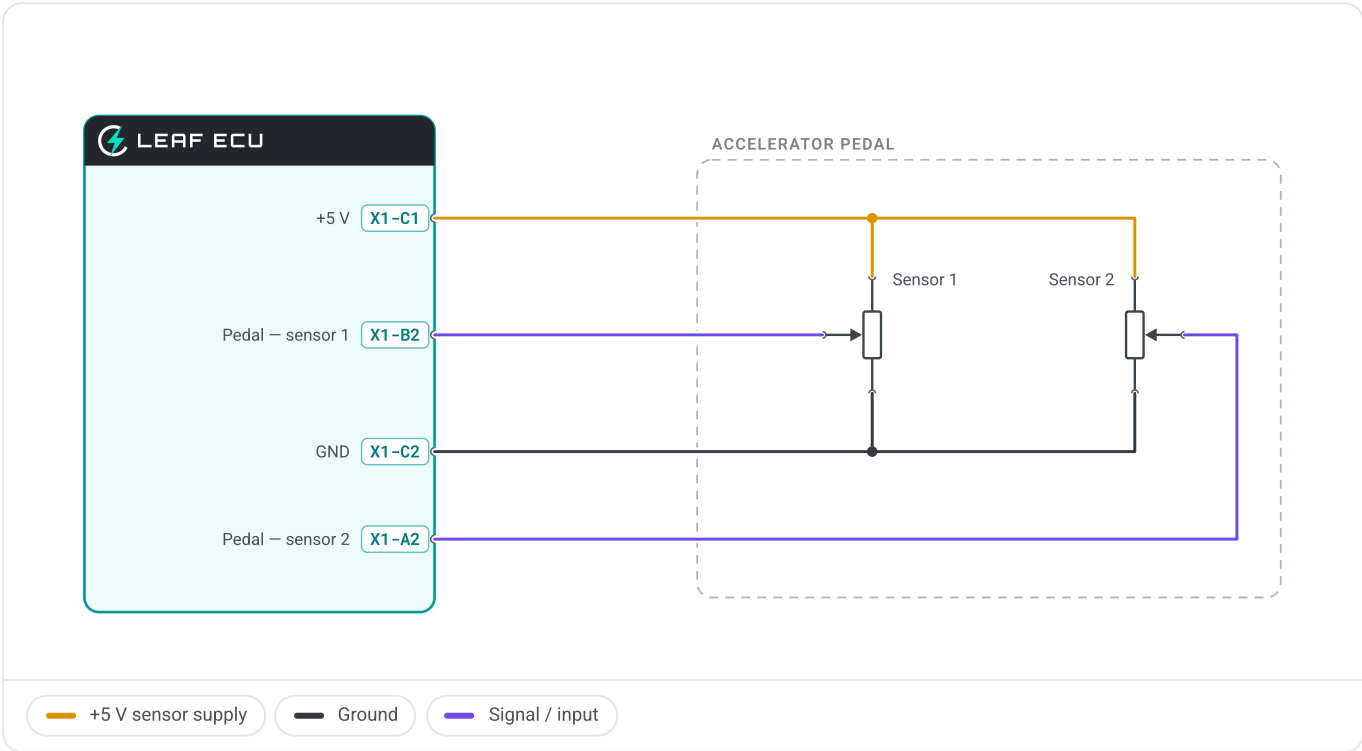


Fig. 26 Accelerator pedal wiring (two potentiometer tracks)



Fig. 27 Nissan Leaf accelerator pedal



The source does not state the view side (Nissan drawing) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION
1		Sensor 2 — signal
2		Sensor 2 — supply "+"
3		Sensor 1 — supply "+"
4		Sensor 1 — signal
5		Sensor 1 — ground "−"
6		Sensor 2 — ground "−"

Fig. 28 Nissan Leaf accelerator pedal connector (the same connector is used in other Nissan and Renault models)

The pedal MIN and MAX positions must be set in the web configurator (see 17.2).

8 BMS

The controller supports the Chevrolet Volt Gen 1 BMS, Orion 2 and the Nissan Leaf BMS.

8.1 Chevrolet Volt Gen 1

The BMS consists of 4 satellite modules marked 1, 2/3 (two modules) and 4. With the satellites the battery can be split into up to four separate packs.

> **NOTE**
> Satellites 2/3 have a jumper that sets the satellite's position in the battery. Without it the satellite will not work.

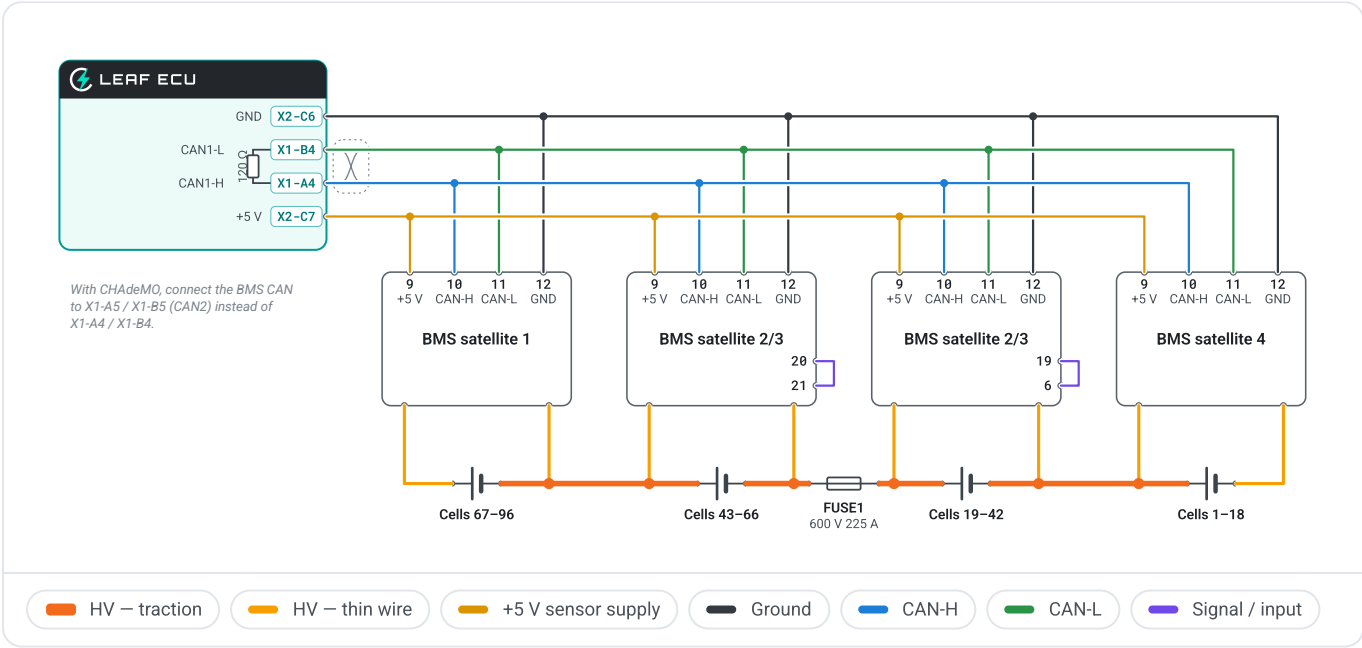
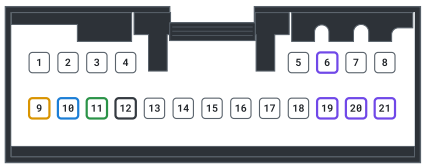


Fig. 29 Chevrolet Volt Gen 1 BMS: connecting the four satellites to the ECU and the battery cell split

The **black connector** on a satellite is the interface connector. We recommend wiring CAN and power as shown in the diagram rather than through the Volt factory harness, where wires pass through the satellites in a daisy chain, which makes diagnostics harder.



View from the BMS (satellite) side

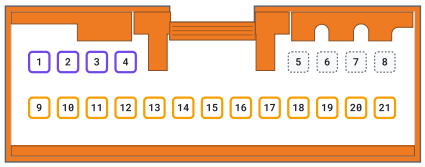
PIN	WIRE	FUNCTION	DIAGRAMS
6		Jumper 19–6 — sets the position of satellite 2/3 (cell group 19–42)	29
9		+5 V — satellite supply (ECU X2-C7)	29
10		CAN-H (ECU X1-A4; with CHAdeMO — X1-A5)	29
11		CAN-L (ECU X1-B4; with CHAdeMO — X1-B5)	29
12		Ground (ECU X2-C6)	29
19		Jumper 19–6 — sets the position of satellite 2/3 (cell group 19–42)	29
20		Jumper 20–21 — sets the position of satellite 2/3 (cell group 43–66)	29
21		Jumper 20–21 — sets the position of satellite 2/3 (cell group 43–66)	29

Fig. 30 Volt BMS satellite — interface connector (black)

The **orange connectors** on the satellites are for the cell balancing wires and temperature sensors. This BMS requires **NTC 10 kΩ** temperature sensors.

There are two connector types: 12-cell (satellites 2/3 have only this type) and 6-cell (found on satellites 1 and 4).

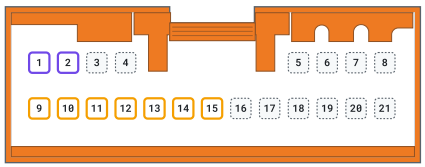
Volt BMS satellite — cell / temperature connector, 12 cells (orange)



The source does not state the view side (GM drawing) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION
1		Battery module temperature signal (NTC 10 kΩ)
2		Low reference — temperature sensor return
3		Battery module temperature signal (NTC 10 kΩ)
4		Low reference — temperature sensor return
5		Not used
6		Not used
7		Not used
8		Not used
9		Battery voltage signal (12)
10		Battery voltage signal (11)
11		Battery voltage signal (10)
12		Battery voltage signal (9)
13		Battery voltage signal (8)
14		Battery voltage signal (7)
15		Battery voltage signal (6)
16		Battery voltage signal (5)
17		Battery voltage signal (4)
18		Battery voltage signal (3)
19		Battery voltage signal (2)
20		Battery voltage signal (1)
21		Low reference — bottom sense point of the cell voltages

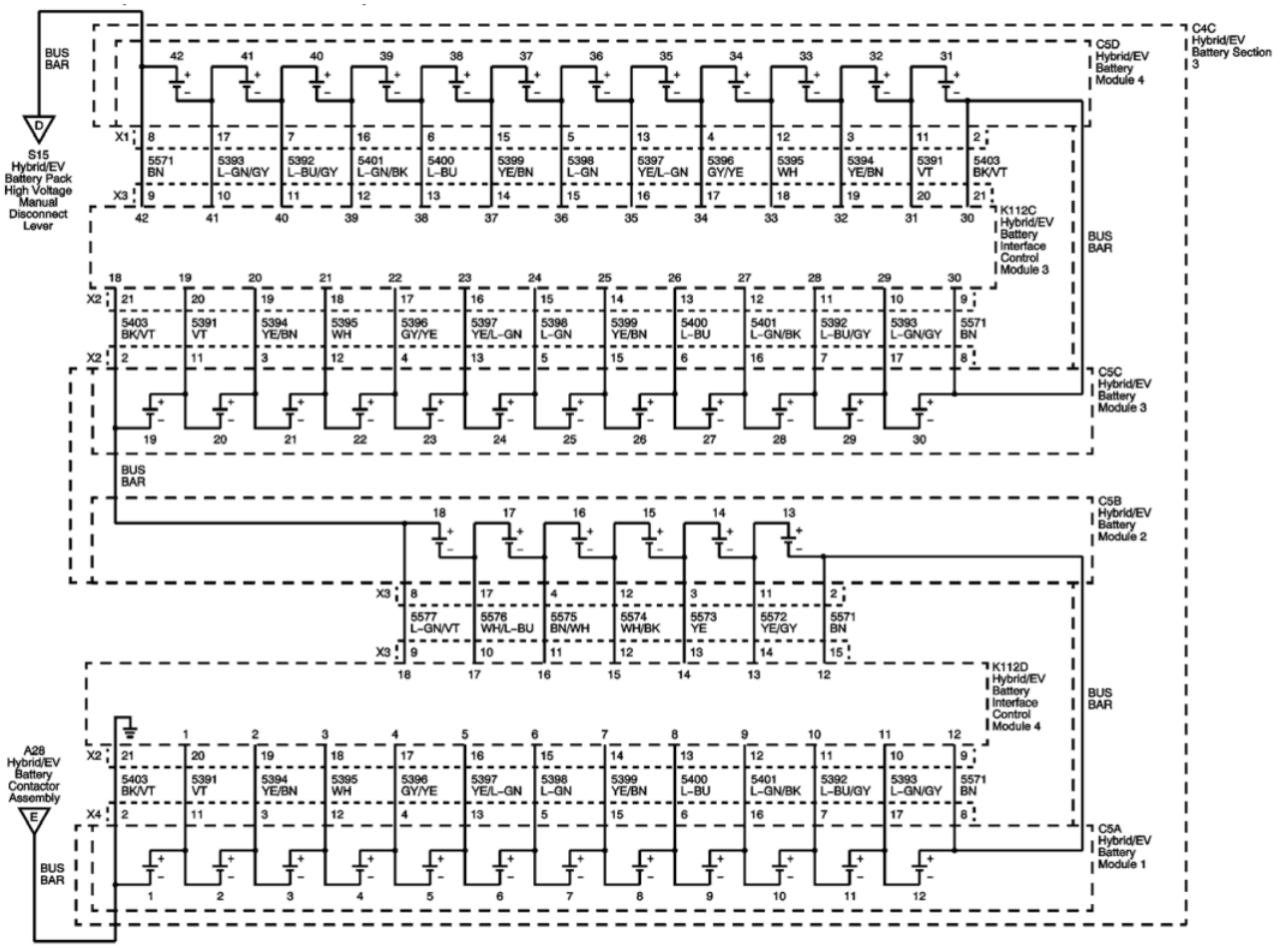
Volt BMS satellite — cell / temperature connector, 6 cells (orange)

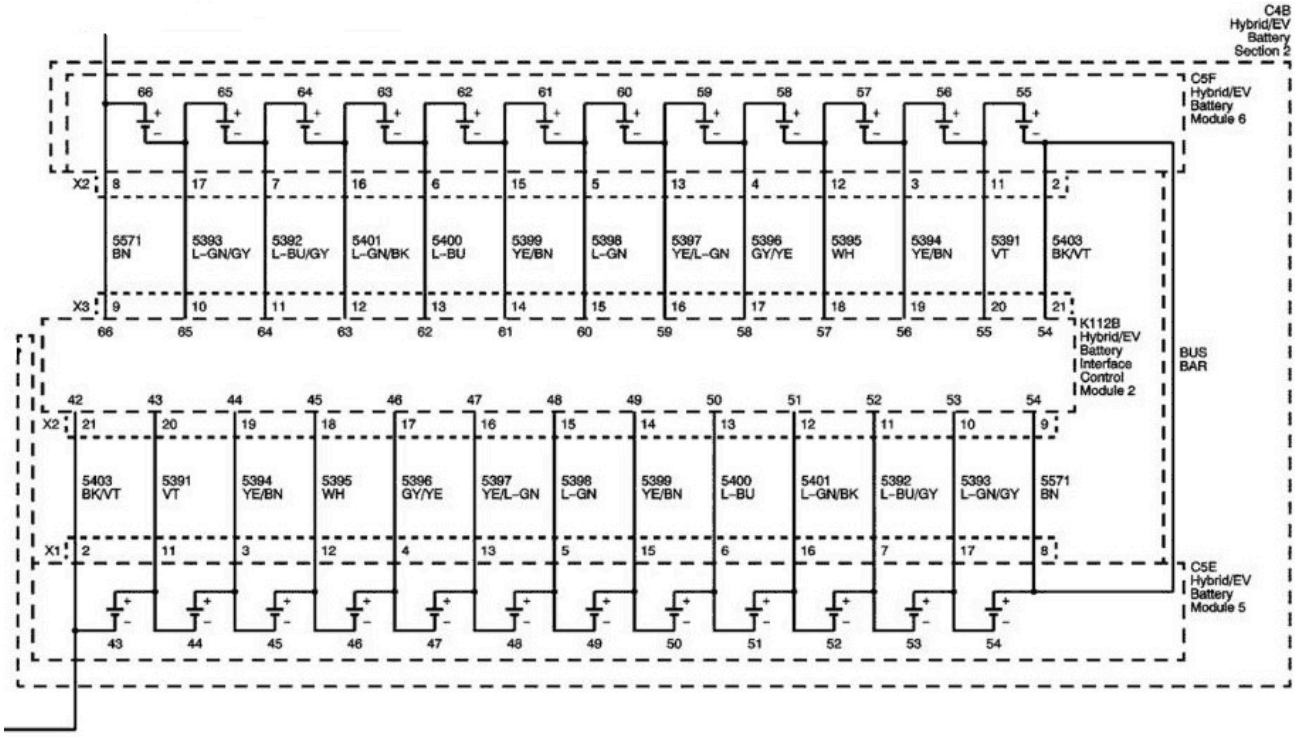


The source does not state the view side (GM drawing) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION
1		Battery module temperature signal (NTC 10 kΩ)
2		Low reference — temperature sensor return
3		Not used
4		Not used
5		Not used
6		Not used
7		Not used
8		Not used
9		Battery voltage signal (18)
10		Battery voltage signal (17)
11		Battery voltage signal (16)
12		Battery voltage signal (15)
13		Battery voltage signal (14)
14		Battery voltage signal (13)
15		Battery voltage signal (12)
16		Not used
17		Not used
18		Not used
19		Not used
20		Not used
21		Not used

Fig. 31 Volt satellite cell connectors: 12-cell and 6-cell variants


BUS BAR
S15 Hybrid/EV Battery Pack High Voltage Manual Disconnect Lever
C4C Hybrid/EV Battery Section 3
C5D Hybrid/EV Battery Module 4
C5C Hybrid/EV Battery Module 3
C5B Hybrid/EV Battery Module 2
C5A Hybrid/EV Battery Module 1
K112C Hybrid/EV Battery Interface Control Module 3
K112D Hybrid/EV Battery Interface Control Module 4
A28 Hybrid/EV Battery Contactor Assembly
X1
X2
X3
X4
5571 BN
5393 L-GN/GY
5392 L-BU/GY
5401 L-GN/BK
5400 L-BU
5399 YE/BN
5398 L-GN
5397 YE/L-GN
5396 GY/YE
5395 WH
5394 YE/BN
5391 VT
5403 BK/VT
5577 L-GN/VT
5576 WH/L-BU
5575 BN/WH
5574 WH/BK
5573 YE
5572 YE/GY


C4B Hybrid/EV Battery Section 2
C5F Hybrid/EV Battery Module 6
C5E Hybrid/EV Battery Module 5
K112B Hybrid/EV Battery Interface Control Module 2
BUS BAR
X1
X2
X3
5571 BN
5393 L-GN/GY
5392 L-BU/GY
5401 L-GN/BK
5400 L-BU
5399 YE/BN
5398 L-GN
5397 YE/L-GN
5396 GY/YE
5395 WH
5394 YE/BN
5391 VT
5403 BK/VT

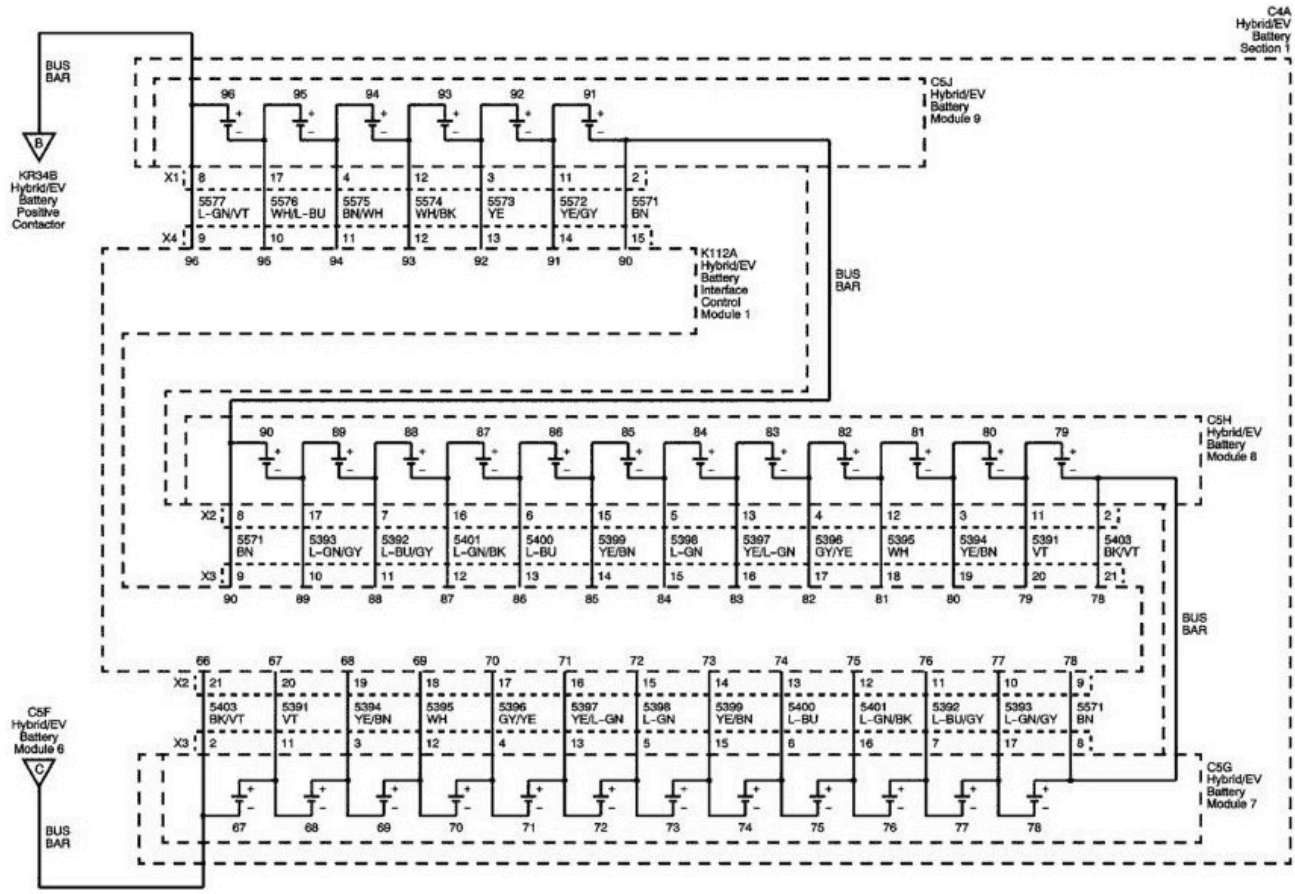


Fig. 32 Satellite placement sequence in the battery — original GM drawing

> **NOTE**
>
> The satellite sequence can be changed, but the cell map will then show positions incorrectly.

8.2 Nissan Leaf (ZE0 / AZE0)

The Nissan Leaf AZE0 and ZE0 BMS can be used (ZE1 is being tested). The BMS connects over CAN:

CONFIGURATION	CAN-H	CAN-L
Without CHAdeMO	X1-A4	X1-B4
With CHAdeMO	X1-A5	X1-B5

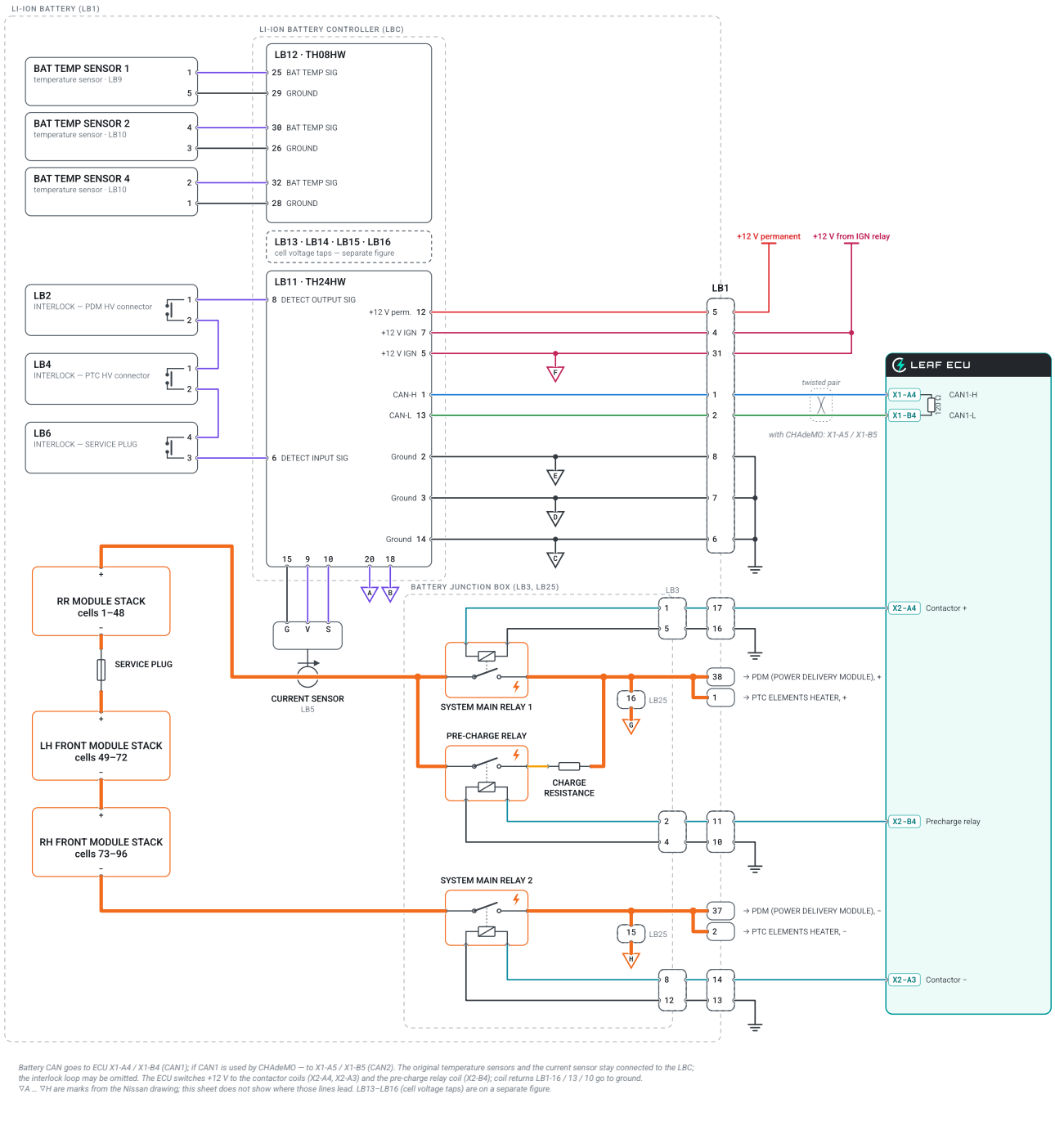
The original temperature sensors and the current sensor must also be connected to the BMS. The interlocks can be left out.

The Leaf battery connects to the controller through its LB1 connector:

LB1	GOES TO
5	+12 V permanent
4, 31	+12 V from the IGN relay
1 / 2	CAN-H X1-A4 / CAN-L X1-B4 (with CHAdeMO — X1-A5 / X1-B5)

LB1	GOES TO
17	X2-A4 — "+" contactor coil (SYSTEM MAIN RELAY 1)
14	X2-A3 — "−" contactor coil (SYSTEM MAIN RELAY 2)
11	X2-B4 — pre-charge relay coil
6, 7, 8, 10, 13, 16	Ground

The contactors and the pre-charge relay sit in the battery junction box, so the separate power relays of section 6 are not needed — the ECU drives the stock ones from the same outputs.



Battery CAN goes to ECU X1-A4 / X1-B4 (CAN1); if CAN1 is used by CHAdeMO — to X1-A5 / X1-B5 (CAN2). The original temperature sensors and the current sensor stay connected to the LBC; the interlock loop may be omitted. The ECU switches +12 V to the contactor coils (X2-A4, X2-A3) and the pre-charge relay coil (X2-B4); coil returns LB1-16 / 13 / 10 go to ground. ▽A … ▽H are marks from the Nissan drawing; this sheet does not show where those lines lead. LB13–LB16 (cell voltage taps) are on a separate figure.

Fig. 33 Leaf AZE0 traction battery wired to the Leaf ECU: 12 V supply and CAN of the battery controller (LBC), contactor and pre-charge relay control, ground; inside the pack — temperature and current sensors, interlocks, battery junction box and LB1 pack-connector pins

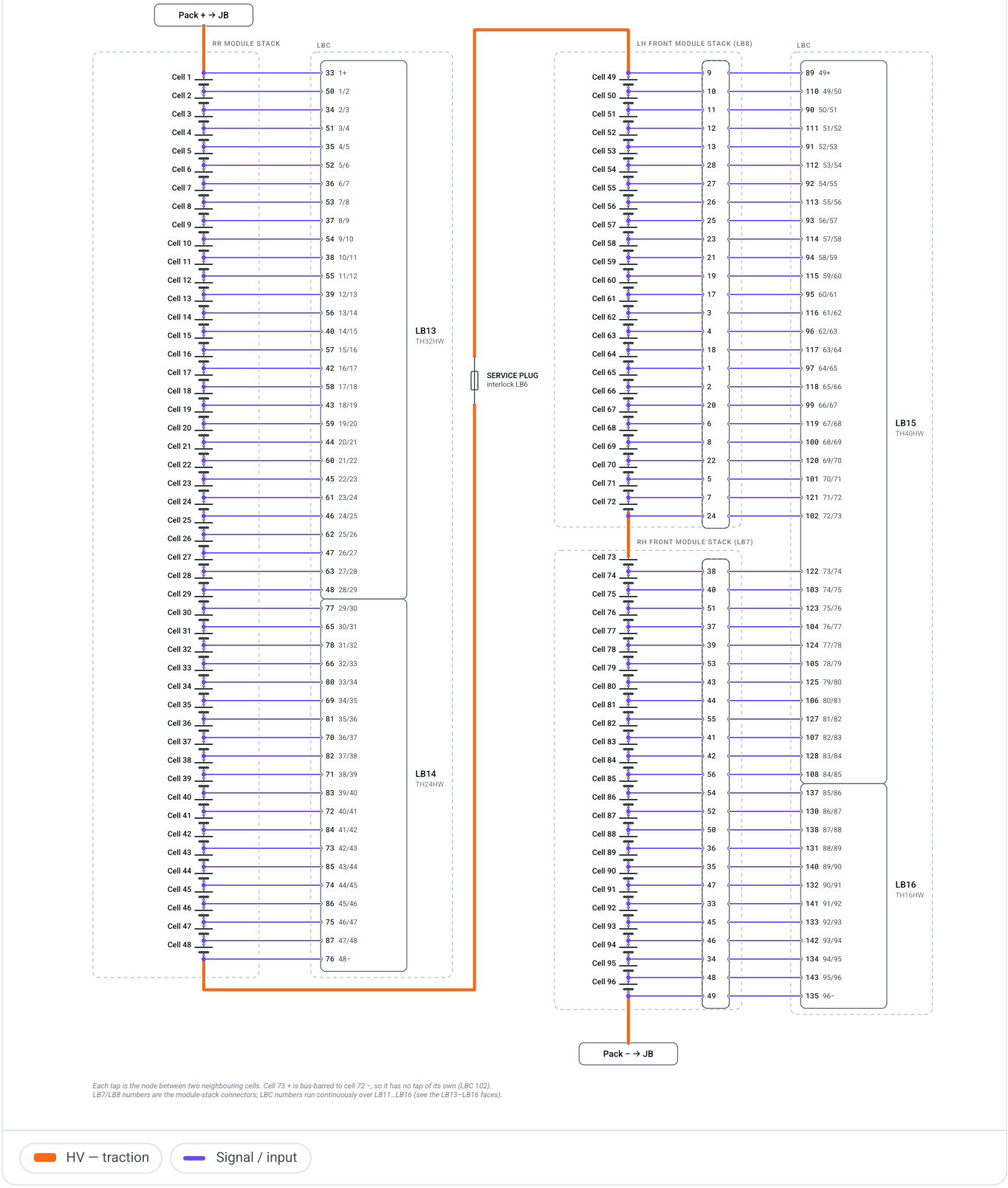


Fig. 34 Leaf AZE0 traction battery: voltage taps of the 96 cells → LBC (LB13–LB16) via stack connectors LB8 and LB7

LBC — connector LB11 (TH24HW): supply, CAN, ground, current sensor, interlock



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
1		CAN-H → LB1-1 → ECU X1-A4 (with CHAdeMO — X1-A5)	33
2		Ground → LB1-8 (branch ▽E)	33
3		Ground → LB1-7 (branch ▽D)	33
5		+12 V from the M/C RELAY via fuse No.74 10 A → LB1-31 (branch ▽F)	33
6		DETECT INPUT SIG — interlock loop input (from LB6-3)	33
7		+12 V "PWR" via fuse No.57 10 A → LB1-4	33
8		DETECT OUTPUT SIG — interlock loop output (to LB2-1)	33
9		VOLTAGE — current sensor supply (LB5 "V")	33
10		CURRENT SIG — current sensor signal (LB5 "S")	33
12		+12 V "12 V BAT" via fuse No.79 20 A → LB1-5	33
13		CAN-L → LB1-2 → ECU X1-B4 (with CHAdeMO — X1-B5)	33
14		Ground → LB1-6 (branch ▽C)	33
15		GROUND — current sensor ground (LB5 "G")	33
18		Line to ▽B (destination not shown on the drawing)	33
20		Line to ▽A (destination not shown on the drawing)	33

LBC — connector LB12 (TH08HW): battery temperature sensors



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
25		BAT TEMP SIG — sensor 1 (LB9-1)	33
26		GROUND — sensor 2 (LB10-3)	33
28		GROUND — sensor 4 (LB10-1)	33
29		GROUND — sensor 1 (LB9-5)	33
30		BAT TEMP SIG — sensor 2 (LB10-4)	33
32		BAT TEMP SIG — sensor 4 (LB10-2)	33

LBC — connector LB13 (TH32HW): cell voltage taps (RR stack, cells 1–29)



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
33		Cell 1 + (pack +) — voltage tap (RR module stack)	34
34		Cells 2/3 — voltage tap (RR module stack)	34
35		Cells 4/5 — voltage tap (RR module stack)	34
36		Cells 6/7 — voltage tap (RR module stack)	34
37		Cells 8/9 — voltage tap (RR module stack)	34
38		Cells 10/11 — voltage tap (RR module stack)	34
39		Cells 12/13 — voltage tap (RR module stack)	34
40		Cells 14/15 — voltage tap (RR module stack)	34
42		Cells 16/17 — voltage tap (RR module stack)	34
43		Cells 18/19 — voltage tap (RR module stack)	34
44		Cells 20/21 — voltage tap (RR module stack)	34
45		Cells 22/23 — voltage tap (RR module stack)	34
46		Cells 24/25 — voltage tap (RR module stack)	34
47		Cells 26/27 — voltage tap (RR module stack)	34
48		Cells 28/29 — voltage tap (RR module stack)	34
50		Cells 1/2 — voltage tap (RR module stack)	34
51		Cells 3/4 — voltage tap (RR module stack)	34
52		Cells 5/6 — voltage tap (RR module stack)	34
53		Cells 7/8 — voltage tap (RR module stack)	34
54		Cells 9/10 — voltage tap (RR module stack)	34
55		Cells 11/12 — voltage tap (RR module stack)	34
56		Cells 13/14 — voltage tap (RR module stack)	34
57		Cells 15/16 — voltage tap (RR module stack)	34
58		Cells 17/18 — voltage tap (RR module stack)	34
59		Cells 19/20 — voltage tap (RR module stack)	34
60		Cells 21/22 — voltage tap (RR module stack)	34
61		Cells 23/24 — voltage tap (RR module stack)	34
62		Cells 25/26 — voltage tap (RR module stack)	34
63		Cells 27/28 — voltage tap (RR module stack)	34

LBC — connector LB14 (TH24HW): cell voltage taps (RR stack, cells 29–48)



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
65		Cells 30/31 — voltage tap (RR module stack)	34
66		Cells 32/33 — voltage tap (RR module stack)	34
69		Cells 34/35 — voltage tap (RR module stack)	34
70		Cells 36/37 — voltage tap (RR module stack)	34
71		Cells 38/39 — voltage tap (RR module stack)	34
72		Cells 40/41 — voltage tap (RR module stack)	34
73		Cells 42/43 — voltage tap (RR module stack)	34
74		Cells 44/45 — voltage tap (RR module stack)	34
75		Cells 46/47 — voltage tap (RR module stack)	34
76		Cell 48 – (to the service plug) — voltage tap (RR module stack)	34
77		Cells 29/30 — voltage tap (RR module stack)	34
78		Cells 31/32 — voltage tap (RR module stack)	34
80		Cells 33/34 — voltage tap (RR module stack)	34
81		Cells 35/36 — voltage tap (RR module stack)	34
82		Cells 37/38 — voltage tap (RR module stack)	34
83		Cells 39/40 — voltage tap (RR module stack)	34
84		Cells 41/42 — voltage tap (RR module stack)	34
85		Cells 43/44 — voltage tap (RR module stack)	34
86		Cells 45/46 — voltage tap (RR module stack)	34
87		Cells 47/48 — voltage tap (RR module stack)	34

LBC — connector LB15 (TH40HW): cell voltage taps (LH stack cells 49–72 and RH stack cells 73–85)



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
89		Cell 49 + (after the service plug) — voltage tap (LH front stack, via LB8-9)	34
90		Cells 50/51 — voltage tap (LH front stack, via LB8-11)	34
91		Cells 52/53 — voltage tap (LH front stack, via LB8-13)	34
92		Cells 54/55 — voltage tap (LH front stack, via LB8-27)	34
93		Cells 56/57 — voltage tap (LH front stack, via LB8-25)	34
94		Cells 58/59 — voltage tap (LH front stack, via LB8-21)	34
95		Cells 60/61 — voltage tap (LH front stack, via LB8-17)	34
96		Cells 62/63 — voltage tap (LH front stack, via LB8-4)	34
97		Cells 64/65 — voltage tap (LH front stack, via LB8-1)	34
99		Cells 66/67 — voltage tap (LH front stack, via LB8-20)	34
100		Cells 68/69 — voltage tap (LH front stack, via LB8-8)	34
101		Cells 70/71 — voltage tap (LH front stack, via LB8-5)	34
102		Cells 72/73 — voltage tap (LH front stack, via LB8-24)	34
103		Cells 74/75 — voltage tap (RH front stack, via LB7-40)	34
104		Cells 76/77 — voltage tap (RH front stack, via LB7-37)	34
105		Cells 78/79 — voltage tap (RH front stack, via LB7-53)	34
106		Cells 80/81 — voltage tap (RH front stack, via LB7-44)	34
107		Cells 82/83 — voltage tap (RH front stack, via LB7-41)	34
108		Cells 84/85 — voltage tap (RH front stack, via LB7-56)	34
110		Cells 49/50 — voltage tap (LH front stack, via LB8-10)	34
111		Cells 51/52 — voltage tap (LH front stack, via LB8-12)	34
112		Cells 53/54 — voltage tap (LH front stack, via LB8-28)	34
113		Cells 55/56 — voltage tap (LH front stack, via LB8-26)	34
114		Cells 57/58 — voltage tap (LH front stack, via LB8-23)	34
115		Cells 59/60 — voltage tap (LH front stack, via LB8-19)	34
116		Cells 61/62 — voltage tap (LH front stack, via LB8-3)	34
117		Cells 63/64 — voltage tap (LH front stack, via LB8-18)	34
118		Cells 65/66 — voltage tap (LH front stack, via LB8-2)	34
119		Cells 67/68 — voltage tap (LH front stack, via LB8-6)	34
120		Cells 69/70 — voltage tap (LH front stack, via LB8-22)	34
121		Cells 71/72 — voltage tap (LH front stack, via LB8-7)	34
122		Cells 73/74 — voltage tap (RH front stack, via LB7-38)	34
123		Cells 75/76 — voltage tap (RH front stack, via LB7-51)	34
124		Cells 77/78 — voltage tap (RH front stack, via LB7-39)	34
125		Cells 79/80 — voltage tap (RH front stack, via LB7-43)	34
127		Cells 81/82 — voltage tap (RH front stack, via LB7-55)	34

PIN	WIRE	FUNCTION	DIAGRAMS
128		Cells 83/84 — voltage tap (RH front stack, via LB7-42)	34

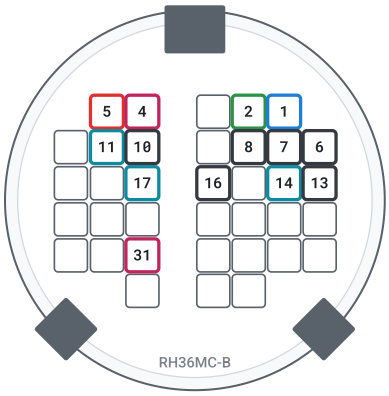
LBC — connector LB16 (TH16HW): cell voltage taps (RH stack, cells 85–96)



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
130		Cells 86/87 — voltage tap (RH front stack, via LB7-52)	34
131		Cells 88/89 — voltage tap (RH front stack, via LB7-36)	34
132		Cells 90/91 — voltage tap (RH front stack, via LB7-47)	34
133		Cells 92/93 — voltage tap (RH front stack, via LB7-45)	34
134		Cells 94/95 — voltage tap (RH front stack, via LB7-34)	34
135		Cell 96 − (pack −) — voltage tap (RH front stack, via LB7-49)	34
137		Cells 85/86 — voltage tap (RH front stack, via LB7-54)	34
138		Cells 87/88 — voltage tap (RH front stack, via LB7-50)	34
140		Cells 89/90 — voltage tap (RH front stack, via LB7-35)	34
141		Cells 91/92 — voltage tap (RH front stack, via LB7-33)	34
142		Cells 93/94 — voltage tap (RH front stack, via LB7-46)	34
143		Cells 95/96 — voltage tap (RH front stack, via LB7-48)	34

Fig. 35 Li-ion battery controller (LBC) connectors LB11–LB16



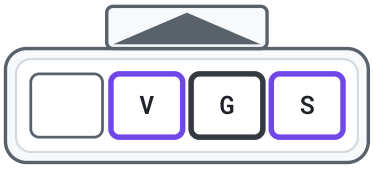
The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
1		CAN-H (LBC 1) → ECU X1-A4 (with CHAdeMO — X1-A5)	33
2		CAN-L (LBC 13) → ECU X1-B4 (with CHAdeMO — X1-B5)	33
4		+12 V from the IGN relay → LBC 7 ("PWR")	33
5		+12 V permanent → LBC 12 ("12 V BAT")	33
31		+12 V from the IGN relay → LBC 5	33
6		Ground (LBC 14)	33
7		Ground (LBC 3)	33
8		Ground (LBC 2)	33
17		SYSTEM MAIN RELAY 1 coil, "+" contactor (LB3-1) ← ECU X2-A4	33
16		SYSTEM MAIN RELAY 1 coil ground (LB3-5)	33
14		SYSTEM MAIN RELAY 2 coil, "−" contactor (LB3-8) ← ECU X2-A3	33
13		SYSTEM MAIN RELAY 2 coil ground (LB3-12)	33
11		Pre-charge relay coil (LB3-2) ← ECU X2-B4	33
10		Pre-charge relay coil ground (LB3-4)	33

Fig. 36 Battery connector LB1

The stock Leaf current sensor (LB5) stays connected to the LBC: G — ground (LBC 15), V — sensor supply (LBC 9), S — signal (LBC 10).

LB5



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
V		V — sensor supply ← LBC 9 (VOLTAGE)	33
G		G — sensor ground ← LBC 15 (GROUND)	33
S		S — current signal → LBC 10 (CURRENT SIG)	33

Fig. 37 Leaf battery current sensor — connector LB5

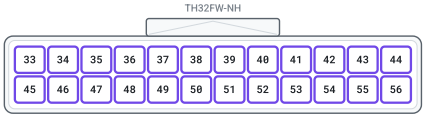
LB8 (TH24FW-NH) — LH front module stack connector (cells 49–72)



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
1		Cells 64/65 → LBC 97	34
2		Cells 65/66 → LBC 118	34
3		Cells 61/62 → LBC 116	34
4		Cells 62/63 → LBC 96	34
5		Cells 70/71 → LBC 101	34
6		Cells 67/68 → LBC 119	34
7		Cells 71/72 → LBC 121	34
8		Cells 68/69 → LBC 100	34
9		Cell 49 + (after the service plug) → LBC 89	34
10		Cells 49/50 → LBC 110	34
11		Cells 50/51 → LBC 90	34
12		Cells 51/52 → LBC 111	34
13		Cells 52/53 → LBC 91	34
17		Cells 60/61 → LBC 95	34
18		Cells 63/64 → LBC 117	34
19		Cells 59/60 → LBC 115	34
20		Cells 66/67 → LBC 99	34
21		Cells 58/59 → LBC 94	34
22		Cells 69/70 → LBC 120	34
23		Cells 57/58 → LBC 114	34
24		Cells 72/73 → LBC 102	34
25		Cells 56/57 → LBC 93	34
26		Cells 55/56 → LBC 113	34
27		Cells 54/55 → LBC 92	34
28		Cells 53/54 → LBC 112	34

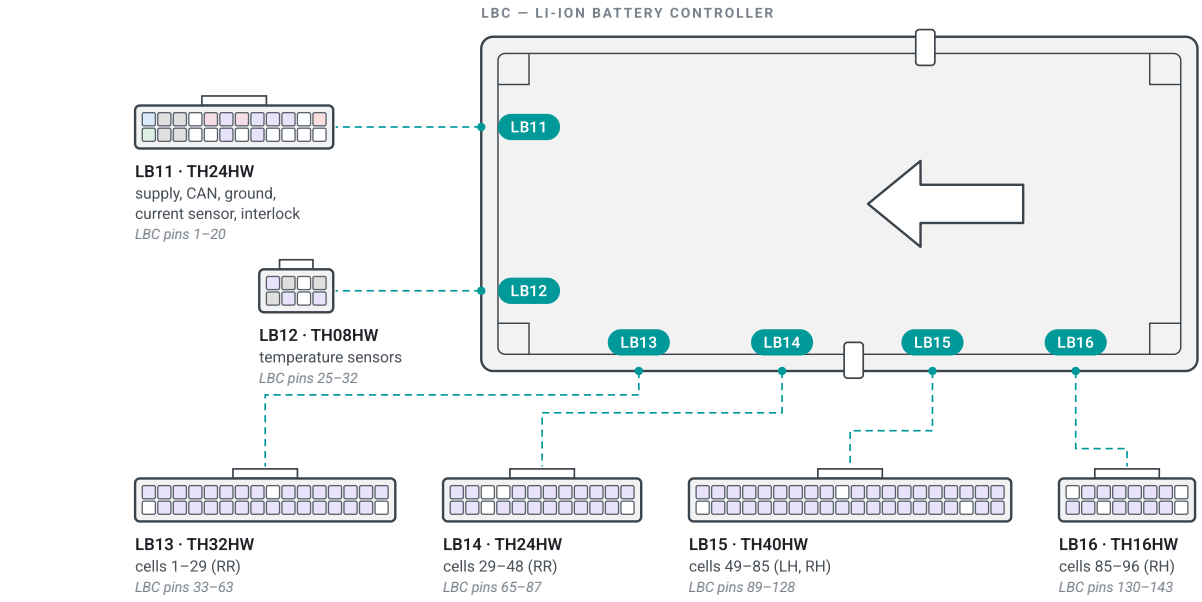
LB7 (TH32FW-NH) — RH front module stack connector (cells 73–96)



The source does not state the view side (Nissan factory drawing without an H.S. / T.S. mark) — check against the numbers on the connector housing

PIN	WIRE	FUNCTION	DIAGRAMS
33		Cells 91/92 → LBC 141	34
34		Cells 94/95 → LBC 134	34
35		Cells 89/90 → LBC 140	34
36		Cells 88/89 → LBC 131	34
37		Cells 76/77 → LBC 104	34
38		Cells 73/74 → LBC 122	34
39		Cells 77/78 → LBC 124	34
40		Cells 74/75 → LBC 103	34
41		Cells 82/83 → LBC 107	34
42		Cells 83/84 → LBC 128	34
43		Cells 79/80 → LBC 125	34
44		Cells 80/81 → LBC 106	34
45		Cells 92/93 → LBC 133	34
46		Cells 93/94 → LBC 142	34
47		Cells 90/91 → LBC 132	34
48		Cells 95/96 → LBC 143	34
49		Cell 96 − (pack −) → LBC 135	34
50		Cells 87/88 → LBC 138	34
51		Cells 75/76 → LBC 123	34
52		Cells 86/87 → LBC 130	34
53		Cells 78/79 → LBC 105	34
54		Cells 85/86 → LBC 137	34
55		Cells 81/82 → LBC 127	34
56		Cells 84/85 → LBC 108	34

Fig. 38 Module stack connectors: LB8 (left front) and LB7 (right front)



Layout and positions follow the Nissan factory drawing (the arrow is copied from it). Cavities match the connector pinouts below; colours match the wire colours of the diagrams.

Fig. 39 Nissan Leaf Li-ion battery controller (LBC): where connectors LB11–LB16 sit

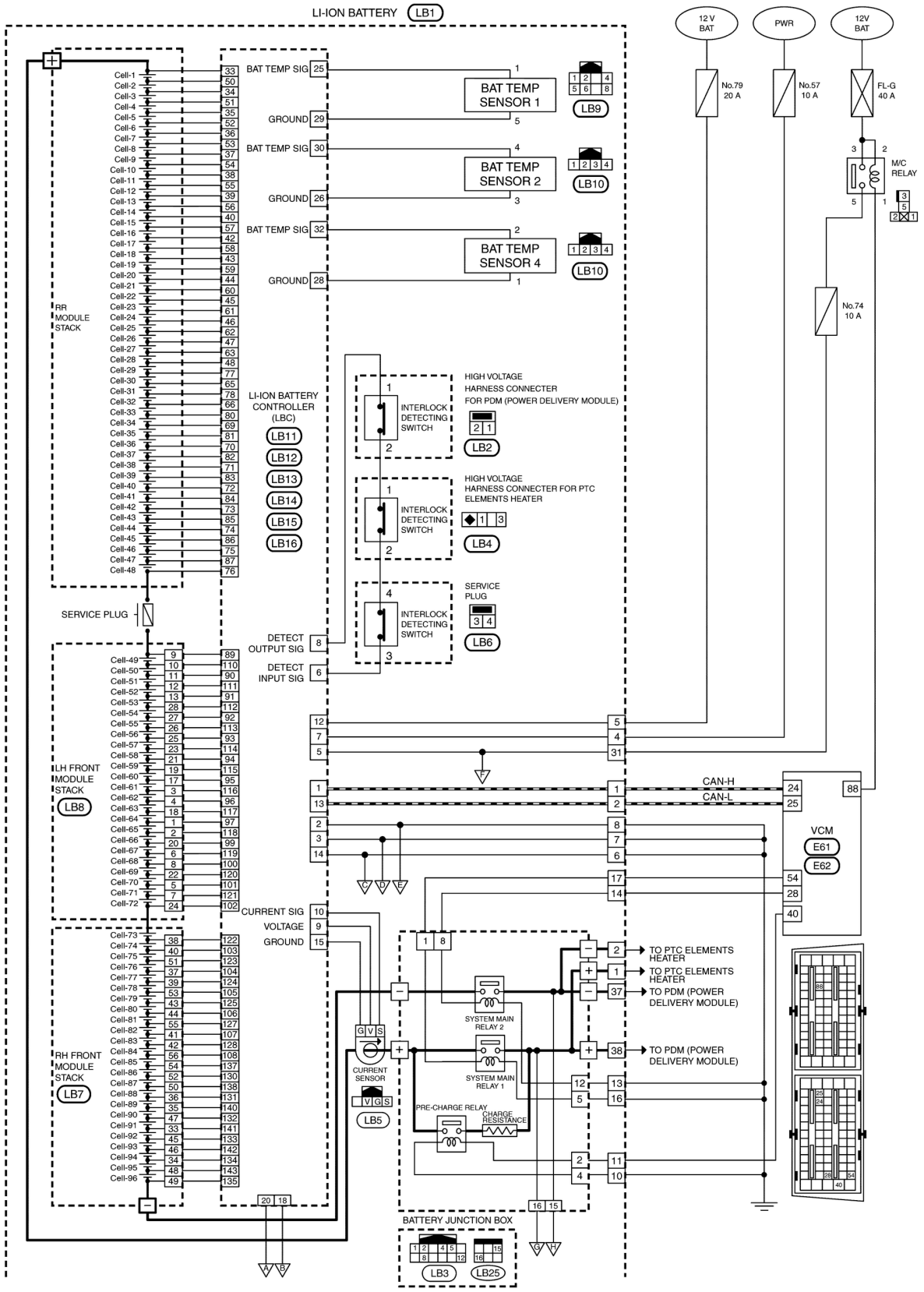


Fig. 40 Nissan Leaf AZE0 battery wiring — original factory drawing

> **NOTE**
> The diagrams are for the AZE0 BMS; other versions differ slightly.

8.3 Orion 2

Supported by the controller. Detailed documentation will be added in future revisions.

8.4 JK BMS (satellites)

The controller uses JK-DZ08-B1A24S active balancers as BMS satellites. Each balancer measures its own group of cells and, when the controller asks, sends the cell voltages and its temperature over CAN.

PARAMETER	VALUE
Bus	CAN1: CAN-H X1-A4, CAN-L X1-B4, 250 kbps
Satellites	Up to 5 supported (addresses 1–5); the diagram shows an example of 4 satellites × 24 cells = 96
Cells	2–24 per satellite, 112 in total at most
Polling	The controller polls addresses 1–5 in turn, each one every 0.5 s

The controller takes the pack voltage as the sum of all cell voltages, and the over-discharge and overcharge protections work on the lowest and highest cell voltage (the cell voltage thresholds are set on the **BMS → Parameters** page).

What the controller does not do:

- **It does not control balancing.** The balancing threshold and current, the balancing on/off switch and the cell count are stored in the balancer itself; the controller only sends the satellites a data request. The balancing settings on the **BMS → Parameters** page have no effect on JK balancers.
- **It does not limit current by temperature.** The controller receives each satellite's temperature, but there is no protection based on it.
- **It gets no pack current from JK.** The balancer does not measure current, so JK-BMS always needs the LEM CAB-500 current sensor on the same bus (see below).

> **WARNING**
> JK-BMS works on CAN1 only. If **CHAdeMO type** is set to **Internal (CAN1)**, CAN1 is handed over to CHAdeMO and the JK satellites are not polled; unlike the other BMS types, JK has no fallback to CAN2. Do not use this CHAdeMO type with JK-BMS.

Wiring

The diagram shows a 96-cell battery as an example: 4 satellites with 24 cells each. The controller supports up to 5 satellites — a fifth one connects to the same bus in the same way.

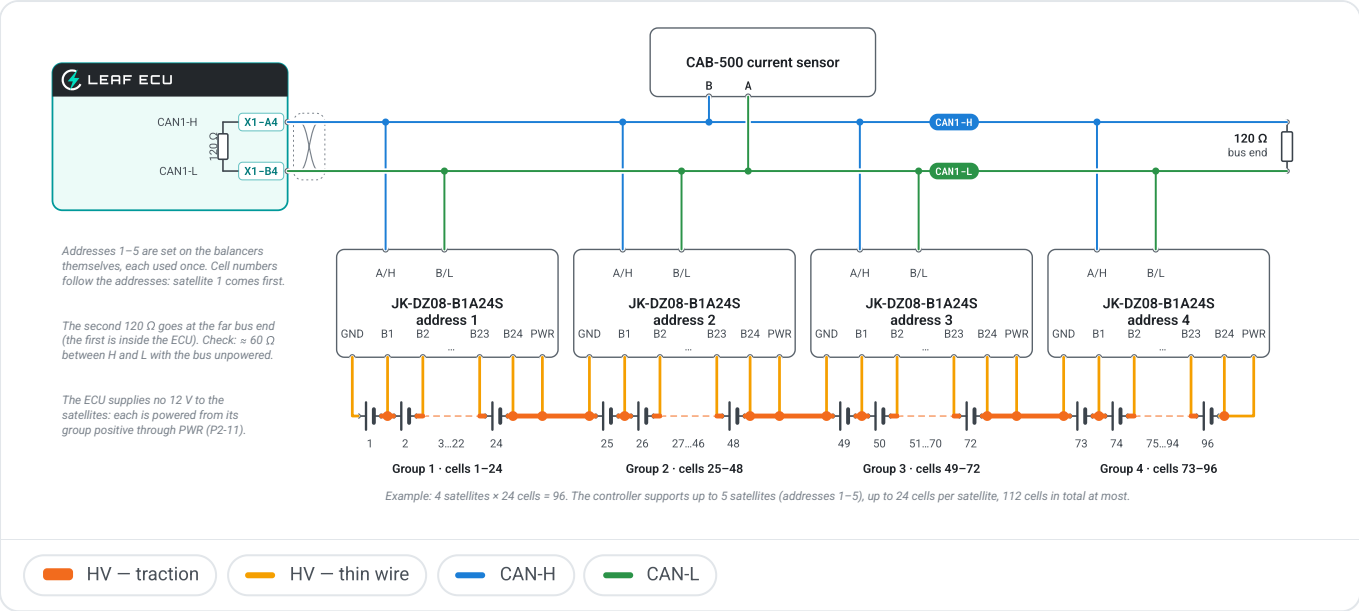
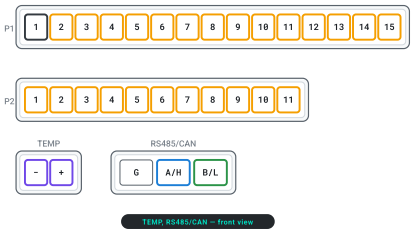


Fig. 41 JK BMS: JK-DZ08-B1A24S satellites on the ECU CAN1 bus and the battery cell split (example: 4 × 24 = 96 cells; up to 5 satellites supported)

- The CAN bus goes to the **RS485/CAN** connector on the balancer's front edge (3 contacts: G, A/H, B/L). Connect A/H (CAN-H) of every satellite to `X1-A4` and B/L (CAN-L) to `X1-B4`, using twisted-pair wire. Contact G is left unconnected.
- Run the bus as one line from the controller to the farthest satellite, with a short stub to each satellite.
- The controller does not supply 12 V to the satellites: each balancer is powered from its own group through the **PWR** pin (`P2-11`), connected to the group positive. Per the JK documentation the balancer supply is 40–100 V, and PWR may go to the positive of any cell from the 13th to the 24th. For 24 NMC cells (up to 100.8 V), connect PWR to the positive of cell 23.
- Balancing leads: connector **P1** — GND (group negative) and B1–B14, connector **P2** — B15–B24 and PWR. Each lead Bn goes to the positive of cell n of its group, in order, with no gaps.
- Next to it is the **TEMP** connector (2 contacts, − and +) for the supplied temperature sensor; it is not wired to the controller.



TEMP and RS485/CAN — front view of the balancer edge, as marked on the case. P1 and P2 — pin numbers per the JK tables for the JK-B1A24S (CAN); the maker does not state their view side.

PIN	WIRE	FUNCTION	DIAGRAMS
P1-1		GND — group negative (B0)	41
P1-2		B1 — positive of cell 1 of the group	41
P1-3		B2 — positive of cell 2 of the group	41
P1-4		B3 — positive of cell 3 of the group	
P1-5		B4 — positive of cell 4 of the group	
P1-6		B5 — positive of cell 5 of the group	
P1-7		B6 — positive of cell 6 of the group	
P1-8		B7 — positive of cell 7 of the group	
P1-9		B8 — positive of cell 8 of the group	
P1-10		B9 — positive of cell 9 of the group	
P1-11		B10 — positive of cell 10 of the group	
P1-12		B11 — positive of cell 11 of the group	
P1-13		B12 — positive of cell 12 of the group	
P1-14		B13 — positive of cell 13 of the group	
P1-15		B14 — positive of cell 14 of the group	
P2-1		B15 — positive of cell 15 of the group	
P2-2		B16 — positive of cell 16 of the group	
P2-3		B17 — positive of cell 17 of the group	
P2-4		B18 — positive of cell 18 of the group	
P2-5		B19 — positive of cell 19 of the group	
P2-6		B20 — positive of cell 20 of the group	
P2-7		B21 — positive of cell 21 of the group	
P2-8		B22 — positive of cell 22 of the group	
P2-9		B23 — positive of cell 23 of the group	41
P2-10		B24 — positive of cell 24 of the group	41
P2-11		PWR — balancer supply 40–100 V: group positive (anywhere from cell 13 to cell 24) or an external supply	41
T-		TEMP - — the supplied temperature sensor; not wired to the controller	
T+		TEMP + — the supplied temperature sensor; not wired to the controller	
G		G — not wired to the controller	
H		A/H — CAN-H, CAN1 bus to ECU X1-A4 (250 kbps)	41
L		B/L — CAN-L, CAN1 bus to ECU X1-B4	41

Fig. 42 JK-DZ08-B1A24S balancer — connectors P1, P2 (balancing leads), TEMP and RS485/CAN

> **NOTE**
> The satellites are connected to different sections of the traction battery. Before joining them into one CAN bus with the controller, make sure from the JK documentation that the balancer's CAN interface is galvanically isolated from the cells.

Addresses and cell numbering

- Give every satellite its own address from 1 to 5. Addresses must not repeat: the reply's CAN ID equals the address, and two balancers with the same address would answer at the same time.
- The controller does not poll addresses above 5.
- The CAN protocol has no command to change the address — it is set on the balancer itself; see the JK documentation for how.

The controller numbers the cells by address: first come the cells of satellite 1, then those of satellite 2, and so on. Each satellite adds as many cells as it has detected itself; a skipped address leaves no gap in the numbering. For example, four satellites with 24 cells each give cell numbers 1–24, 25–48, 49–72 and 73–96.

Assign the addresses in the order of the groups along the battery — the cell map in the web configurator will then match the physical cell layout.

> **NOTE**
> If a satellite detects fewer cells than are connected (for example, because of a broken balancing lead), the cell numbers of all following satellites shift and the total cell count drops.

Termination resistors

CAN1 must be terminated with two 120 Ω resistors, one at each end of the bus. The controller already has a 120 Ω resistor built in on every CAN channel, so none is needed at the controller. Fit the second 120 Ω at the far end of the bus — at the farthest satellite. If the device at the far end already has a built-in terminator, no separate resistor is needed there.

Check: with the power off and the bus connected, measure the resistance between `X1-A4` and `X1-B4`:

RESISTANCE	MEANING
≈ 60 Ω	Correct — two terminators
≈ 120 Ω	One terminator missing
≈ 40 Ω or less	One terminator too many

Current sensor

With JK-BMS the pack current is measured **only by the LEM CAB-500 sensor** — it is required. The sensor connects to the same CAN1 bus as the satellites (see 9.2).

On the **BMS → Parameters** page, set **Current sens type** to **CAB-500**, then press **Train** (the **CAB-500 Training** row): the controller finds the sensor and switches it to the JK bus speed, 250 kbps.

> **NOTE**
> The CAB 500-C/SP5-012 sensor has a built-in 120 Ω terminator; the other CAB-500 versions do not. Take this into account when checking the bus resistance.

Web configurator settings

1. **ECU → Settings:** set **Type BMS** to **JK-BMS** and press **Write**. Check the **current value**. CAN1 switches to 250 kbps immediately.
2. **BMS → Parameters:** set **Cells in pack** to the total number of cells across all satellites. The controller derives the pack voltage limit for overcharge protection from it.
3. **BMS → Parameters:** set the cell voltage thresholds and the current sensor type.

Checks after power-up

- [ ] Before power-up: resistance between `X1-A4` and `X1-B4` ≈ 60 Ω.
- [ ] **BMS → Cells:** all cell voltages are non-zero, numbers follow the addresses, and the last filled cell matches the total cell count.
- [ ] **BMS → Parameters:** the pack voltage matches a multimeter reading, and the lowest and highest cell voltages are plausible.
- [ ] The current changes under load and has the correct sign; for CAB-500 the direction is changed with the **Toggle** button in the **Sensor direction** row.

> **NOTE**
> The **BMS → Cells** page shows cells 1–96; cells 97–112 are not visible in the web configurator.

> **WARNING**
> If a satellite does not answer for 35 s, the controller sets its cell voltages to zero (0 V on the cell map) and no longer counts them in the lowest and highest cell voltage, and the pack voltage drops by their sum. Driving is not blocked. The exception is the satellite holding cell 1: a zero voltage on that cell is treated as over-discharge and stops the car. Cells showing 0 V on the map indicate a lost satellite.

9 Current Sensor

The controller works with two current sensor types: the analog LEM DHAB (Chevrolet Volt) and the CAN sensor LEM CAB 500. The type is chosen in the web configurator: **BMS → Parameters → Current sens type**. The sensor can be placed anywhere inside the battery before the power relays.

9.1 Chevrolet Volt Current Sensor (LEM DHAB)

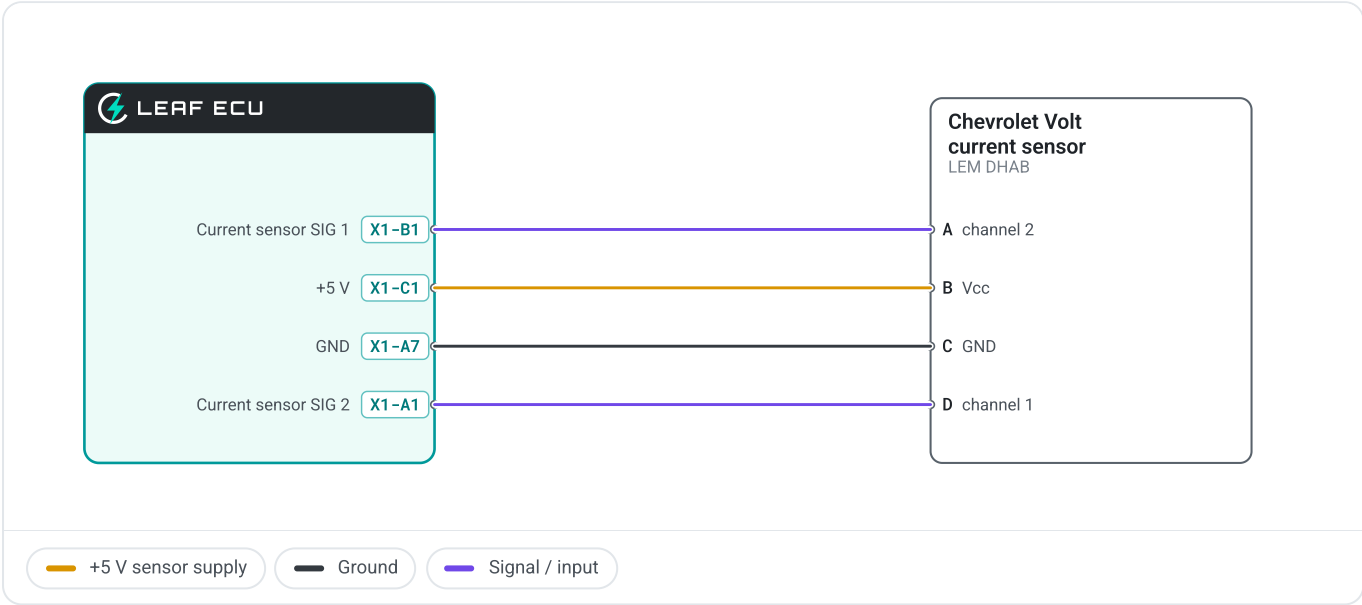
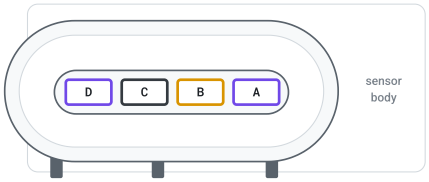


Fig. 43 Chevrolet Volt current sensor wiring



Front view into the sensor connector (as in the LEM drawing)

PIN	WIRE	FUNCTION	DIAGRAMS
A		Channel 2 → ECU X1-B1 (SIG 1)	43
B		Vcc, +5 V ← ECU X1-C1	43
C		Gnd, ground ← ECU X1-A7	43
D		Channel 1 → ECU X1-A1 (SIG 2)	43

Fig. 44 Chevrolet Volt current sensor connector (LEM DHAB)

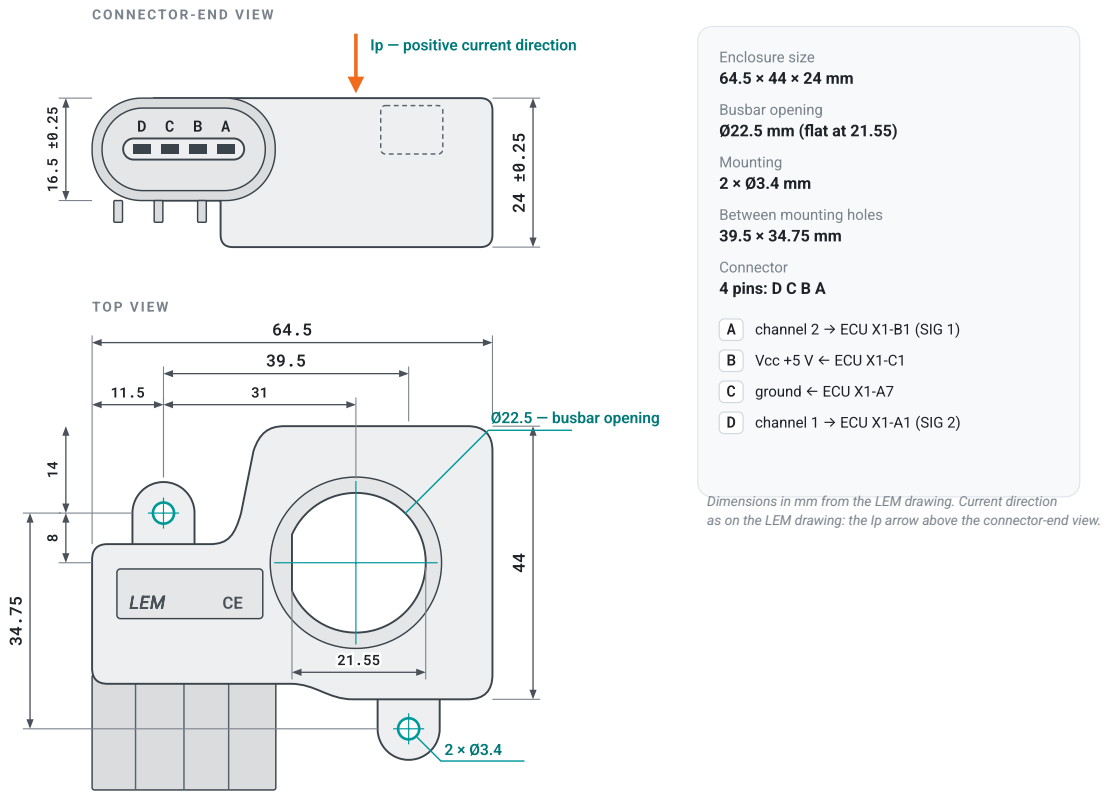


Fig. 45 LEM DHAB current sensor: dimensions, mounting, busbar opening, pinout and current direction



Fig. 46 Current sensor connector

9.2 LEM CAB 500 Current Sensor

The LEM CAB 500 (CAB 500-C series) is a galvanically isolated fluxgate current sensor with a digital CAN output. A battery power cable is passed through the sensor's opening. Nominal range is ±500 A, total error 0.3 % at 25 °C, offset below 10 mA. The sensor sends the measured current by itself in frame `0x3C2` every 10 ms, and the controller receives it on the BMS bus — CAN1 — in parallel with the BMS.

The controller works with the current in whole amperes: charge current (including regeneration) is positive, discharge current is negative. The controller counts the battery charge from this value and switches on the brake light during regeneration.

> **NOTE**
>
> The controller reads the CAB 500 only when CAN1 serves the BMS: **Type BMS** set to Volt Gen 1, Leaf, Orion 2, VW ID or JK-BMS. With **Type BMS** "Deye HV" or "OFF", or with **CHAdeMO type** "Internal (CAN1)", the sensor is not read. For Deye HV select the current sensor type "In BMS" — the BMS itself reports the current.

Wiring

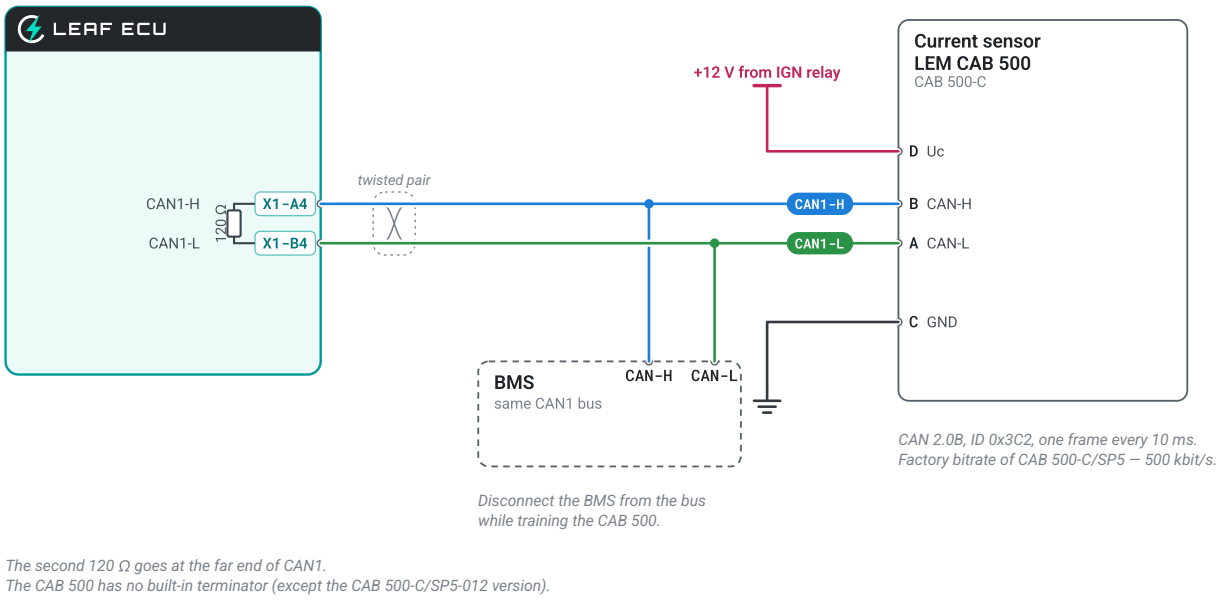
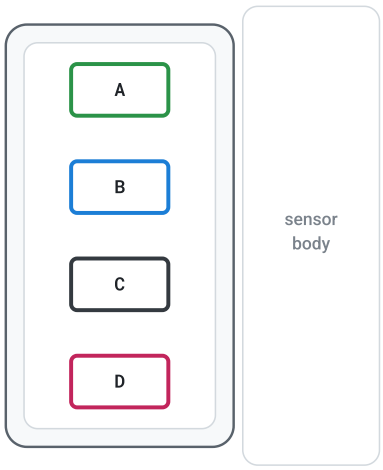


Fig. 47 Connecting the LEM CAB 500 current sensor to the CAN1 bus



Connector on the back of the sensor, near the bottom edge. Contacts in a row along the body: A nearest the cable opening, D at the bottom edge (A and D are moulded on the connector housing)

PIN	WIRE	FUNCTION	DIAGRAMS
A		CAN-L → ECU X1-B4 (CAN1-L)	41 47
B		CAN-H → ECU X1-A4 (CAN1-H)	41 47
C		GND — ground	47
D		Uc — +12 V supply from the IGN relay (8…16 V)	47

Fig. 48 LEM CAB 500 current sensor connector (TE 1565749)

- **CAN-H and CAN-L** — twisted pair to the CAN1 bus, in parallel with the BMS: X1-A4 → pin B (CAN-H), X1-B4 → pin A (CAN-L).
- **Supply** — +12 V from the IGN relay to pin D (Uc). Operating range 8…16 V, consumption 40…160 mA. The sensor has no sleep mode, so do not feed it from permanent +12 V.
- **Ground** — to pin C, common with the ECU ground.
- **Connector** on the sensor — TE 1565749; the mating harness plug is TE 1473672-1.
- **Termination.** Each end of the CAN1 bus needs a 120 Ω resistor between CAN-H and CAN-L. The controller end already has 120 Ω built in. The CAB 500 has no built-in terminator (except the CAB 500-C/SP5-012 version), so if the sensor is at the far end of the bus, fit the 120 Ω next to it.
- **Required with JK-BMS.** JK balancers do not measure current, so with JK-BMS (see 8.4) the CAB 500 is always needed.

Mounting

- Pass **one** battery power cable through the Ø24.2 mm opening — either the positive or the negative. Location — as for any current sensor: inside the battery, before the power relays.
- The sensor is fixed through two Ø6.5 mm holes (63.5 mm between centres). Versions SP2 and SP5 are fixed with plastic rivets, maximum load 70 N. SP1 and SP10 have metal inserts for M6 screws (8 N·m), SP7 and SP77 for M4 (2 N·m).
- The positive current direction Ip on the LEM drawing is through the opening from the label side towards the connector side. The sensor may face either way: the current sign is corrected in the configurator (see "Checking the current direction" below).

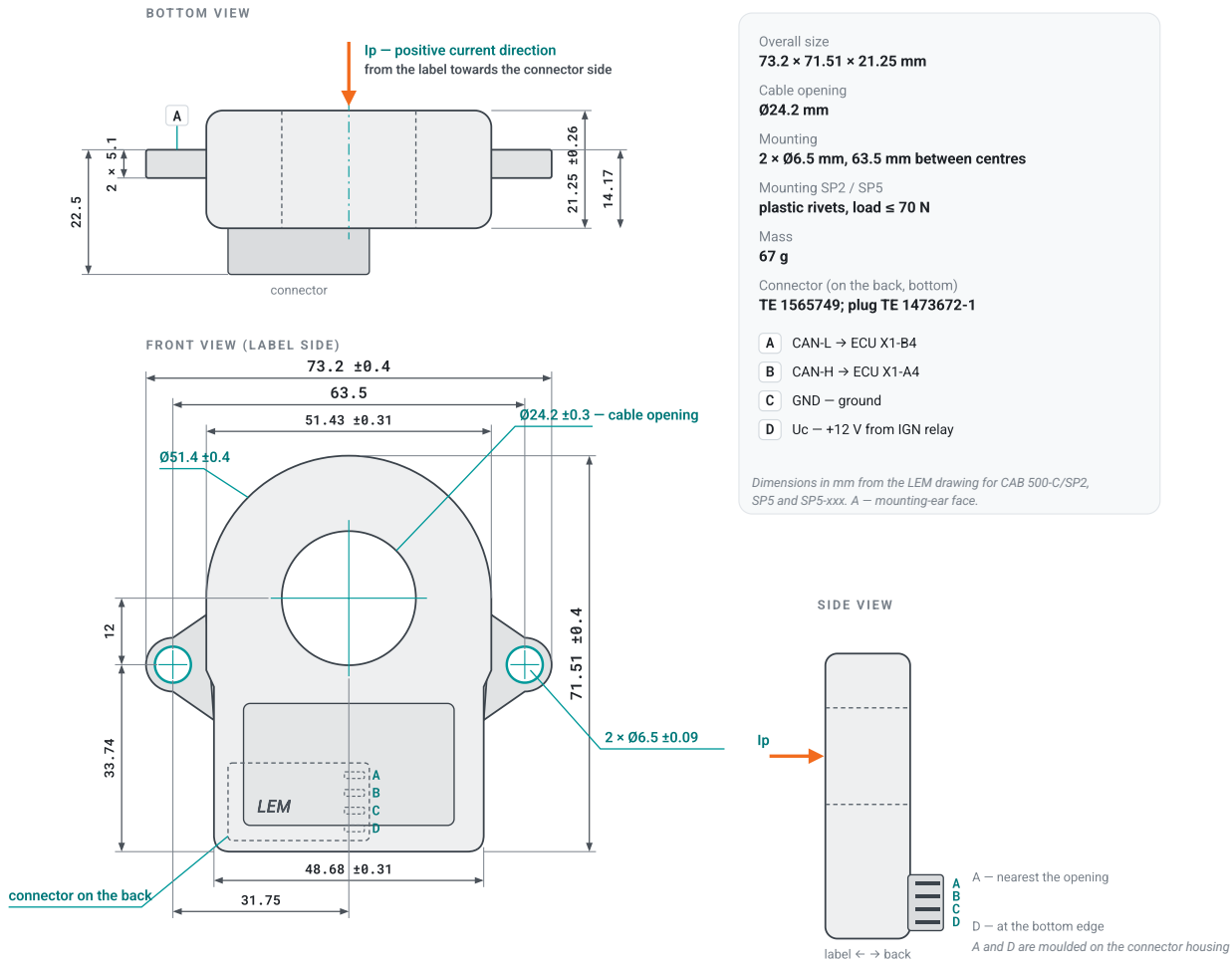


Fig. 49 LEM CAB 500 current sensor: dimensions, mounting, cable opening, pinout and current direction

> **DANGER — HIGH VOLTAGE**
>
> Fit the sensor and route the power cable through its opening only with the traction battery de-energised.

Setup in the web configurator

1. Open **ECU → Settings**, set **Type BMS** and press **Write**. Training (below) sets the sensor to the bitrate of exactly this BMS, so set the BMS type first.
2. On **BMS → Parameters**, in the **Current sens type** row select **CAB-500** and press **Write**. Make sure the current value has changed to "CAB-500". The **CAB-500 Training** and **Sensor direction** rows then become active.

The sensor must run at the same bitrate as the BMS:

TYPE BMS	CAN1 AND SENSOR BITRATE
Volt Gen 1	125 kbit/s
JK-BMS	250 kbit/s
Leaf, Orion 2, VW ID	500 kbit/s

The factory bitrate of the CAB 500-C/SP5 is 500 kbit/s, of the CAB 500-C/SP5-001 250 kbit/s. If it already matches the BMS bitrate, training only confirms it.

CAB 500 training

Training looks for the sensor on the bus at 500, 250 and 125 kbit/s. If its bitrate differs from the BMS bitrate, the controller writes the new bitrate into the sensor, restarts it and checks that the sensor answers at the new bitrate.

1. The car must be stationary and not in drive mode: the controller receives no BMS data during training.
2. Switch the ignition on. The sensor must be connected and powered.
3. Disconnect the BMS from the CAN1 bus (open its CAN-H and CAN-L) — only the CAB 500 must remain on the bus.
4. On **BMS → Parameters**, in the **CAB-500 Training** row press **Train** and confirm the prompt "Disconnect BMS from CAN1 bus! Leave only the CAB-500 sensor. Continue?".
5. Watch the status field. `SCAN_500K` → `SCAN_250K` → `SCAN_125K` — searching for the sensor (up to 1.5 s per bitrate). `WRITE_BAUD` → `RESET` → `VERIFY` → `RESTORE` — writing the bitrate, restart and check. The whole procedure takes up to 10 s.
6. Wait for the result and reconnect the BMS.

STATUS	MEANING
`OK`	Bitrate changed, the sensor answers at the new bitrate
`ALREADY_OK`	The sensor already runs at the BMS bitrate, nothing changed
`ERR_NOT_FOUND`	The sensor was not found at any bitrate — check the supply, CAN-H/CAN-L and that the BMS is disconnected
`ERR_UDS_FAIL`	The sensor did not confirm writing the new bitrate
`ERR_VERIFY`	After the restart the sensor does not answer at the new bitrate

After any result the controller returns CAN1 to the BMS bitrate. Training can be started again.

Checking the current direction

1. With no load, **Current** on **BMS → Parameters** should be about 0 A.
2. While discharging (driving, a load switched on) the current must be negative; while charging and regenerating it must be positive.
3. If the sign is the other way round, press **Toggle** in the **Sensor direction** row. The status switches between "Normal" and "Reverse" and is stored in the controller's memory. There is no need to turn the sensor round.

> **WARNING**
> Check the current sign before the first drive and the first charge. With a wrong sign the controller counts the battery charge the wrong way and switches the brake light on at the wrong time.

10 Cooling System

A liquid cooling system is required for the powertrain to work properly.

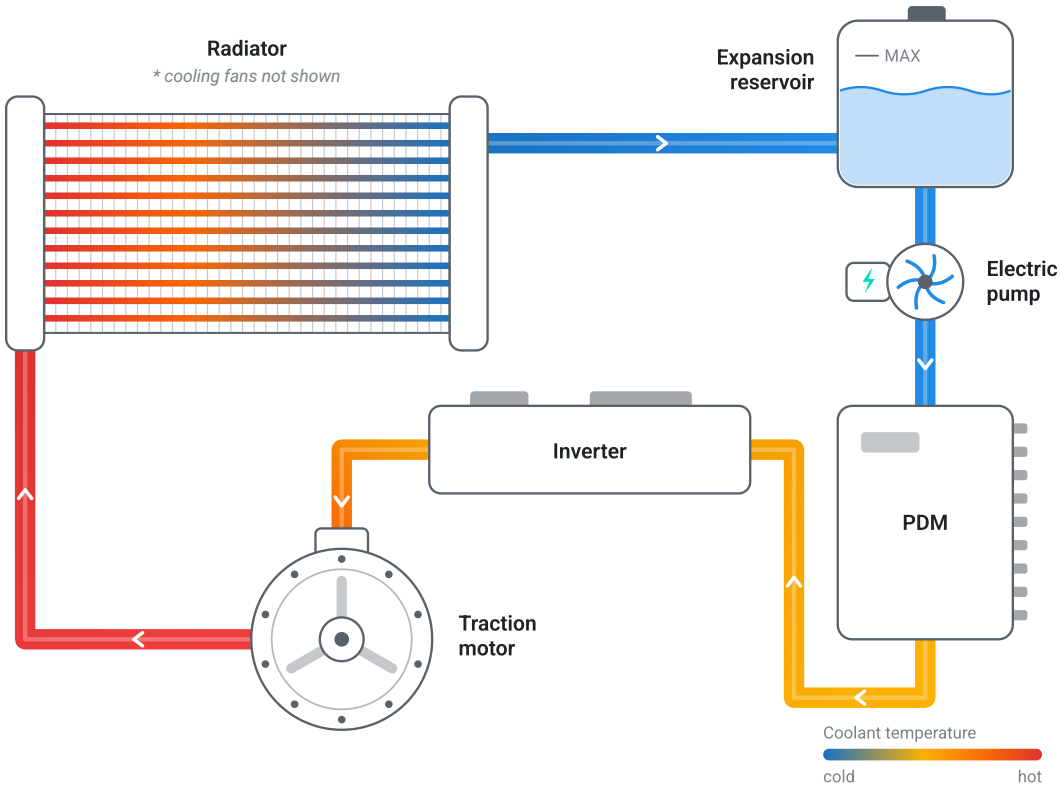


Fig. 50 Liquid cooling loop: radiator → reservoir → pump → PDM → inverter → motor

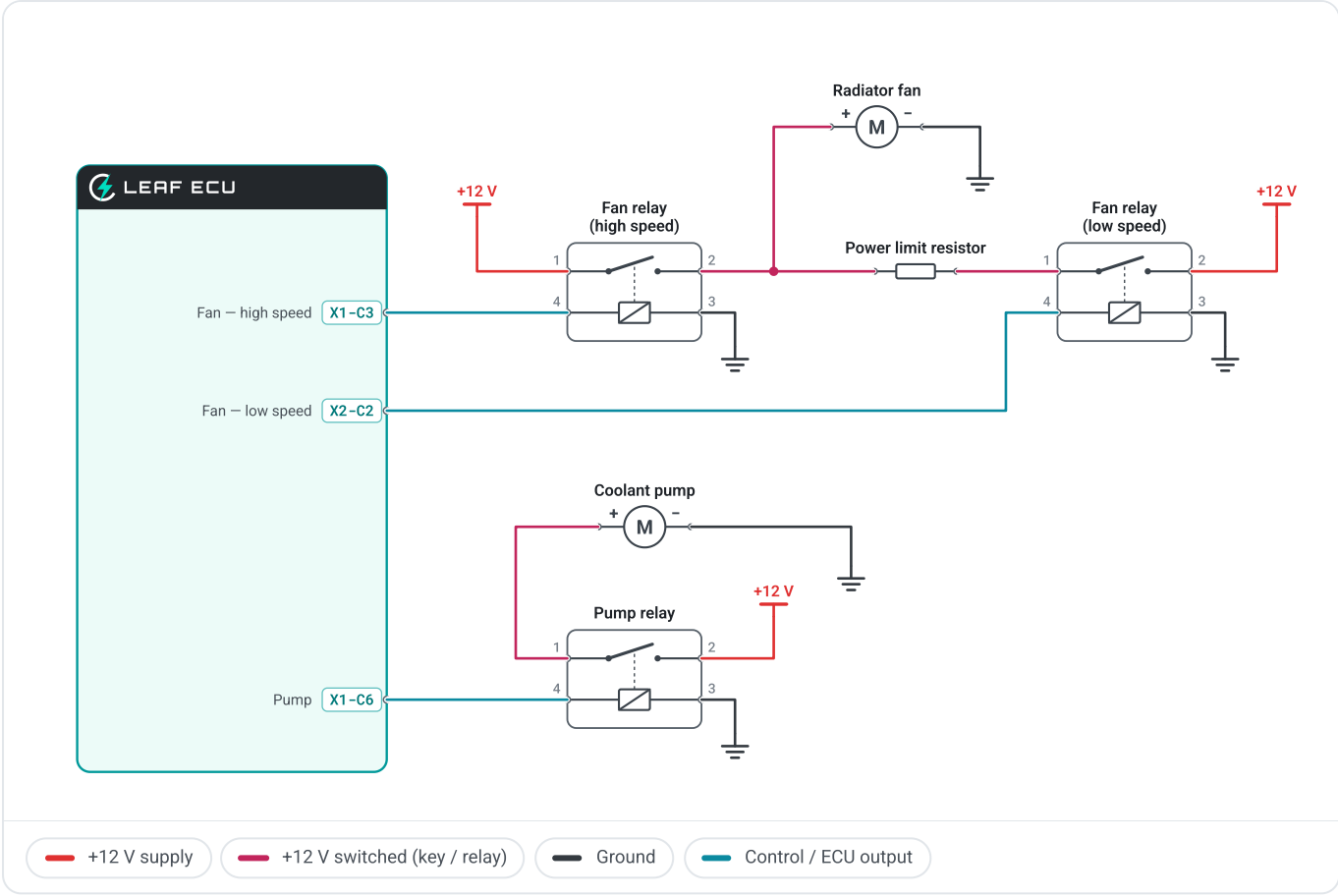


Fig. 51 Radiator fan (two speeds) and coolant pump control

Pump and fan switch-on temperatures are set in the web configurator on **ECU → Settings**. The controller can run the fan at two speeds:

- full;
- reduced — through a power-limiting resistor.

The fan can also be driven by an external PWM controller (without the resistor).

11 Brake Light (Regeneration)

The controller can switch on the brake light during regenerative braking. To do this, install a relay in parallel with the brake pedal switch.

Feed the brake-light signal to pin X2-B8 so the controller can see the brake pedal being pressed.

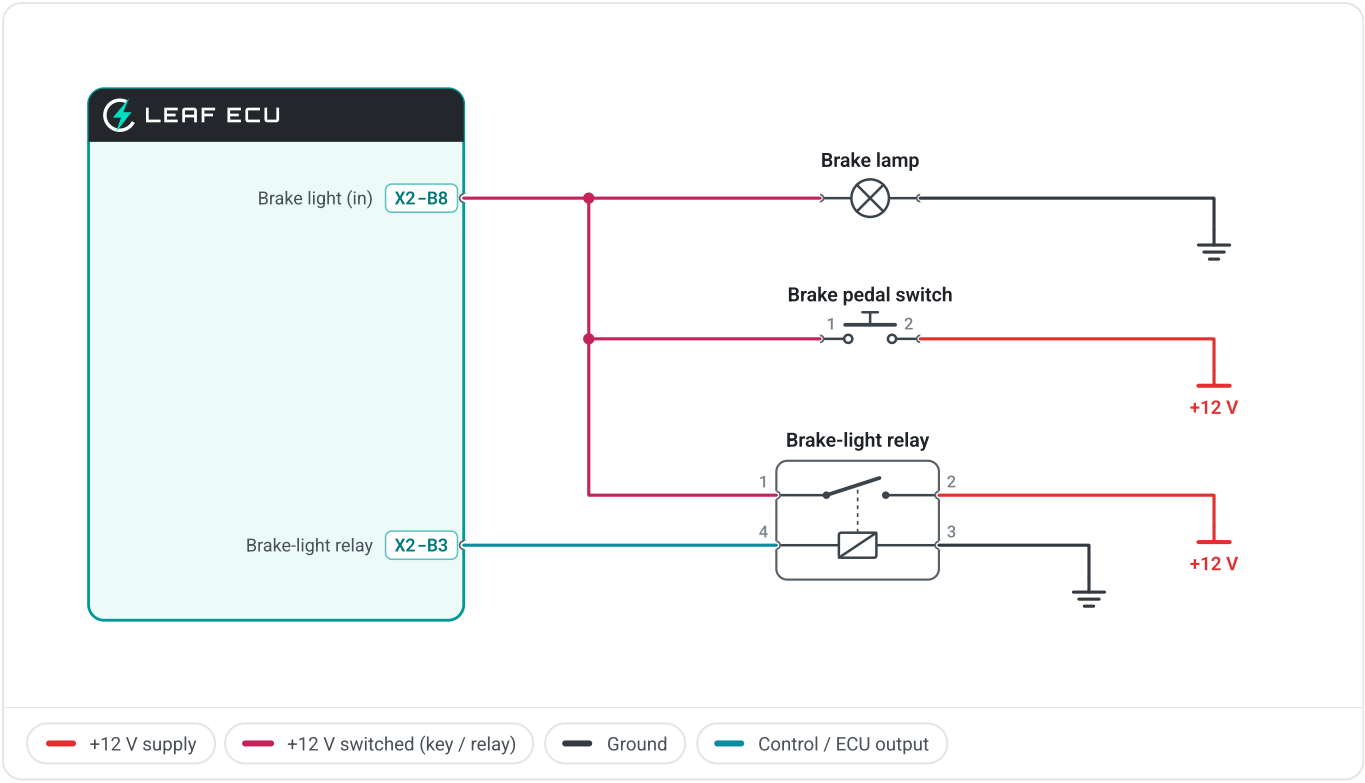


Fig. 52 Brake light during regeneration: relay in parallel with the brake pedal switch

12 CHAdeMO

If the PDM has a CHAdeMO port, its internal power contactors (CHAdeMO Power Relay + and −) can be used by running separate wires from them. If the PDM has no CHAdeMO port, two additional power contactors must be installed.

The Enable Relay is installed separately regardless of the PDM version. It does not need to be high-power — only strong enough to switch the power contactor coils.

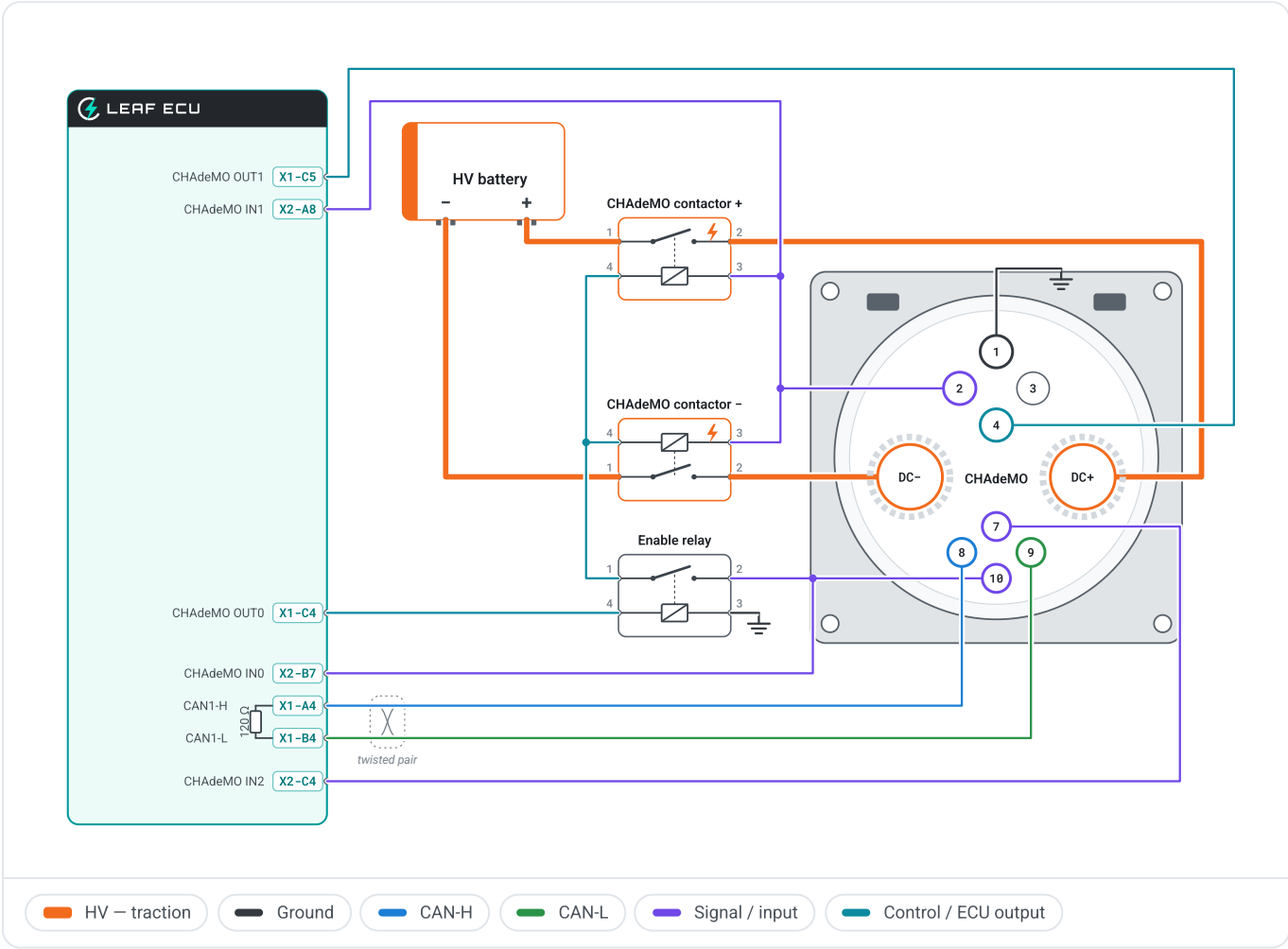
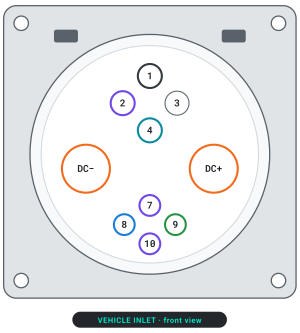


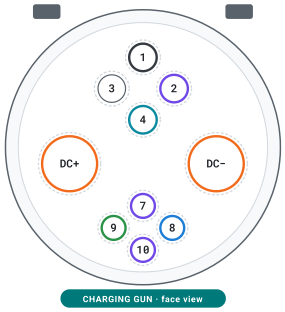
Fig. 53 CHAdeMO inlet wiring: power contactors, enable relay and charge sequence signals

CHAdeMO — vehicle inlet



Front view of the vehicle inlet (the side the gun plugs into); the mounting flange with its holes is schematic

CHAdeMO — charger gun



View onto the face of the charging gun — the mirror image of the inlet

PIN	WIRE	FUNCTION	DIAGRAMS
1		FG, protective earth → ground	53
2		charge sequence signal 1 → ECU X2-A8	53
3		not used	
4		vehicle charge permission ← ECU X1-C5	53
5		DC− → CHAdeMO contactor −	53
6		DC+ → CHAdeMO contactor +	53
7		proximity detection → ECU X2-C4	53
8		CAN-H → ECU X1-A4	53
9		CAN-L → ECU X1-B4	53
10		charge sequence signal 2 → ECU X2-B7	53

Fig. 54 CHAdeMO: vehicle inlet and charger gun

DC POLARITY

Pin **5 is DC−**, pin **6 is DC+**: on the vehicle inlet (front view) DC− is on the left, DC+ on the right. The diagram in earlier revisions of this manual showed DC+ and DC− on the inlet the other way round. Before the first charge, check with a continuity tester that inlet pin 6 goes to the CHAdeMO "+" contactor and pin 5 to the "−" contactor.

13 Display

The display controller sits on CAN0 Powertrain and reads the warning-lamp states through its own inputs; it drives a Nextion touch display.

13.1 Display Controller

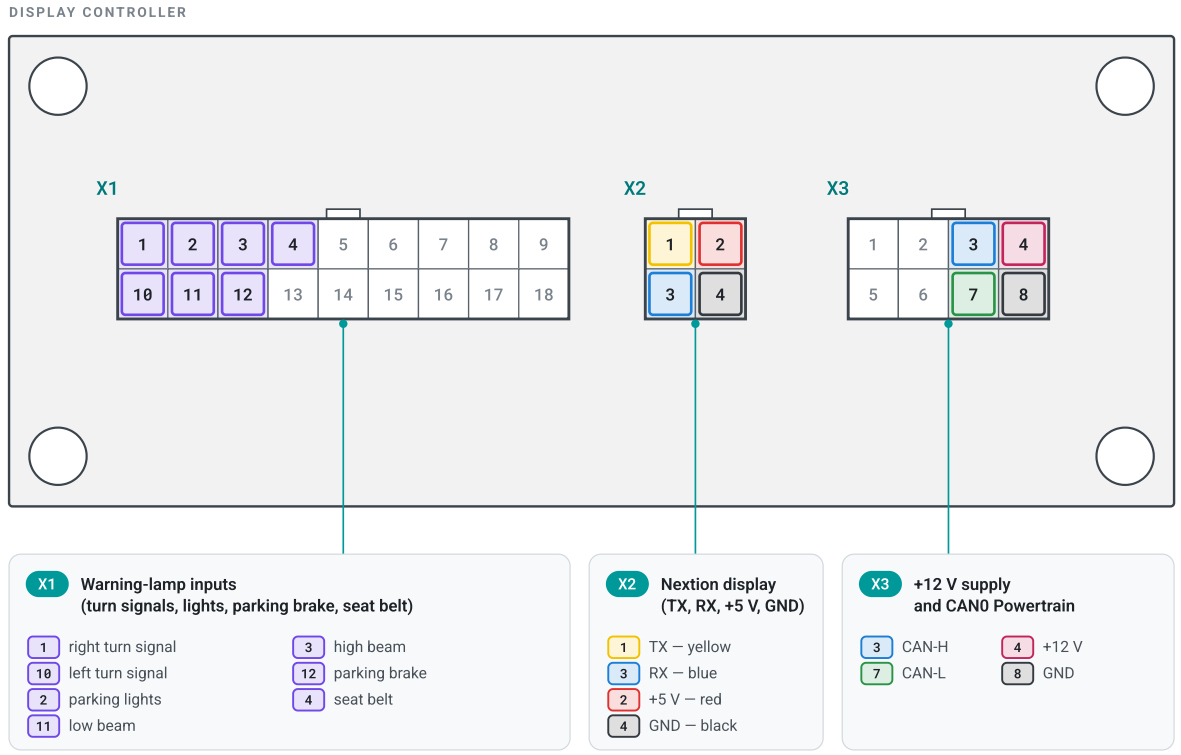
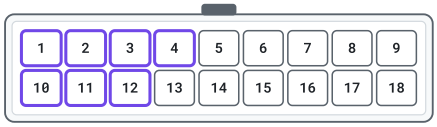


Fig. 55 Display controller: connectors X1–X3 and mounting holes

Display controller — connector X1 (signal-lamp inputs)



Board top view — looking into the connector on the board (as in the board drawing)

PIN	WIRE	FUNCTION
1		Right turn signal (IN1)
10		Left turn signal (IN2)
2		Parking lights (IN3)
11		Low beam (IN4)
3		High beam (IN5)
12		Parking brake (IN6)
4		Seat belt (IN7)

Display controller — connector X2 (Nextion display)



Board top view — looking into the connector on the board (as in the board drawing)

PIN	WIRE	FUNCTION	DIAGRAMS
1		TX — data out to the Nextion display (yellow wire → display RX)	58
3		RX — data in from the Nextion display (blue wire ← display TX)	58
2		+5 V — display supply output (red wire)	58
4		Display ground (GND) (black wire)	58

Display controller — connector X3 (interface)



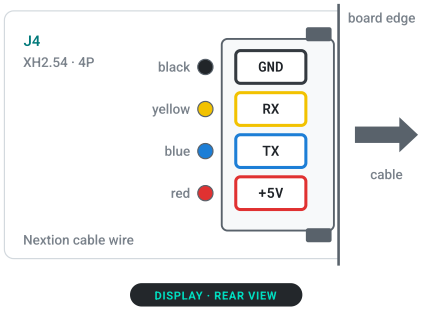
Board top view — looking into the connector on the board (as in the board drawing)

PIN	WIRE	FUNCTION
3		CAN-H — Powertrain bus (CAN0)
7		CAN-L — Powertrain bus (CAN0)
4		+12 V — controller supply
8		Ground (GND)

Fig. 56 Display controller connectors: X1 — warning-lamp inputs, X2 — display, X3 — interface

13.2 Nextion Intelligent Display

The display connects to the controller's X2 connector with four wires: +5 V supply, ground and the serial port (controller TX to display RX, RX to TX).



Rear view onto the display board (as on the Nextion drawing): J4 at the right board edge

PIN	WIRE	FUNCTION	DIAGRAMS
+5V		+5 V — display supply, red wire ← controller X2-2. Current 7": 430 mA, 5": 220 mA (typ., 100 % brightness); Nextion recommends a 5 V / 1 A supply	58
TX		TX — data out of the display, blue wire → controller RX, X2-3	58
RX		RX — data into the display, yellow wire ← controller TX, X2-1	58
GND		GND — ground, black wire ← controller X2-4	58

Fig. 57 Nextion Intelligent 7" (and 5") display — interface connector J4

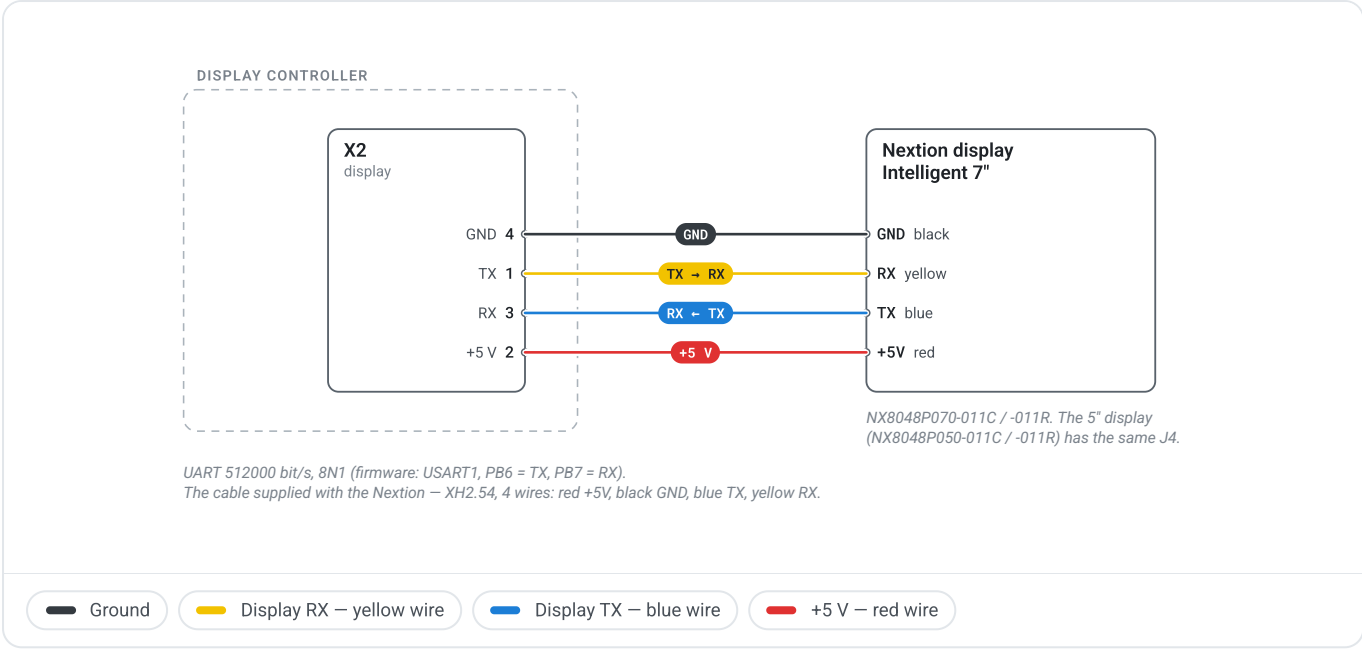


Fig. 58 Connecting the Nextion Intelligent 7" display to connector X2 of the display controller

The controller works with the Nextion Intelligent 7" and 5" displays. Dimensions and mounting holes of both:

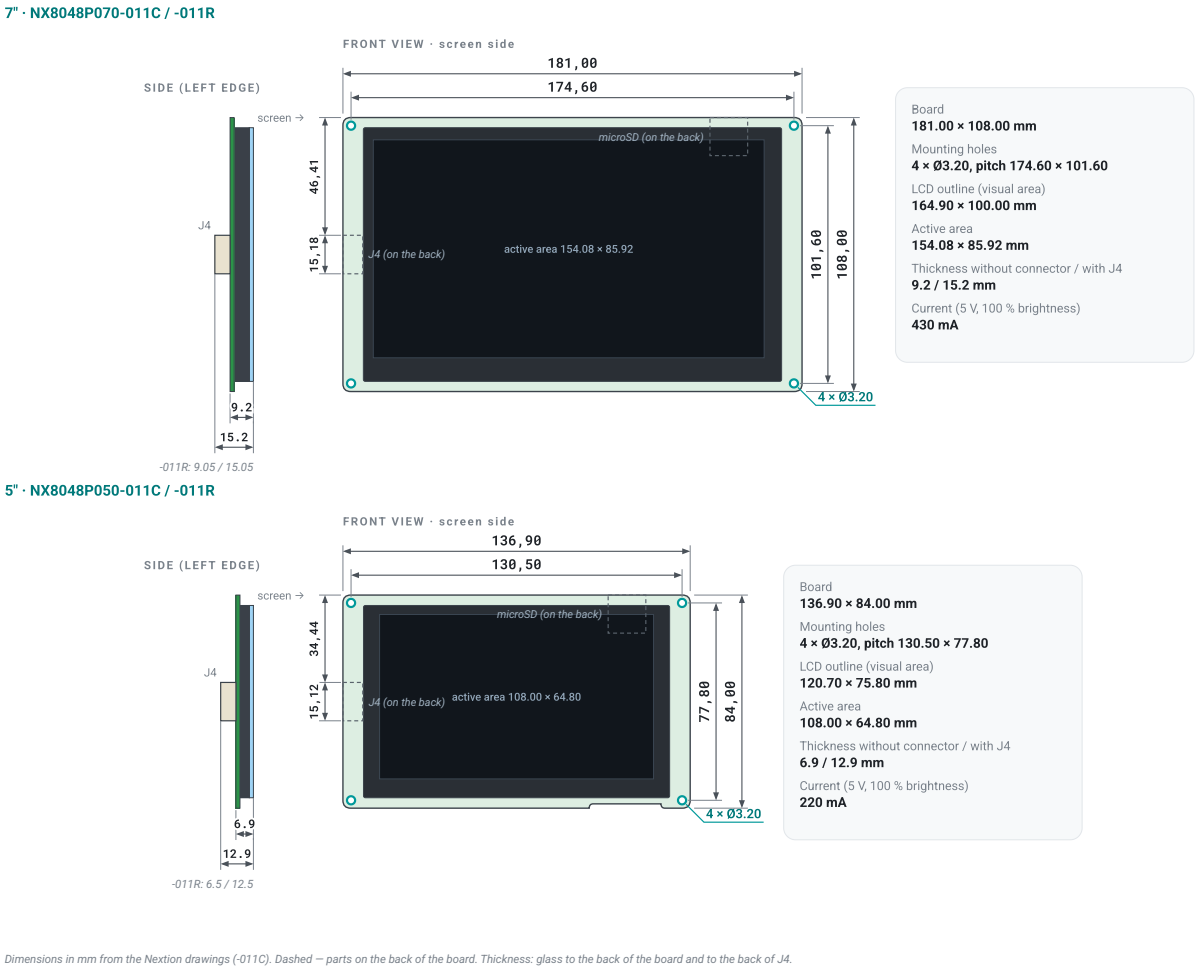


Fig. 59 Nextion Intelligent 7" and 5" displays: screen-side and side views

14 Selector

To use the Leaf selector or separate momentary buttons with position backlighting, an additional selector controller is required. Like the display, it connects to the main Powertrain CAN bus; its +12 V comes from the relay that powers the ECU.

Pin `X1-17` of the selector controller is the reverse-light signal (in both wiring options).

> **NOTE**
> Pin numbering on the selector connector itself may be mirrored. Wire according to the diagrams below — they show the connectors from the wire side. The interface connector (power and CAN) is X3; some older drawings label it X2.

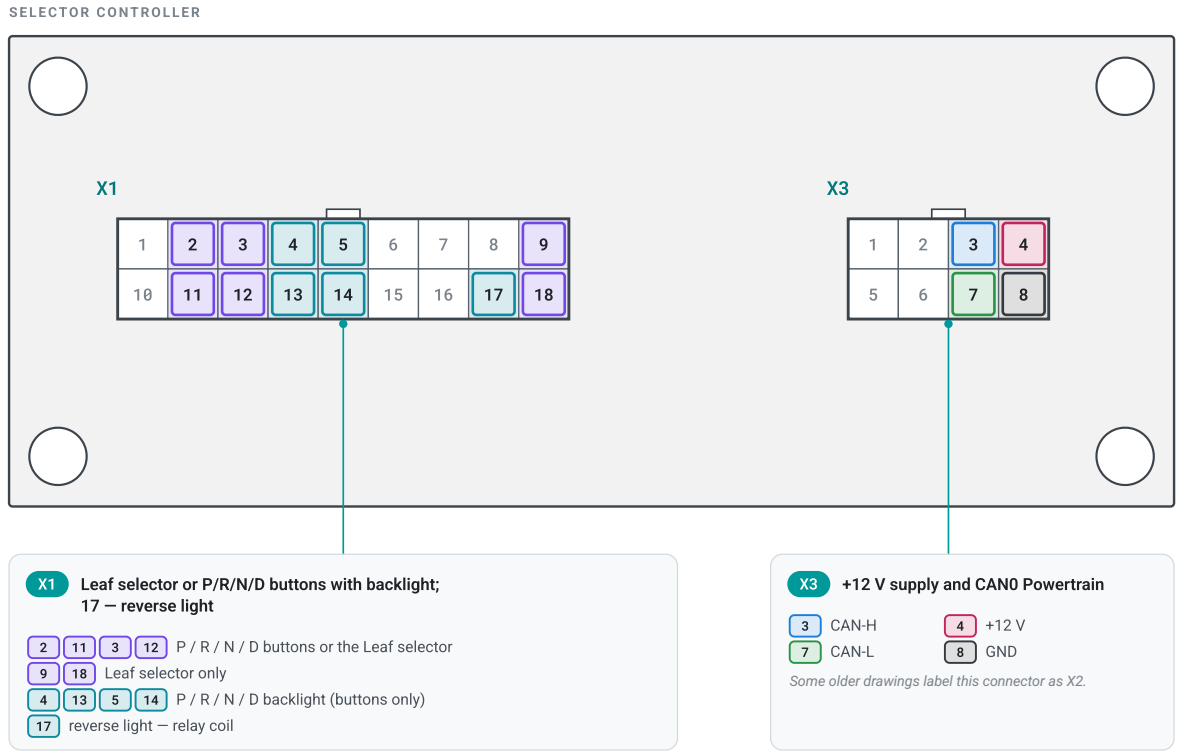
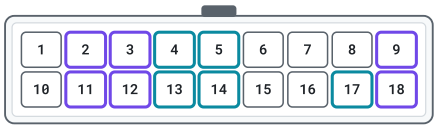


Fig. 60 Selector controller: connectors X1, X3 and mounting holes

Selector controller — connector X1 (selector / buttons)



Board top view — looking into the connector on the board (as in the board drawing)

PIN	WIRE	FUNCTION	DIAGRAMS
9		Leaf selector: → selector pins 1, 7	62
18		Leaf selector: → selector pins 6, 12	62
2		Leaf selector: → selector pin 8 · Buttons: button P	62 64
11		Leaf selector: → selector pin 9 · Buttons: button R	62 64
3		Leaf selector: → selector pin 3 · Buttons: button N	62 64
12		Leaf selector: → selector pin 4 · Buttons: button D	62 64
4		Buttons: P backlight (LED anode)	64
13		Buttons: R backlight (LED anode)	64
5		Buttons: N backlight (LED anode)	64
14		Buttons: D backlight (LED anode)	64
17		Reverse light — relay coil (both configurations)	62

Selector controller — connector X3 (interface)



Board top view — looking into the connector on the board (as in the board drawing)

PIN	WIRE	FUNCTION	DIAGRAMS
4		+12 V — from the relay that powers the ECU	62 64
8		Ground (GND)	62 64
3		CAN-H — Powertrain bus (CAN0)	62 64
7		CAN-L — Powertrain bus (CAN0)	62 64

Fig. 61 Selector controller connectors

14.1 Leaf Selector

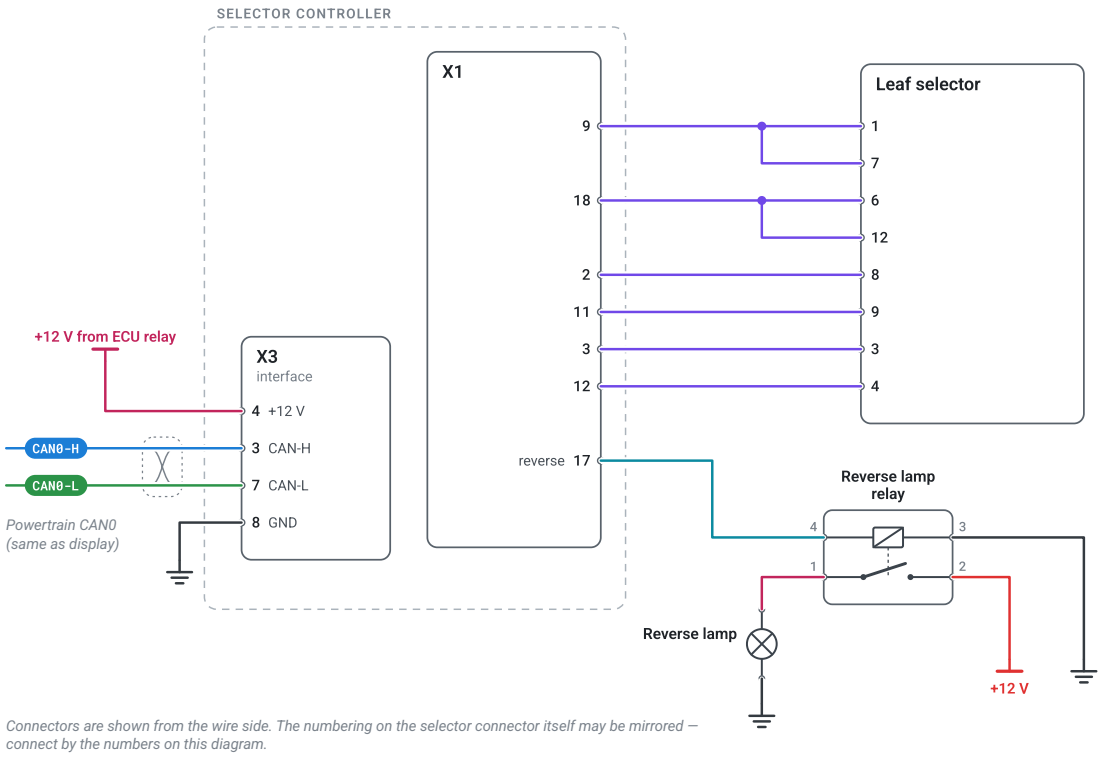
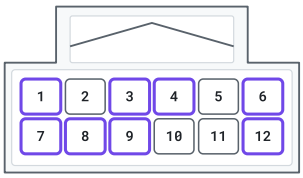


Fig. 62 Selector controller with the Nissan Leaf selector and the reverse-lamp relay



H.S. — harness (wire) side view

PIN	WIRE	FUNCTION	DIAGRAMS
1		To selector controller X1-9 (together with pin 7)	62
7		To selector controller X1-9 (together with pin 1)	62
6		To selector controller X1-18 (together with pin 12)	62
12		To selector controller X1-18 (together with pin 6)	62
8		To selector controller X1-2	62
9		To selector controller X1-11	62
3		To selector controller X1-3	62
4		To selector controller X1-12	62

Fig. 63 Nissan Leaf selector — 12-way connector

14.2 Momentary Buttons

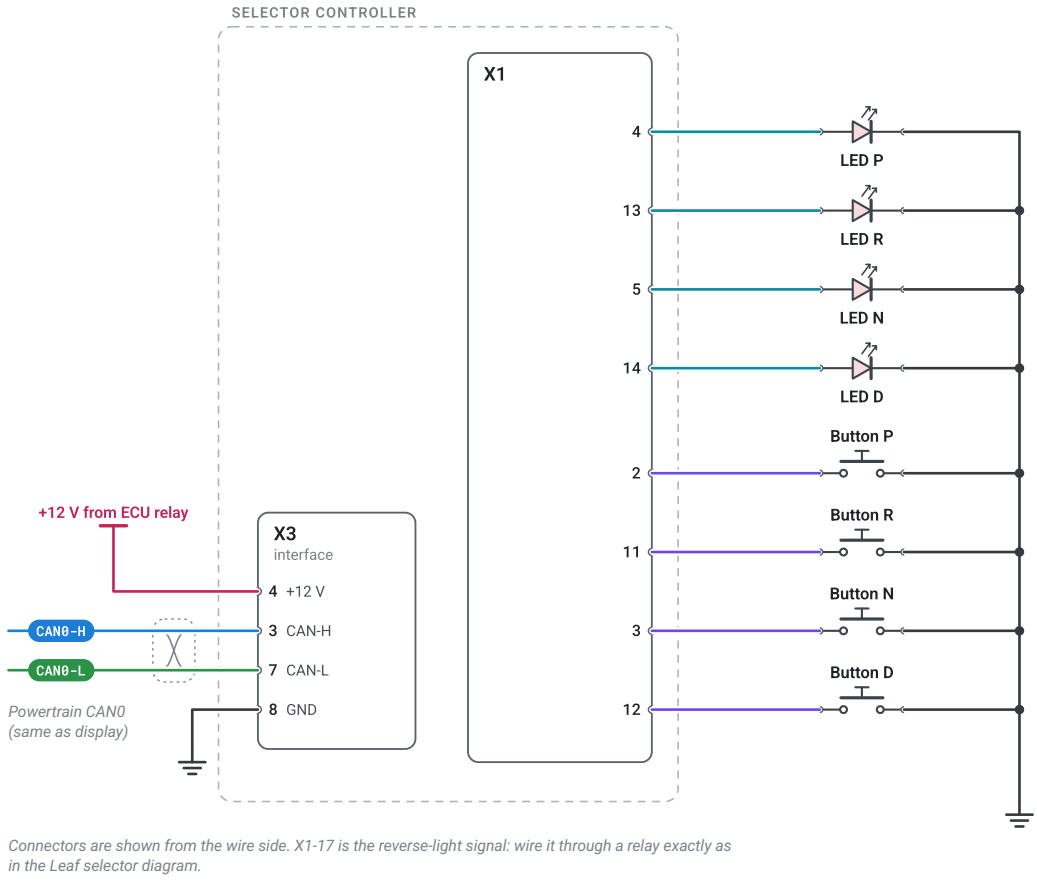
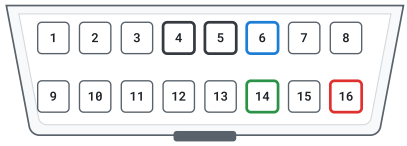


Fig. 64 Selector controller: momentary P/R/N/D buttons with backlight LEDs

15 OBD2 Diagnostics

15.1 OBD2 Connector

An OBD2 socket must be installed for controller setup and diagnostics. Use an ELM327 v1.5 adapter (USB or Bluetooth) or an OBDLink. The web configurator does not support Wi-Fi adapters.



Front view, from the scan-tool side

PIN	WIRE	FUNCTION	DIAGRAMS
4		Chassis ground	
5		Signal ground (GND)	7 9 11 13
6		CAN-H (Powertrain, CAN0)	7 9 11 13
14		CAN-L (Powertrain, CAN0)	7 9 11 13
16		+12 V permanent	7 9 11 13

Fig. 65 OBD2 socket (vehicle side)

15.2 Software

- **E-Engines web configurator** — the main tool: configuration, live parameters, cell map, firmware update. Runs in the browser, see section 16.
- **CanZE** — Android app: cell map and battery temperature.
- **DDT4ALL** — alternative configurator for Windows and Linux.

16 Web Configurator

The E-Engines web configurator runs in the browser at diag.electricengines.com.ua — nothing to install. Through an OBD2 adapter it reads the controller's live parameters, writes settings, saves them to a file and updates the firmware.

16.1 Requirements

- **Chrome or Edge** on Windows, macOS, Linux or ChromeOS. On Android — Chrome with a Bluetooth LE adapter. Firefox, Safari and iOS browsers are not supported.
- **Adapter:**
  - USB or classic Bluetooth — ELM327 v1.5 (a dual-board one is recommended) or OBDLink; connect with the **BT** button;
  - Bluetooth LE (vLinker, Vgate iCar Pro BLE and compatible) — with the **BLE** button;
  - Wi-Fi adapters are not supported.
- After the first visit the site also works offline — you still need the adapter.

16.2 Connecting

1. Switch the ignition on and plug the adapter into the OBD2 socket.
2. Open the configurator. Pair a classic Bluetooth adapter in the system settings first.
3. Press **BT** (USB or Bluetooth) or **BLE** and choose the adapter in the browser's dialog.
4. When the connection is up, the button is highlighted and the log on the Terminal page shows "✓ Стан: Підключено" (connected). The "CAN Bus" indicator flashes yellow on every controller response.
5. Open any page — values start updating automatically. To disconnect, press the highlighted button again.

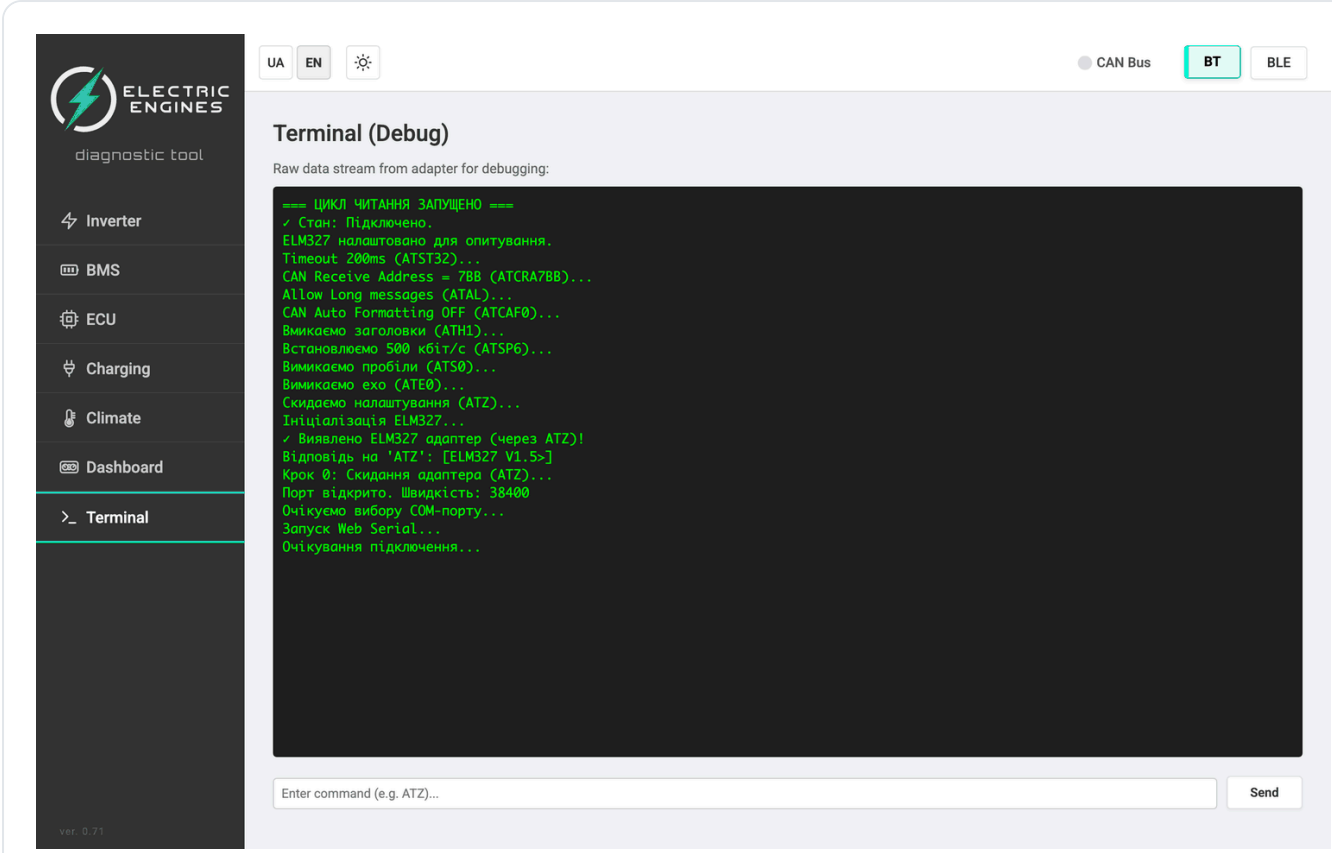


Fig. 66 Web configurator after connecting: adapter log on the Terminal page

> **NOTE**
>
> If the BT/BLE button is highlighted but the "CAN Bus" indicator stays grey, the adapter works but the controller does not answer: check the controller power and the CAN bus. Over BLE the indicator does not flash — go by the values on the pages.

16.3 Configurator Pages

SECTION	CONTENTS
Inverter	HV voltage, status, motor and inverter temperatures, selector position, rpm, pedal position (mV), torque request. Pedal parameters and chart — see 17.2
BMS → Parameters	Pack voltage, SOC, min/max cell voltage, current. Charge/discharge current limits, balancing, **recuperation**, cell voltage thresholds, SOC/SOH, current sensor type
BMS → Cells, Temp Map, SOC Map	All cell voltages, temperatures, SOC calibration by cell voltage
ECU → Settings	Inverter, BMS, selector, start button, charger and CHAdeMO types; wheel circumference; contactor ON voltage; pump and fan temperatures — see 17.3
ECU → Internal	Analog input voltages, input and output states, firmware and bootloader versions
ECU → Brake	Vacuum sensor, vacuum pump on/off thresholds
ECU → Cruise Control	Speed, kP/kI/kD gains, chart and CSV log recording

SECTION	CONTENTS
ECU → Update	Firmware update, export and import of parameters
Charging → AC, CHAdeMO, CCS	Charging state, maximum AC charge current
Climate	Climate system states, temperatures and pressure
Dashboard	Speed / RPM display mode, turn signals, odometer and trip, clock sync
Terminal	Adapter log; manual commands are sent with the Enter key

16.4 Writing Parameters

Each parameter is a row: **Current Value** (what the controller holds now), a field for the new value and a **Write** button. The value is applied and stored in the controller's memory immediately — there is no separate "save" command.

- After writing, check the **Current Value**: that is what confirms the controller accepted the parameter. The "✓" in the log only means the command was sent.
- Enter whole numbers in the unit shown in the field. Cell voltage thresholds and vacuum pump thresholds are entered **in mV** although they are displayed in volts; cruise gains are entered ×10 (15 = 1.5).
- Drop-down lists always open on the first option, not on the current value — pick the option explicitly before pressing Write.
- After changing the inverter type, restart the controller (ignition off and on).

DANGER — HIGH VOLTAGE

Change parameters only with the car stationary in P or N. Writing is not blocked while driving, and, for example, "Max torque" takes effect immediately.

16.5 Saving and Restoring Settings

On the **ECU → Update** page, **Export Parameters** reads the controller settings and downloads `ecu_preset_<date>.json`. **Import Parameters** writes every parameter from the file and reads them back for verification; the result is shown in a message and in the log.

The file contains the pedal and torque parameters, BMS and current sensor settings, inverter and selector types, temperatures, thresholds and cruise control. Charger type, CHAdeMO type, cell count, SOC map and dashboard settings are not included.

WARNING

Importing does not ask for confirmation and does not check which controller the file came from. The SOC charge counter (Ah) is restored to its value at export time as well.

16.6 Firmware Update

1. Put the car in P or N and stop; make sure the 12 V supply is stable. USB is preferable.
2. On **ECU → Update**, click the file area and choose the firmware file (`.ota`) from E-Engines.
3. Leave **Update target** at **Application (FLASH0)**.
4. Press **Upload Firmware** and confirm. Flashing takes about 6–15 min over USB or 15–30 min over BLE.

5. Do not close, reload or hide the tab: in a background tab the browser slows the upload down.

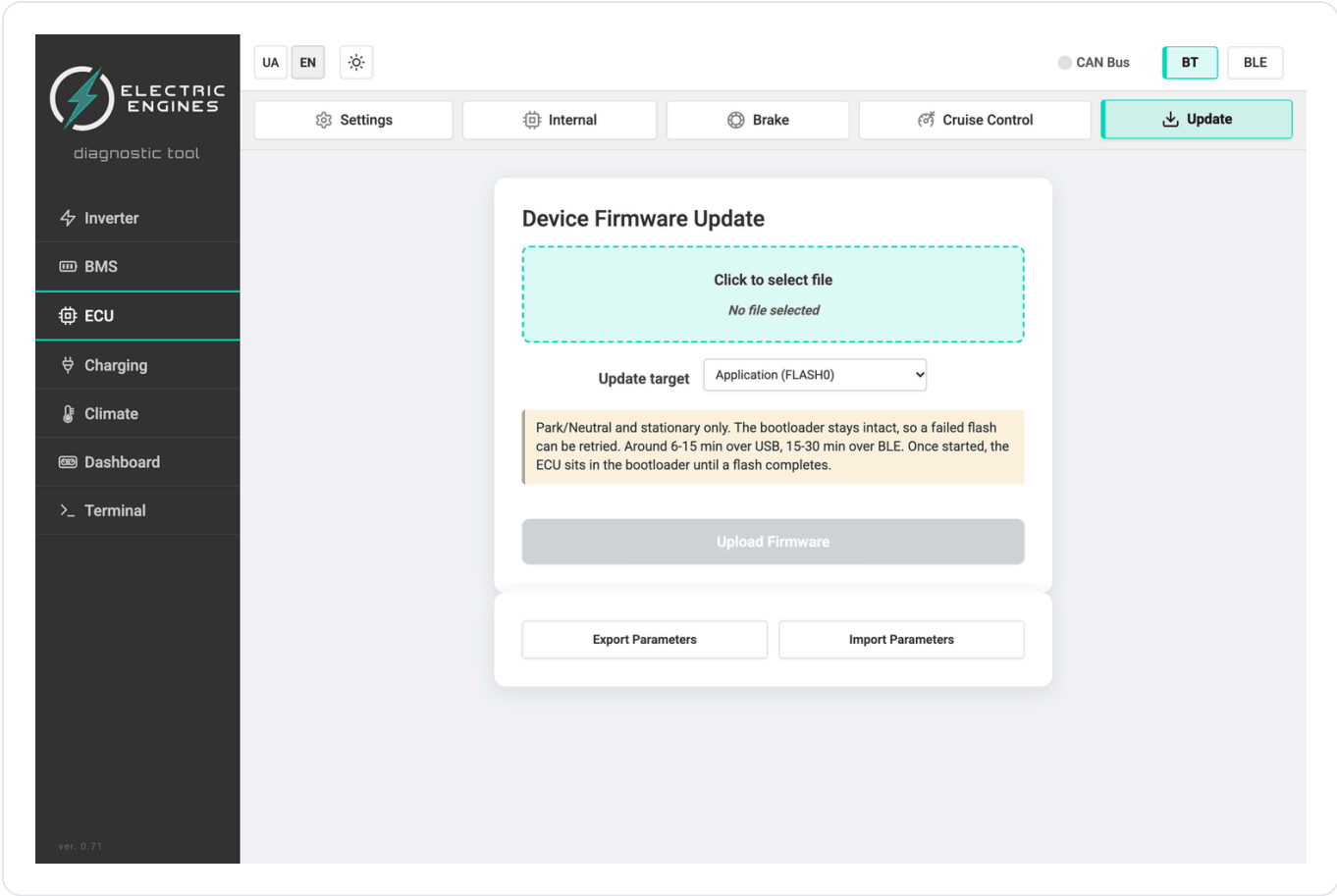


Fig. 67 Firmware update page: file selection, update target, parameter export and import

An application update leaves the bootloader untouched, so a failed upload can simply be repeated.

BOOTLOADER

Choose **Bootloader (FLASH1)** only when E-Engines asks you to: losing power while the bootloader is written leaves the controller unusable, with USB-only recovery.

17 Initial Setup

Once all wiring is connected and carefully checked (make sure no connector is pinned mirrored), proceed with the setup.

17.1 Start-up Procedure

Step 1. Power checks (before connecting the controller to the car)

- ☐ Permanent +12 V on pin `X1-A8` relative to `X1-A7` (ground).
- ☐ +12 V from the ignition switch on pin `X1-C8`.
- ☐ Charge inlet without a gun: resistance between `X1-B7` and `X1-A7` is **4.7 kΩ** or **2.7 kΩ** (depending on inlet type).
- ☐ Charge inlet with the gun plugged in: resistance between `X1-B7` and `X1-A7` is **infinite (OL)**.

Step 2. Wake-up and sleep checks

- ☐ Connect the controller to the car.
- ☐ Turn the ignition on — power appears on `X1-B8`, the IGN relay closes.
- ☐ Turn the ignition off and wait 10 s — power on `X1-B8` disappears, the IGN relay opens.
- ☐ Check wake-up and sleep by plugging and unplugging the AC charging gun.

Step 3. Configuration in the web configurator

> **DANGER — HIGH VOLTAGE**
> Do not continue without configuring the controller — the motor may spin at full speed or not spin at all!

- ☐ Connect to the controller with the web configurator (16.2).
- ☐ **ECU → Settings:** set the inverter, BMS and selector types and the other parameters (17.3).
- ☐ **Inverter:** motor and inverter temperatures, no inverter error.
- ☐ **Charging → AC Charging:** PDM module temperature.
- ☐ Selector: the position must change on the **Inverter** page.
- ☐ Pedal: **Pedal position** (mV) on the **Inverter** page must rise as you press the pedal.

Step 4. Power relay and battery polarity checks

1. Disconnect the traction battery from the powertrain (pull the service disconnect).
2. Connect a voltmeter to the battery output after the contactors.
3. Press Start — the voltmeter must show voltage of the correct polarity.
4. The controller enters protection mode and reports a precharge error — this is expected.
5. Restart the controller (let it go to sleep and switch it on again).
6. Repeat the test measuring resistance across the positive contactor with a multimeter:
   - before start — infinite;
   - at start — 30–50 Ω;
   - after one second — 0 Ω.

17.2 Accelerator Pedal Setup

The pedal parameters are on the **Inverter** page, recuperation is on **BMS → Parameters**. Pedal position is the pedal signal voltage in millivolts; the live value is shown in the **Pedal position** row.

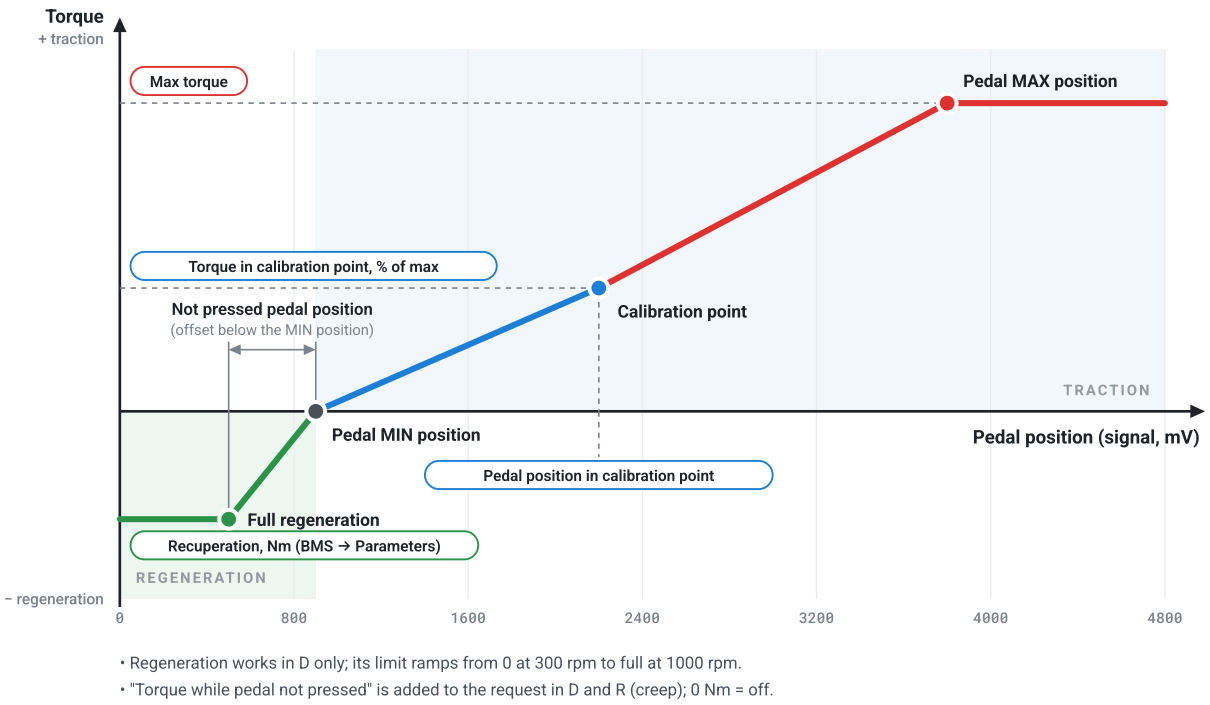


Fig. 68 Accelerator pedal map: how the setup parameters define torque and regeneration

PARAMETER	DESCRIPTION
Pedal MIN position, mV	Where traction starts. Below it is the regeneration zone
Pedal MAX position, mV	Where "Max torque" is reached
Pedal position in calibration point, mV	The break point of the curve: two linear segments before and after it
Torque in calibration point, %	Torque at the calibration point as a percentage of "Max torque"
Not pressed pedal position, mV	How many millivolts **below** the MIN position full regeneration is reached. This is an offset, not an absolute pedal position
Torque while pedal not pressed, Nm	Torque added to the request in D and R (creep); 0 = off
Max torque, Nm	Torque at the MAX position and beyond
Recuperation, Nm (BMS → Parameters)	Full regeneration torque. Works in D only and ramps from 0 at 300 rpm to the full value at 1000 rpm

Setup order:

1. Without touching the pedal, note the **Pedal position** reading; then press the pedal to the floor and note it again.

2. Set **Pedal MIN position** above the released reading: the span between them becomes the regeneration zone — the wider it is, the more gradually regeneration fades as you press. Set **Pedal MAX position** slightly below the fully pressed reading.
3. Set **Not pressed pedal position** to the difference between the MIN position and the released reading.
4. Use the calibration point to tune the sensitivity at the start of the pedal travel, then set "Max torque" and "Recuperation".

> **WARNING**
>
> Do not set "Not pressed pedal position" lower than the difference from step 3: the firmware does not clamp regeneration, and with the pedal released it would exceed the set value.

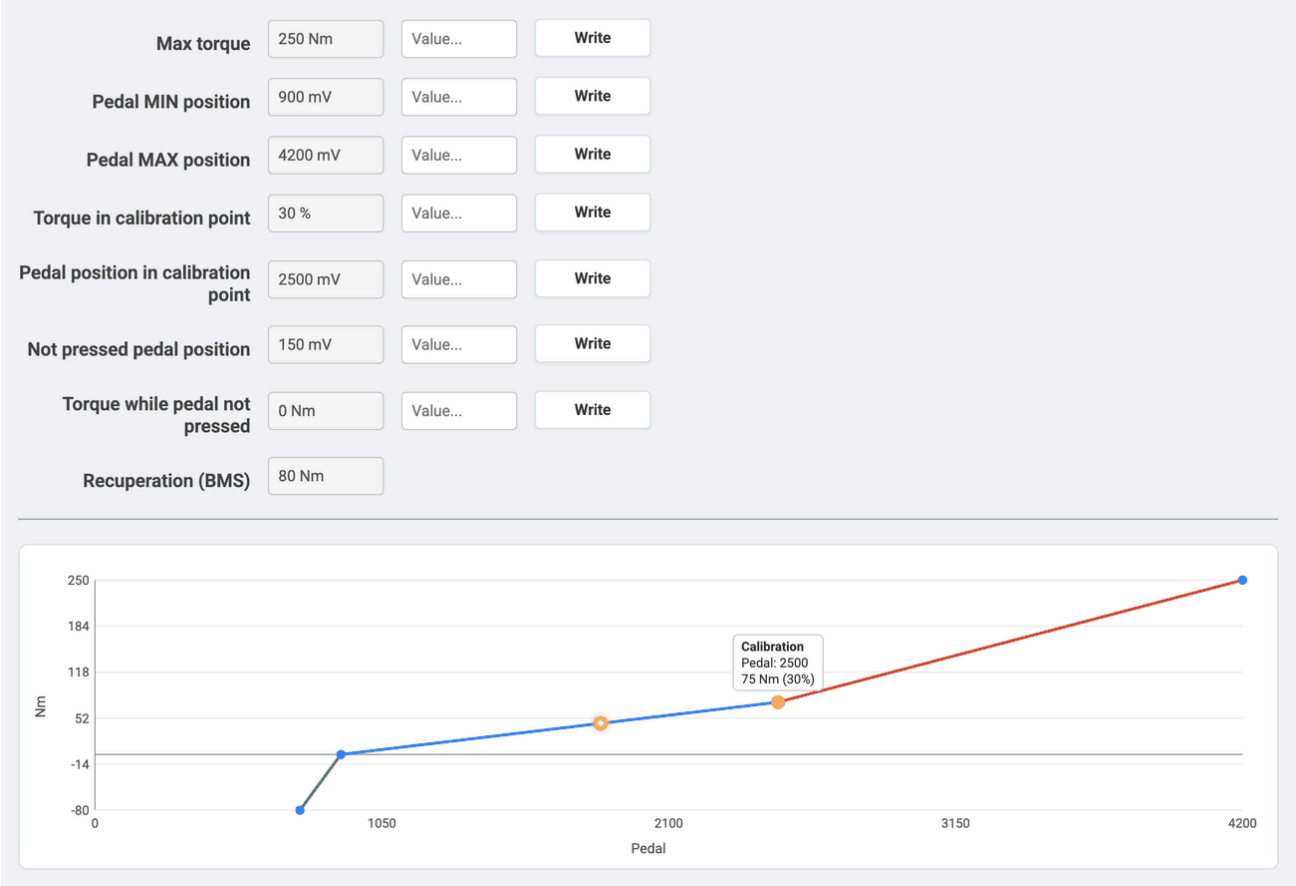


Fig. 69 Pedal parameters and chart in the web configurator (Inverter page)

17.3 Controller Configuration

After the pedal is set up, open **ECU → Settings** and check every parameter.

PARAMETER	DESCRIPTION
Wheel circumference	Wheel circumference in mm (used for the speed calculation)
Contactor ON voltage, V	The main contactor will not close if the inverter voltage after precharge is below this value — protection against a broken power wire or precharge resistor
Type selector	**Button** — toggle switch or buttons on the controller inputs (5.1); **CAN** — separate selector controller (14); **CAN inverted** — the same with D and R swapped; **PSA**

PARAMETER	DESCRIPTION
Type start button	With or without fixation (latching / momentary)
Type inverter	Leaf AZE0, ZE0, ZE1 or Tesla M3 — select exactly the one installed. Restart the controller after changing it
Charger type	Leaf PDM, Tesla PCS or OFF
CHAdeMO type	OFF, internal (CAN1 or CAN3) or external CHAdeMO controller
Pump start temperature, °C	See section 10
Fan ON temperature, °C	Separately for high and low speed
Type BMS	OFF, Volt Gen 1, Leaf, Orion 2, VW ID, JK-BMS, Deye HV

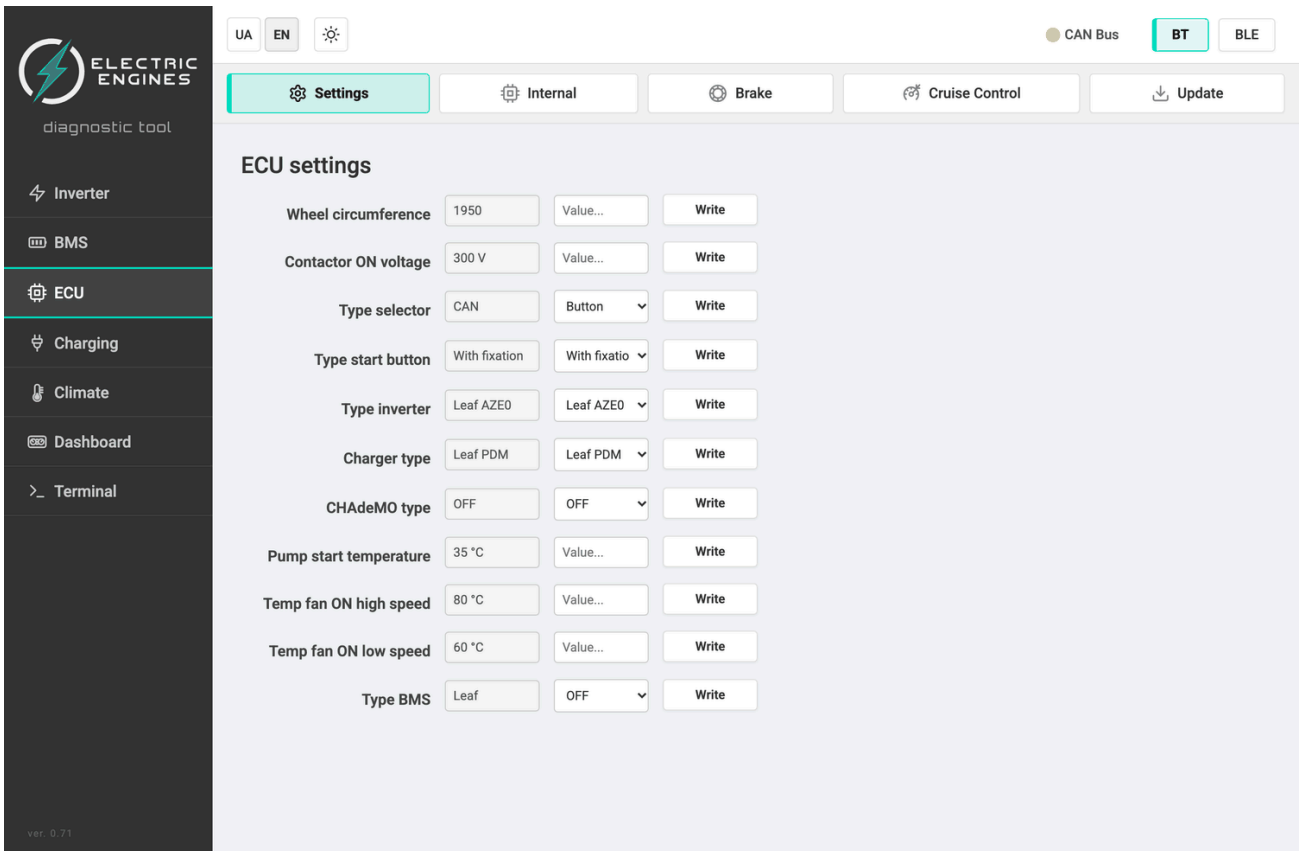


Fig. 70 The ECU → Settings page in the web configurator

18 FAQ / Troubleshooting

The web configurator does not connect to the adapter

- Use Chrome or Edge; Firefox, Safari and iOS are not supported.
- Pair a classic Bluetooth adapter in the system settings first and connect with BT. A Bluetooth LE adapter connects with BLE.
- Wi-Fi adapters are not supported.
- The adapter must be ELM327 v1.5 compatible (a dual-board one is recommended) or an OBDLink. The error reason is shown in the log on the Terminal page.

Values do not update, the "CAN Bus" indicator is grey

The adapter is connected but has no CAN communication with the controller. Possible causes:

- no power to the controller;
- CAN bus wired incorrectly (CAN-H and CAN-L swapped);
- controller fault.

All voltages from the BMS = 0 V

The controller does not see the BMS. Check the CAN bus wiring and the BMS power supply.

One cell reads much higher and its neighbour much lower

Most likely a poor balancing-wire contact with the cell. If it happens under load — a poor contact in the power busbars between cells. Check every connection in the battery.

The motor whines but does not turn

With the original gearbox, check that it is not locked: turn both driven wheels in the same direction. If they don't turn, rotate the parking-lock shaft counter-clockwise.

The motor turns only in reverse

This happens when a ZE0 inverter is installed but a different inverter type is selected in the controller settings.

The car briefly accelerates or regenerates by itself while driving

Seen with resistive pedals: over time the resistor tracks wear and give an unstable voltage. **Solution:** replace the pedal.

The traction battery contactors weld

1. Check the contactor wiring.
2. Check the precharge resistor value.
3. Check the battery activation voltage parameter (above 200 V) in the configurator — it protects against a failed relay or precharge resistor.

The DC-DC converter in the PDM burned out

A ground fault (especially when the traction battery negative is tied to the 12 V system negative) can destroy the PDM.

DANGER — HIGH VOLTAGE

Always check the insulation resistance of the traction battery to the chassis.

Revision History

REVISION	CHANGES
2.0	New format: interactive web edition and PDF in Ukrainian and English. All diagrams redrawn in one style, connector views with pinouts added. Web configurator chapter added; setup moved from DDT4ALL to the web configurator. Enclosure drawing (dimensions, mounting holes, label), display and selector controller housings, the Nextion 7" and 5" pinout and dimensions and the LEM current sensor drawing added; full wiring diagrams for each inverter — Gen 1, Gen 2, Gen 3 (12 V supply, CAN0, PDM, charge inlet, resolver and motor temperature sensor) instead of the basic diagram and the cross table; Gen 3 uses Nissan's numbering (the old picture numbered the connector differently and wrongly marked pins 3 and 9 as ground); the resolver connector on the motor (the same for Gen 1–3); the built-in 120 Ω resistors are drawn inside the controller (every CAN channel) and the Leaf inverter; mini ISO automotive relay pinout; LEM CAB 500 current sensor (the drawing shows the connector and its pin numbering); JK BMS (satellites, connectors P1/P2 and PWR supply; CAB 500 is required with JK). Added the Tesla Model 3 / Model Y rear inverter (Highland firmware): wiring diagram and logic-connector pinout. LBC connector locations redrawn. ECU pinout moved to section 1.1, appendices removed. Leaf battery wiring now shows the connection to the Leaf ECU; current sensor connector LB5 added. Type 1, Type 2 and CHAdeMO pinouts from both the inlet and the gun side, the inlet drawn with its mounting flange and holes; every connector states its view side; the controller's built-in 120 Ω resistor is drawn inside the ECU on the diagrams; DC polarity on the CHAdeMO inlet corrected. New pedal chart matching the firmware: "Not pressed pedal position" is an offset from the MIN position. Wheel circumference in mm. Figure captions corrected; the ECU pinout is compiled from the wiring diagrams.
1.13	Translated to English. Improved structure: contents, numbered sections, figure captions, full pinout tables.
1.12	Added information on the selector and reverse light.
1.11	Added the selector CAN diagram.
1.10	Added FAQ and start-up procedure. Added the selector wiring diagram.
1.9	Fixed an error in the fan diagram. Added more information about IGN KEY.
1.8	Added the Volt BMS pinout.
1.7	Added Leaf BMS information. Fixed an error in the 12 V supply diagram with the charge-inlet resistor. Fixed an error in the current sensor diagram (pin A7 X1 → C7 X1). Added revision history.

ELECTRIC ENGINES

www.electricengines.com.ua

Interactive version with circuit highlighting:

https://diag.electricengines.com.ua/manual/en/

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